

Carbohydrates

Q.1 What is the composition of Benedict's qualitative reagent?

Benedict's qualitative reagent contains:

1. Copper sulphate: This supplies cupric (Cu^{++}) ions.
2. Sodium carbonate: This makes the medium alkaline.
3. Sodium citrate: It prevents the precipitation of Cu^{++} ions as $\text{Cu}(\text{OH})_2$ or CuCO_3 by, forming loosely bound complex with Cu^{++} ions, i.e. (Cu^{++} sodium citrate complex) which on dissociation gives a continuous supply of Cu^{++} ions.

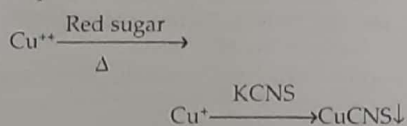
Q.2 What is the composition of Benedict's quantitative reagent?

Benedict's quantitative reagent contains:

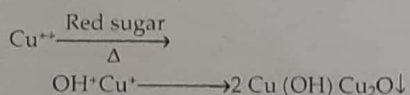
1. Copper sulphate.
2. Sodium carbonate.
3. Sodium citrate.
4. Potassium ferrocyanide: It keeps cuprous oxide (Cu_2O) in the solution.
5. Potassium thiocyanate: It precipitates Cu^+ ions as cuprous thiocyanate (CuCNS).

Q.3 Why Benedict's quantitative reagent gives a white precipitate while Benedict's qualitative reagent gives a brick-red precipitate?

Benedict's quantitative reagent gives a white precipitate of CuCNS .



whereas Benedict's qualitative reagent gives a red precipitate of Cu_2O .



Q.4 What are the substances which give false Benedict's test?

Glucuronates, salicylates (Aspirin), vitamin C, homogentisic acid, etc.

Q.5 What is the difference between Benedict's qualitative test and Barfoed's test?

Benedict's qualitative test is a reduction test carried out in alkaline medium, whereas Barfoed's test is a reduction test carried out in acidic medium.

Q.6 Define carbohydrates in chemical term.

Carbohydrates are defined chemically as aldehyde or ketone derivatives of the higher polyhydric alcohols or compounds which yield these derivatives on hydrolysis.

Q.7 How will you classify carbohydrates?

- Carbohydrates are classified into four major groups:
- Monosaccharide (simple sugars): They cannot be hydrolyzed into simpler forms.
- Disaccharides: They yield two molecules of same or different monosaccharide units on hydrolysis.
- Oligosaccharides: They yield three to six molecules of monosaccharides on hydrolysis.
- Polysaccharides (glycans): They yield more than 6 molecules of monosaccharides on hydrolysis.

Q.8 How are monosaccharides further classified?

- Monosaccharides are further classified into 2 groups depending on:
- The number of carbon atoms they possess, e.g. trioses, tetroses, pentoses, hexoses, etc.
- Whether aldehyde ($-\text{CHO}$) or ketone ($-\text{CO}$) group is present, e.g. aldoses, ketoses.

Q.9 Give an example of an aldohexose and a ketohexose which is of biological importance.

- Aldohexose: D-Glucose
- Ketohexose: D-Fructose

Q.10 How will you classify polysaccharides?

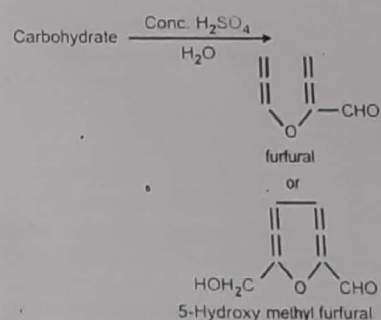
- Polysaccharides are classified into 2 main groups:
- Homopolysaccharides (homoglycans): Polymer of same monosaccharide units, e.g. starch, glycogen, insulin, dextrans, cellulose, etc.
- Heteropolysaccharides (heteroglycans): Polymer of different monosaccharide units or their derivatives, e.g. mucopolysaccharides (MPS).

Q.11 What is the general test of carbohydrates?

Molisch reaction, Anthrone test.

Q.12 What is the principle of Molisch test?

Carbohydrates on treatment with concentrated sulphuric acid undergoes dehydration to give furfural or furfural derivatives which on condensation with β -naphthol gives a characteristic purple or violet color ring at the junction.



Q.13 Is iodine test for polysaccharides a physical or chemical reaction?

It is physical reaction in which iodine molecules get adsorbed on the surface of polysaccharides.

Q.14 What are reducing sugars?

Reducing sugars are those which possess free aldehydic or ketonic group in their structure.

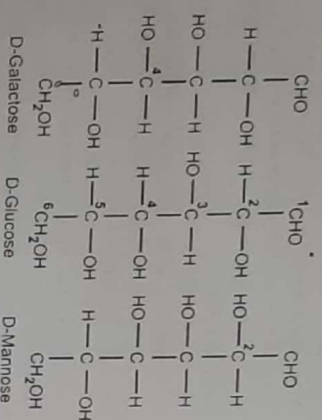
OR

Sugars having free anomeric carbon atom in their structure are called reducing sugars.

Q.15 What are epimers? Give examples.

- Carbohydrates that differ in their configuration around a specific carbon atom other than the carbonyl carbon atom are called epimers.

Glucose and galactose are epimers as they differ in their configuration around C-4 carbon atom. Similarly, glucose and mannose are epimers as they differ around C-2 carbon atom.



Q.16 What is an asymmetric carbon?

A carbon atom to which four different atoms or groups of atoms are attached is said to be an asymmetric carbon.

Q.17 What are the effects of presence of asymmetric carbon in a compound?

- The presence of asymmetric carbon atoms in a compound produces the following effects:
- Gives rise to the formation of stereoisomers of that compound.
- Also confers optical activity to the compound.

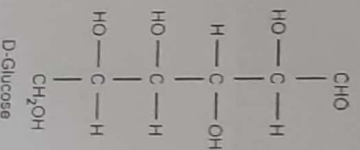
Q.18 What are stereoisomers?

The compounds which are identical in composition and differs only in spatial configuration are called stereoisomers. Two such stereoisomers of glucose, D-glucose and L-glucose are mirror-images of each other.

Q.19 What are optical isomers?

When a beam of plane polarized light is passed through sugar solution exhibiting optical activity, it will be rotated to the right or left according to the type of the compound. Such compounds are called "optical isomers" (or "enantiomorphs").

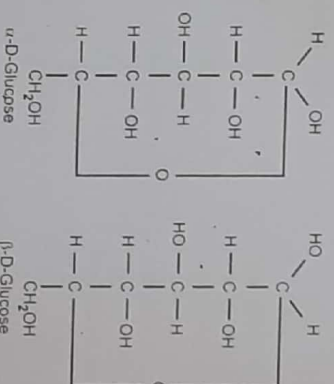
Q.20 Draw the structure of L-glucose.



Q.21 What are anomers? Give examples.

Carbohydrates that differ only in their configuration around the carbonyl carbon atom are called anomers. The carbonyl carbon atom is called the anomeric carbon atom.

α -D-glucose and β -D-glucose are the anomeric form of D-glucose.



Q.22 How will you determine the D and L series of the sugars?

If the hydroxyl group on the highest asymmetric carbon atom is right oriented then the sugar belongs to D-series and if the hydroxyl group is left oriented then it belongs to L-series.

Q.23 What is mutarotation?

The change in specific rotation of an optically active solution without any change in other properties is known as mutarotation.

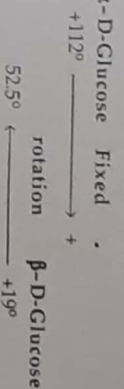
Q.24 Give details of mutarotation.

Crystalline glucose is α -D-glucopyranose. The cyclic structure is retained in solution but isomerism takes place about position-1, i.e. about anomeric carbon atom to give a mixture of α -glucopyranose (38%) and β -glucopyranose (62%). This equilibrium is

accompanied by optical rotation, i.e. mutarotation, as hemiacetal ring opens and reforms with change of position of H and -OH group on carbon-1.

Q.25 What are the optical rotations shown by glucose in solution?

The glucose solution shows rotation according to its form: a form shows +112° and b-form shows +19°. When the solution has an equilibrium mixture of a and b forms, it shows fixed rotation of +52.5°.



Q.26 What is meant by pyranose form?

The pyranose forms of the sugars are internal hemiacetals formed by combination of the aldehyde or ketone group of the sugar with the -OH group on the 5th carbon atom from the aldehyde or ketone group.

Q.27 What is meant by the furanose form of sugar?

The furanose forms of sugars are formed by reaction between the aldehyde or ketone group with the -OH group on the 4th carbon from the aldehyde or ketone group.

Q.28 Which property of reducing sugars best explains the ring or cyclic structure of the carbohydrates?

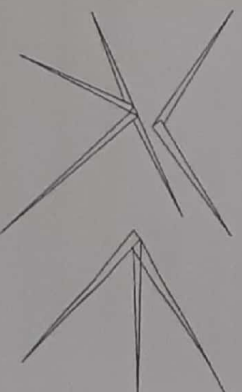
Mutarotation.

Q.29 Why glucose and fructose give the same osazones?

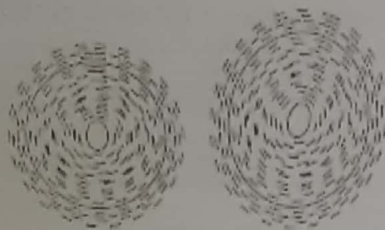
Glucose and fructose differ in their structure at first two carbon atoms, i.e. C₁ and C₂ only. In osazone formation these two carbon atoms takes part in the reaction and during osazone formation the structural dissimilarity at C₁ and C₂ disappears. Hence, they give the same osazone.

Q.30 What are the shapes of osazones of glucose, fructose, lactose and maltose?

Glucose: Needle shaped.
Fructose: Needle shaped.
Lactose: Cotton ball shaped.
Maltose: Sunflower shaped.



NEEDLE SHAPED



LACTOSE (COTTON BALL-SHAPED)



LACTOSE (COTTON BALL-SHAPED)

Q.31 What are the reduction products of glucose, mannose, galactose and fructose?

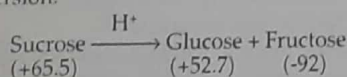
- On reduction the monoaccharides produce sugar alcohols. Thus,
- D-Glucose → D-Sorbitol
- D-Galactose → D-Dulcitol
- D-Mannose → D-Mannitol
- D-Fructose → D-mannitol + D-Sorbitol

Q.32 Why sucrose is a non-reducing sugar?

Sucrose consists of glucose and fructose which are linked through their reducing sugars, i.e. aldehyde group of glucose is linked to keto group of fructose. As a result of this linkage, both the reducing groups are blocked. Hence, sucrose is a nonreducing sugar.

Q.33 What is inversion?

The process by which dextrorotatory sucrose is converted to a levorotatory mixture of glucose and fructose is called inversion.



Q.34 Why sucrose is called an invert sugar?

Sucrose on hydrolysis gives glucose and fructose. Fructose has greater specific rotation than glucose. The resulting mixture is levorotatory. The mixture is known as invert sugar.

Q.35 Name one biological fluid which is rich in fructose. What is the source of fructose in this fluid and its importance?

- Seminal fluid is rich in fructose.
- **Source:** It is formed from glucose in the seminiferous tubular epithelial cells.
- **Importance:** Spermatozoa utilizes fructose for energy.

Q.36 What is the fate of disaccharide (i.e. sucrose) when injected into the blood?

Sucrose will be excreted as such in the urine as there is no enzyme sucrase present in the blood to hydrolyse it.

Q.37 What are the components of lactose?

Lactose contains glucose and galactose.

Q.38 What are the components of sucrose?

Sucrose contains glucose and fructose.

Q.39 What are the components of maltose?

Maltose contains two molecules of glucose only.

Q.40 What is the nature of linkage in lactose?

β-(1,4) linkage
β-Galactose → C₄ glucose.

Q.41 What is the nature of linkage in maltose?

α-(1,4) linkage.

Q.42 What is the nature of linkage in sucrose?

α-(1,2).

Q.43 What is the nature of linkage in starch?

Starch is formed by amylose and amylopectin. Amylose is nonbranching. Amylopectin consists of 1,4 linkage but at branching point 1,6 linkage is present.

Q.44 What is the nature of linkage in glycogen?

Glycogen has α-(1,4) linkage but at branching point α-(1,6) linkage is present.

Q.45 What is the nature of linkage in cellulose?

β-(1,4).

Q.46 Name the components of starch.

Starch contains two components:

1. Amylose
2. Amylopectin.

Q.47 What is difference between amylose and amylopectin?

Amylose	Amylopectin
1. Linear molecule containing α-(1,4) linkages	Branched molecule containing α-(1,4) linkages and α-(1,6) linkages at branching.
2. Blue color with iodine	Violet color with iodine.
3. Water soluble	Sparingly soluble.

Q.48 What are hydrolytic products of starch and state their reaction with I₂ solution?

Hydrolysis of starch yields succession of polysaccharides of diminishing molecular size as follows:

Course of hydrolysis	Reaction with I ₂ solution
Starch	Blue
↓	
Soluble starch	Blue
↓	
Amylodextrin	Purple
↓	
Erythrodestrin	Red
↓	
Achroodestrin	Colorless
↓	
Maltose	

Q.49 What is aglycone?

The noncarbohydrate portion of a glycoside is called aglycone.

Q.50 What is the difference between starch and cellulose?

In starch, the glucose units are linked by α-(1,4) glucosidic linkages whereas in the cellulose, the glucose units are linked by β-(1,4) glucosidic linkages.

Q.51 Why cellulose is not utilized by human body?

The enzyme responsible for the cleavage of β-(1,4) linkages in the cellulose is absent in the human system. Hence, it cannot be utilised.

Q.52 What are amino sugars?

Sugars containing an NH₂ group in their structure are called amino sugars. The alcoholic OH group on carbon 2 is usually replaced by -NH₂ group. Examples: D-Glucosamine, D-Galactosamine.

Q.53 What are glycosides?

Glycosides are compounds containing a carbohydrate and a non-carbohydrate residue in the same molecule; Carbon 1 of carbohydrate is attached to the non-carbohydrate residue by an acetyl linkage.

Q.54 What are dextrans?

Dextrans are the partially degraded breakdown products of starch.

Q.55 Differentiate dextrans and dextrans.

Dextrins	Dextrans
• Are hydrolytic products of starch	• Synthetic polymer of D-glucose
• Used in infant feeding	• Used as a plasma expander when given IV in cases of haemorrhage (blood loss), it increases the blood volume.

Q.56 What is the difference between starch and glycogen?

Starch	Glycogen
1. Plant origin.	1. Animal origin.
2. It is a branched molecule. Branching occurs after every 20-24 glucose units.	2. Highly branched than starch. Branching occurs after every 8-10 glucose units.
3. Blue color with iodine solution	3. Red color with iodine solution.

Q.57 What are mucopolysaccharides?

Mucopolysaccharides are acidic substances containing uronic acids and N-acetylated amino sugars in combination with proteins.

Q.58 Give few examples of mucopolysaccharides.

Hyaluronic acid, heparin, chondroitin sulphate, dermatin sulphate.

Q.59 What are proteoglycans?

Proteoglycans are conjugated proteins (called "core" proteins) covalently linked to any of the glycosaminoglycans (GAGs). The amount of carbohydrates in proteoglycans is much greater (up to 95%) as compared to glycoproteins.

Q.60 State the type of linkages found in proteoglycans?

- Three types of linkages with core protein and GAG is observed. They are:
- **O-glycosidic linkage:** between N-acetyl galactosamine (Gal NAc) and serine/threonine of core protein.
- **N-glycosyl amine linkage:** Formed between N-acetyl glucosamine (Glc NAc) and amide N of asparagine (Asn) of core protein.
- **O-Glycosidic linkage with xylose:** formed between xylose and serine of core protein.

Q.61 What are the repeating units of hyaluronic acid?

Glucuronic acid: N-acetylglucosamine.

Q.62 What are the various pathways by which glucose is utilized?

1. Glycolysis followed by tricarboxylic acid cycle.
2. Hexose monophosphate shunt pathway.
3. Conversion to glycogen.
4. Conversion to galactose and then to lactose.
5. Give rise to nonessential amino acids.
6. Conversion to fat.

Q.63 What is glycolysis. What are rate-limiting enzymes of glycolysis?

- Glycolysis can be defined as oxidation of glucose or glycogen to pyruvate and/or lactate. The rate-limiting enzymes of glycolysis are:

- Hexokinase
- Phosphofructokinase
- Pyruvate kinase.

Glycolysis is sometimes also known as Embden-Meyerhof Pathway (EM pathway)

Q.64 What is the role of O_2 in glycolysis?

The glycolytic pathway is unique in the sense that it can occur in presence of O_2 if available ("aerobic" phase) and it can function also in absence of O_2 ("anaerobic" phase).

Q.65 In which part of the cell glycolysis occurs and where the enzymes are located?

Glycolysis occurs in cytosol and enzymes involved are cytosolic (extramitochondrial).

Q.66 State the biomedical importance of glycolysis.

- Provides energy.
- Cardiac muscle has poor glycolytic activity and poor survival under conditions of ischemia.
- Rate of glycolysis is very high in fast growing cancer cells. Enhanced glycolysis produces more pyruvic acid than TCA cycle can handle. Hence pyruvic acid accumulates and forms lactic acid producing local lactic acidosis which is congenial for certain cancer therapy.
- Inherited enzyme deficiencies like hexokinase and pyruvate kinase produce hemolytic anemia.

Q.67 Name the steps of glycolysis where ATP is consumed.

ATP is utilized for phosphorylations:

- For conversion of glucose $\rightarrow$ glucose-6-P
- For conversion of fructose-6-P $\rightarrow$ fructose 1,6-bi-P
- Body spends two ATP molecules (-2 ATP)

Q.68 Which enzymes are required for phosphorylations?

- For phosphorylation of glucose: Hexokinase/and/or glucokinase enzymes.
- For phosphorylation of fructose-6-P: Phosphofructokinase enzyme.

Q.69. State Five differences between hexokinase and glucokinase.

Hexokinase	Glucokinase
• Found in all tissues	• Found only in liver
• More stable	• More labile
• Non-specific, can phosphorylate any of the hexoses	• Specific only for glucose
• Km is low, hence high affinity for glucose	• Km is high, hence low affinity for glucose
• Main function to make available glucose to tissues for oxidation at lower blood glucose level	• Main function to clear glucose from blood after meals and at blood levels greater than 100 mg per dl

Q.70 Which is the most energy-yielding step in glycolytic pathway?

Oxidation of glyceraldehyde-3-P, in presence of O_2 , by the enzyme glyceraldehyde-3-P dehydrogenase which is NAD^+ dependent. 2 NADH when oxidized in ETC gives 6 ATP.

Q.71 What is the end-product of glucose oxidation by glycolysis?

Pyruvic acid.

Q.72 Name the inhibitors of glycolysis.

- **Iodoacetate and iodoacetic acid:** Inhibit glyceraldehyde-3-P-dehydrogenase.
- **Arsenite:** Inhibits phosphoglycerate kinase indirectly and no ATP is formed at substrate level.
- **Fluoride:** Inhibits the enzyme enolase.

Q.73 Name the tissues which solely depends for energy from glycolysis.

- Red blood cells and,
- Brain and nervous tissue.

Q.74 Enumerate the sources of pyruvic acid (PA) in the body.

- By glycolysis—principal source.
- Oxidation of LA $\rightarrow$ PA in presence of O_2 .
- Deamination of alanine.
- Other pyruvic acid forming amino acids, e.g. glycine, serine, cysteine/cystine, threonine.
- Decarboxylation of oxalo- acetic acid (OAA)
- From malic acid by malic enzyme.

Q.75 Enumerate the fate of pyruvic acid (PA) in the body.

- Oxidative decarboxylation of PA to form acetyl CoA in presence of O_2 .
- Reduction of PA $\rightarrow$ LA in absence of O_2 .
- Amination to form alanine.
- Conversion to glucose (gluconeogenesis).
- Conversion to malic acid
- Formation of oxaloacetate (OAA) by "CO₂-fixation reaction".

Q.76 State the irreversible steps in glycolysis.

- Glucose-6-P $\rightarrow$ Glucose.
- Fructose-1, 6-bis-P $\rightarrow$ Fructose-6-P
- Phosphoenol pyruvate $\rightarrow$ Enol-pyruvate.

Q.77 What is anaplerotic reaction or anaplerosis?

- A sudden influx of PA or acetyl CoA to the TCA cycle might seriously deplete the supplies of OAA required for the citrate synthase reaction.
- Two reactions that are auxiliary to TCA cycle operate to prevent this situation. These are called *anaplerotic reactions* (or "filling-up" reactions), and the phenomenon is called *anaplerosis*.

Q.78 Name the two anaplerotic reactions.

- Conversion of PA to OAA by CO_2 -fixation reaction by the enzyme *pyruvate carboxylase* which requires biotin, ATP, Mg^{++} and acetyl CoA. Acetyl CoA acts as a +ve modifier; it helps the enzyme to maintain "active" conformation.
- Conversion of PA to OAA through malic acid formation (Fig. 23.1)

Q.79 Name the inhibitor of lactate dehydrogenase enzyme (LDH).
Oxamate It competitively inhibits lactate dehydrogenase (LDH) and prevent reoxidation of NADH.

Q.80 How many ATPs are produced in glycolysis in presence of O_2 (Aerobic phase)? Explain.

- Produces 8 (Eight) ATP.
- Details as follows:

- (a) Loss:
- Phosphorylation of glucose = -1 ATP
 - Phosphorylation of fructose = -1 ATP

- (b) Gain:
- Oxidation of glyceraldehyde-3-P = +6 ATP
 - Phosphoglycerate kinase reaction (Substrate level) = +2 ATP
 - Pyruvate kinase reaction (substrate level) = +2 ATP
 - Net gain = 10 ATP - 2 ATP = 8 ATP

Q.81 How many ATPs are produced in glycolysis in absence of O_2 (Anaerobic phase)?

- In absence of O_2 , NADH + H^+ produced by oxidation of glyceraldehyde-3-P, cannot be oxidized in ETC. NADH is

converted to NAD^+ in reduction of pyruvate to lactate. Hence 6 ATP is not produced.

- In anaerobic phase, per molecule of glucose oxidized, 4 ATP - 2 ATP = 2 ATP will only be produced.

Q.82 What is the ATP yield under aerobic conditions?

38 ATP per molecule of glucose metabolised.
i.e. glycolysis 8 ATP
TCA cycle 30 ATP

Q.83 What is TCA cycle?

- A TCA cycle is the final common pathway for metabolism of carbohydrates, lipids and proteins (IIIrd phase of metabolism). It is a cyclic process and involves a sequence of compounds interrelated by oxidation-reduction and other reactions which finally produce CO_2 and H_2O .
- TCA cycle is also known as Krebs' cycle or citric acid cycle (Fig. 23.2).

Q.84 How many ATPs are formed in the TCA cycle?

- From acetyl CoA
 - From pyruvate.
- 12 ATPs
 - 15 ATPs.

Inhibits the reaction

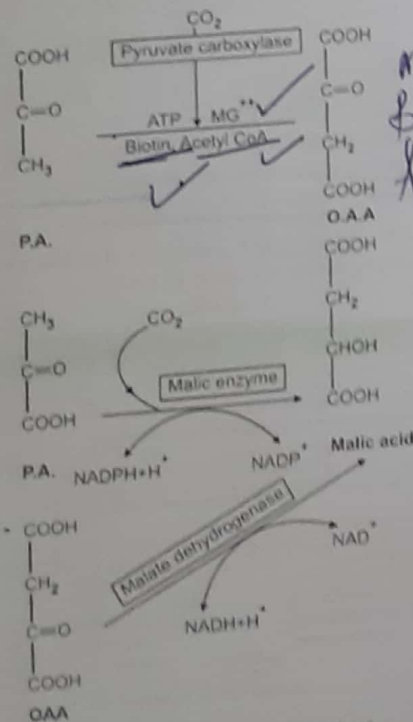


Fig. 23.1: Anaplerotic reactions

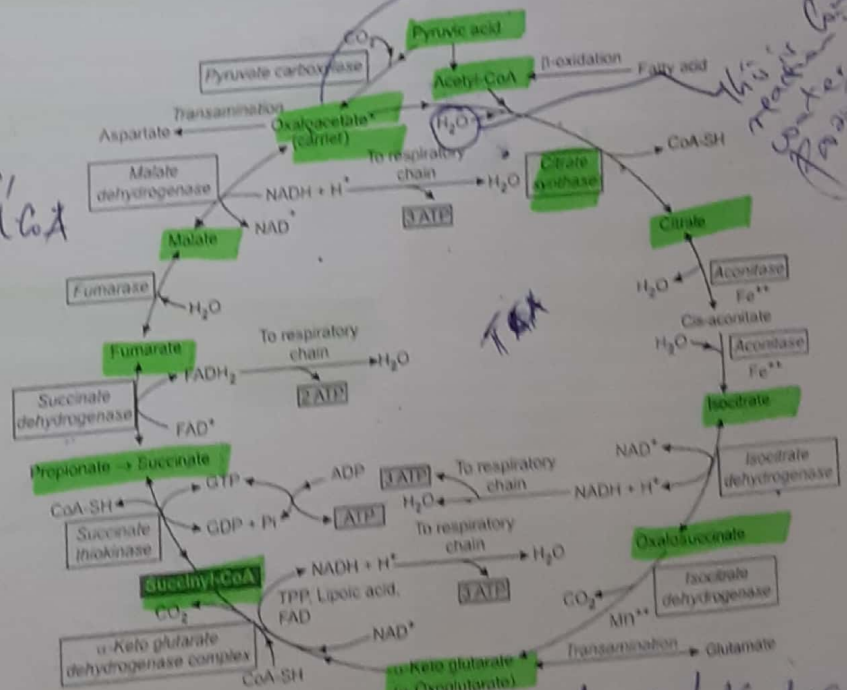


Fig. 23.2: Citric acid cycle

TCA / Krebs cycle

Q.85 State the over-all bioenergetics in complete oxidation of glucose/glycogen in glycolysis-cum-TCA cycle in presence of O_2

A. Glycolysis

ATP yield per
hexose unit

- Glycogen $\rightarrow$ F-1, 6-bi-P -1 ATP
- Glucose $\rightarrow$ F-1, 6-bi-P -2 ATP
- Glyceraldehyde-3-P dehydrogenase (2 NADH $\rightarrow$ 2 NAD $^+$) +6 ATP
- Substrate level phosphorylation:
- Phosphoglycerate kinase +2 ATP
- Pyruvate kinase +2 ATP
- Net gain in glycolysis:
 - For glucose = +8 ATP
 - For glycogen = +9 ATP

B. Oxidative decarboxylation of P.A.:

- PDH complex (2 NADH $\rightarrow$ 2 NAD $^+$) +6 ATP

C. TCA cycle:

- Isocitrate dehydrogenase (2 NADH $\rightarrow$ 2 NAD $^+$) +6 ATP
- α -oxoglutarate dehydrogenase (2 NADH $\rightarrow$ 2 NAD $^+$) +6 ATP
- Substrate level phosphorylation: Succinate thiokinase (2 GTP or ITP $\rightarrow$ 2 ATP + 2 ATP)
- Succinate dehydrogenase (2 FAD $\cdot H_2 \rightarrow$ FAD + 4 ATP)
- Malate dehydrogenase (2 NADH $\rightarrow$ 2 NAD $^+$) +6 ATP
- Total = + 24 ATP

$\therefore$ Total energetics:

- Per mole of Glucose = 24+6+8 ATP = 38 ATPs
- Per mole of Glycogen = 25+6+9 ATP = 39 ATPs

Note: Under anaerobic conditions (in absence of O_2):

- Glucose = +2 ATPs
- Glycogen = +3 ATPs.

Q.86 State the efficiency of complete oxidation of glucose.

- One mole of glucose after complete oxidation produces = 38 ATPs
- Total energy captured in ATP per mole of glucose oxidized = $7600 \times 38 = 2,88,800$ calories.
- Oxidation of one molecule of glucose "in vitro" produces = 6,86,000 calories
- $\therefore$ Hence efficiency = $\frac{2,88,800}{6,86,000} \times 100 = 42\%$.

Q.87 How 2,3-DPG formation takes place?

When bisphosphoglycerate mutase acts upon 1,3 bisphosphoglycerate formation of 2,3 bisphosphoglycerate takes place in erythrocytes.

Q.88 What is Rapoport-Leubering cycle or shunt (RLC or RLS)?

- RLC or RLS is a diversion in glycolytic pathway in red blood cells. Conversion of 1, 3-BPG to 3 PG does not occur and ATP is not formed at substrate level. It forms 2, 3-BPG.
- It is calculated to deplete and waste the energy needed by the RB cells (Fig. 23.3).

Q.89 What is the function of 2,3 DPG? ^{Biphosphoglyc}
2,3-DPG reduces the affinity of oxygen with hemoglobin so, is responsible for normal oxygen delivery to peripheral tissues.

Q.90 What is the net energy change during formation of 2-3-DPG in glycolysis?
No net production of ATP takes place.

Q.91 Why citric acid cycle is considered the common pathway for carbohydrate, fat and protein metabolism?

Citric acid cycle is the common pathway for the metabolism of carbohydrates, fats and proteins since it provides the complete oxidation of acetyl CoA to carbon dioxide and water.

Acetyl CoA comes from all the three metabolism:

1. Carbohydrate metabolism: Glycolysis

2. Fat metabolism: β -oxidation

3. Protein metabolism: Transamination

Hence citric acid cycle is the common pathway for the metabolisms of carbohydrate, fat and protein.

Q.92 What is the inhibitor of aconitase step in Krebs' cycle?

Aconitase, which converts citrate to isocitrate is inhibited by fluoroacetate.

Q.93 Which step of Krebs' cycle is inhibited by arsenite?

Arsenite inhibits the α -keto glutarate dehydrogenase complex thus impeding the conversion of α -keto glutarate to succinyl-CoA.

Q.94 What is the inhibitor of succinate dehydrogenase?

Succinate dehydrogenase is competitively inhibited by malonate and oxaloacetate.

Q.95 What is the peculiarity of succinate dehydrogenase?

It is only enzyme of TCA cycle which is found to inner mitochondrial membrane unlike others which are present in matrix of mitochondria.

Q.96 Enumerate the vitamins which play an important role in TCA cycle?

1. Riboflavin.
2. Niacin.
3. Thiamin.
4. Pantothenic acid.

Q.97 What is oxidative decarboxylation? Oxidation accompanied by decarboxylation is called oxidative decarboxylation.

Q.98 What is the oxidative decarboxylation product of pyruvic acid?

Pyruvic acid $\xrightarrow{\text{Oxidative decarboxylation}}$ Acetyl CoA.
 $-2H, -CO_2$

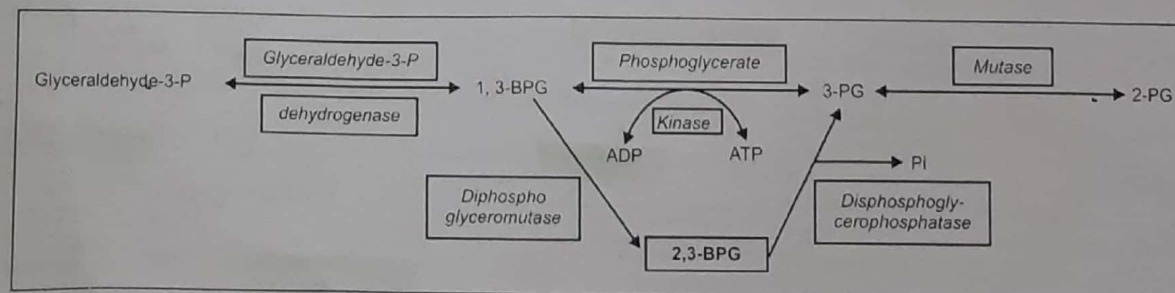


Fig. 23.3: Rapoport-Luebering shunt or cycle

Q.99 What is substrate level phosphorylation?
When energy is liberated without entrance to electron transport system, it is termed as substrate level phosphorylation.

Q.100 When substrate level phosphorylation occurs in TCA cycle?
When succinyl CoA is converted into succinate. One ATP is liberated by substrate level phosphorylation.

Q.101 Can TCA cycle function in absence of O₂?
TCA cycle cannot function in absence of O₂.

Q.102 Where are the enzymes of TCA cycle located?
Enzymes of TCA cycle are located in mitochondrial matrix, either free or attached to the inner surface of the inner mitochondrial membrane, which facilitates the transfer of reducing equivalents to the adjacent enzymes of ETC.

Q.103 Succinyl CoA is an intermediate in TCA cycle. How it is formed?
Succinyl CoA is formed by oxidative decarboxylation of α -oxoglutarate by " α -oxoglutarate dehydrogenase complex" which requires TPP, lipoic acid, CoA-SH, FAD, NAD⁺ and Mg⁺⁺ ions as coenzymes/cofactors (Fig. 23.4).

Q.104 State the inhibitors of TCA cycle.
• **Fluoroacetate:** Inhibitor of "aconitase" and allows citrate to accumulate.
• **Arsentite:** inhibits " α -oxoglutarate dehydrogenase" enzyme complex and allows accumulation of α -oxoglutarate (α -keto glutarate)
• **Malonate/OAA:** inhibits succinate dehydrogenase by competitive inhibition and allows accumulation of succinate.

Q.105 What is the importance of OAA in TCA cycle?
• It is required to start the cycle
• A small quantity is necessary.
• At the end of the cycle, OAA is regenerated by oxidation of malate by malate dehydrogenase.
• Thus OAA acts catalytically to restart the cycle again.

Q.106 What will happen to TCA cycle if OAA is not available?
In absence of OAA, TCA cycle will not operate. Acetyl CoA will accumulate and will be diverted to form ketone bodies and biosynthesis of FA and cholesterol.

Q.107 Why TCA cycle is said to be amphibolic in nature?
• TCA cycle has dual role:
• **Catalytic role:** The acetyl CoA produced by metabolism of carbohydrates, lipids and proteins are completely oxidized to produce CO₂, H₂O and energy.
• **Anabolic role** (Synthetic role): Intermediates of TCA cycle are utilized for synthesis of various biologically important compounds in the body, e.g.
– Synthesis of non-essential amino acid.
– Formation of glucose (gluconeogenesis)
– FA synthesis
– Synthesis of cholesterol and steroids.
– Heme synthesis.

Q.108 What is the enzyme of the above oxidative decarboxylation reaction?
Enzyme involved is pyruvate dehydrogenase complex. This enzyme complex consists of three enzymes:
i. Pyruvate dehydrogenase.
ii. Dihydrolipoyl transacetylase.
iii. Dihydrolipoyl dehydrogenase.

Q.109 What are the cofactors of the above reaction?
1. Thiamine pyrophosphate (TPP).
2. Lipoic acid.
3. Coenzyme A (CoA-SH).
4. Flavin adenine dinucleotide (FAD).
5. Nicotinamide-adenine dinucleotide (NAD⁺).
6. Mg⁺⁺ ions.

Q.110 What is the active form of PDH?
"Active" forms of PDH is the dephosphorylated form. Insulin stimulates phosphatase enzyme and converts 'inactive' to 'active' form by dephosphorylation.

Q.111 What is the inactive form of the PDH enzyme?
• "Inactive" form is the Phosphorylated form catalyzed by the enzyme "PDH kinase": Following favor the formation of "inactive" form:
• Rise in ATP/ADP ratio
• Rise in NADH/NAD⁺ ratio
• Acetyl CoA/CoA. SH ratio
• Increased cyclic AMP level in cells

Q.112 Pyruvic acid is formed in cytosol by glycolysis but oxidative decarboxylation takes place in mitochondrion. Pyruvic acid is impermeable to mitochondrial membrane. How it is done?
Pyruvic acid formed in cytosol is not permeable to mitochondrial membrane, it is transported to mitochondrion by a specific transport protein.

Q.113 What is the importance or metabolic significance of Hexose monophosphate shunt pathway?
1. Provides NADPH which is used in fatty acid synthesis.
2. Provides pentose sugar which is the building block of nucleic acid.

Q.114 Acetyl CoA is formed inside mitochondria but the fatty acid synthesis (de Novo) from acetyl CoA occurs in cytosol. Acetyl CoA is not permeable to mitochondrial membrane. How acetyl CoA made available in cytosol?
• Acetyl CoA is transported out in the form of citrate, an intermediate of TCA cycle, to cytosol, as citrate is readily permeable to mitochondrial membrane.
• In the cytosol, citrate is cleaved by the enzyme citrate cleavage enzyme (ATP-citrate lyase) to acetyl CoA and OAA, so that acetyl CoA can be used for FA synthesis.

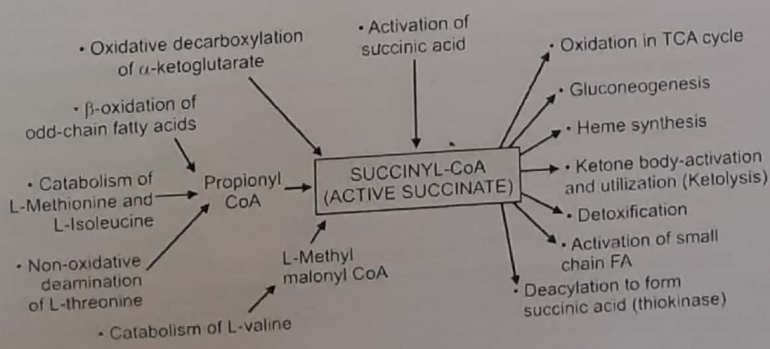


Fig. 23.4: Succinyl CoA: Formation and fate

Fatty acid Biosynthesis takes place in cytosol

Q.115 In glycolysis, NADH is produced in cytosol, but it is oxidized in ETC in mitochondria to produce ATP. NADH is not permeable to mitochondrial membrane. Explain how it is achieved?

NADH produced in cytosol by glycolysis transfer the reducing equivalents through the mitochondrial membrane via substrate pairs linked by suitable dehydrogenases by "shuttle systems". Two such shuttle systems are:

- Glycerophosphate shuttle.
- Malate shuttle.

Q.116 Show schematically glycerophosphate shuttle.

See Figure 23.5

Note: α -glycero-P-dehydrogenase in mitochondrion is Fp-dependent.

Hence produces 2 ATP per mole of glucose oxidized. Hence in this 36 ATP is produced per mole of glucose oxidized.

Q.117 Show schematically malate shuttle.

See Figure 23.6.

Note: Malate shuttle is commonly used by the body.

Use of malate shuttle forms 38 ATP.

Q.118 Name the tissues where hexose monophosphate shunt pathway is active?

HMP shunt pathway is active in liver, adipose tissue, mammary gland, adrenal cortex, testis, etc.

Q.119 What are the clinical problems do G-6-PD deficient persons have to face?

The patients with G-6-PD deficiency when given antimalarials like primaquine, they develop hemolysis.

Q.120 Why is it so?

G-6-PD is responsible for maintenance of glutathione in reduced state. Moreover, when primaquine is given it leads to generation of more free radicals. These two factors contribute to hemolysis.

Q.121 Name two keto acids involved in carbohydrate metabolism.

- Pyruvic acid
- α -keto-glutaric acid.

Q.122 Name two reactions catalysed by kinases.

Reactions catalysed by kinases:

- Glucose $\rightarrow$ Glucose-6- PO_4
- Fructose-6- $\text{PO}_4 \rightarrow$ Fructose-1,6-di PO_4 .

Q.123 Name two reactions catalysed by mutases.

Reactions catalysed by mutases:

- 3-Phosphoglycerate $\rightarrow$ 2-Phosphoglycerate.
- Glucose-6- $\text{PO}_4 \rightarrow$ Glucose 1- PO_4 .

Q.124 What is glycogenesis?

It is the formation of glycogen from glucose in the body.

Q.125 Show schematically the steps of glycogenesis.

See Figure 23.7

Q.126 What happens to liberated UDP by glycogen synthase action?

UDP is converted to UTP again by the enzyme nucleoside diphosphokinase and is reutilized again for UDPG formation.

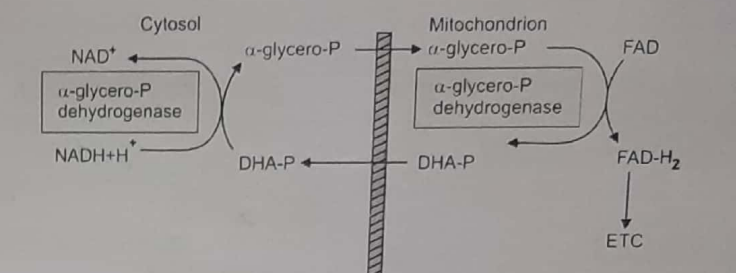


Fig. 23.5: Glycerophosphate shuttle

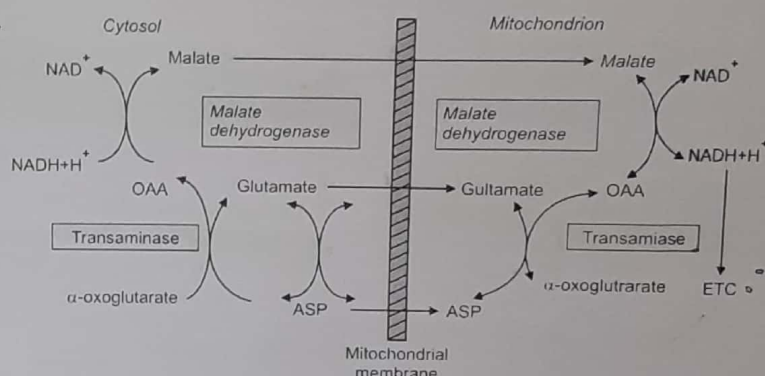


Fig. 23.6: Malate shuttle

Q.127 What is the primer? How the first primer formed?

- The first "primer" originally supposed to be synthesized on a protein backbone which is a process similar to synthesis of other glycoproteins.
- A pre-existing glycogen molecule or "primer" is a must, so that UDPG can add the glucose molecule to the outer end of a chain.

Q.128 How much ATP is spent by the body to add one glucose unit to outerchain?

- In each addition of glucose unit, 2 ATP molecules are expended by the body:
 - One ATP is utilized in phosphorylation of glucose $\rightarrow$ Glucose-6-P.

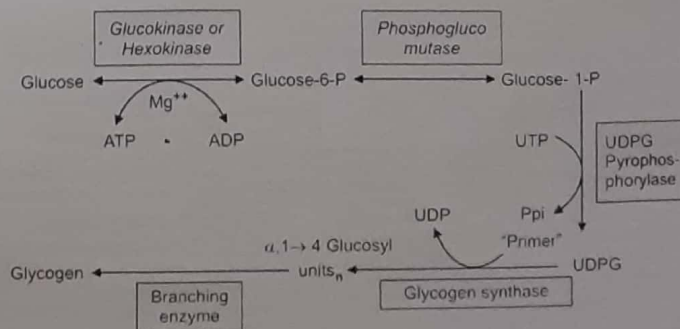


Fig. 23.7: Steps of glycogenesis

- Another ATP is used to convert UDP to UTP again.

Q.129 What is key and rate-limiting enzyme in glycogenesis?

Glycogen synthase (synthetase) enzyme.

Q.130 What are the active and inactive forms of the enzyme glycogen synthase?

- Glycogen synthase enzyme occurs as 'active' GS ' α ' or 'inactive' GS ' β ' forms and both are interconvertible.
- GS ' α ' $\rightarrow$ GS ' β ' by phosphorylation which is modulated by cyclic-AMP dependent protein kinase and glycogenesis is stopped.
- GS ' β ' $\rightarrow$ GS ' α ' is formed by dephosphorylation catalyzed by the enzyme "protein phosphatase-1" when glycogenesis starts.

Q.131 What is UDPG? What are its functions?

- UDPG is uridine diphosphate glucose. It is an intermediate in glycogenesis. It is activated glucose which adds one glucose unit to a chain in 'primer' molecule, by α 1 $\rightarrow$ 4 glycosidic linkage.
- Other functions:
 - It is formed as an intermediate in uronic acid pathway required for formation of D-glucuronic acid.
 - It is also required for synthesis of lactose from galactose in lactating mammary gland.

Q.132 State the factors that bring about stimulation and inhibition of glycogenesis.

- **Stimulators:**
 - Insulin
 - Glucocorticoids
 - High concentration of glucose.
- **Inhibitors:**
 - Increase concentration of glycogen (by "feedback" inhibition).
 - Increased cyclic AMP in the cells, which can be brought about by hormones viz.,
 - Epinephrine,
 - Norepinephrine,
 - Glucagon,
 - Thyroid hormones.

Q.133 How insulin increases glycogenesis? Insulin directly stimulates the enzyme protein phosphatase-1 thus brings about dephosphorylation of glycogen synthetase and forms "active" glycogen synthase, GS ' α ', and increases glycogenesis.

Q.134 How does increased cyclic AMP level in cells inhibits glycogenesis?

- Increased cyclic AMP in the cells converts inactive protein kinase (C_2R_2) $\rightarrow$ active protein kinase (C_2).
- Active protein kinase (C_2) has following two effects:
 - Brings about phosphorylation of 'GS' enzyme with the help of ATP and thus GS ' α ' $\rightarrow$ GS ' β ' inhibiting glycogenesis.
 - Also converts a protein factor inhibitor 1 (inactive) and phosphorylates it to form "active inhibitor 1-P", which in turn inhibits protein phosphatase-1, so that conversion of inactive GS ' β ' $\rightarrow$ to active GS ' α ' does not occur thus inhibiting glycogenesis.

Q.135 What do you know about branching and debranching enzymes?

- Branching enzymes produce a branching in the molecule by establishing an α (1,6) glycosidic linkages.
- Debranching enzymes produce a cleavage of α -(1,6) linkages.

Q.136 What is glycogenolysis?

Breakdown of glycogen to glucose is called as glycogenolysis.

Q.137 Which is the key and rate-limiting enzyme in glycogenolysis?

Phosphorylase enzyme.

Q.138 What are the active and inactive forms of liver phosphorylase?

- Active phosphorylase is the phosphorylated form "active" phosphophosphorylase ' α '.
- Inactive phosphorylase is the dephosphorylated form "inactive" dephosphophosphorylase ' β '.

Q.139 What are the basic differences between liver and muscle phosphorylase?

- There is no cleavage of structure with liver phosphorylase as compared to muscle phosphorylase.
- Four molecules of pyridoxal-P is required for activity of muscle phosphorylase, not so with liver phosphorylase.
- Muscle phosphorylase is not affected by glucagon (no receptor on muscle).

Q.140 Name the hormones which bring about glycogenolysis through increased cyclic AMP levels in cells.

- Catecholamines-epinephrine and norepinephrine
- Glucagon, and
- Thyroid hormones.

Q.141 What is the product of phosphorylase activity on glycogen molecule?

Active phosphorylase in presence of inorganic P_i brings about phosphorolytic cleavage of α 1 $\rightarrow$ 4 glycosidic bond from outermost chain of glycogen molecule and glucose is released as glucose-1-P and not free glucose.

Q.142 What is the fate of glucose-1-P released by phosphorylase activity.

- Glucose-1-P is converted to glucose-6-P by the enzyme phosphogluco mutase.
- In liver and kidney (not in muscle), glucose-6-P is acted upon by the enzyme glucose-6-phosphatase and free glucose is formed.

Q.143 Why glucose is not formed in muscle from glycogen break-down?

In muscle, glucose-6-phosphatase enzyme is absent, hence glucose-6-P enters the glycolytic cycle and forms pyruvates and lactates.

Q.144 How does glucagon action differ from catecholamines in glycogenolysis?

- Catecholamines cause breakdown of liver as well as muscle glycogen.
- But glucagon breaks down only liver glycogen and not muscle glycogen (as receptor for glucagon is not present in muscle).

Q.145 What is calmodulin?

Calmodulin is a Ca^{++} dependent regulatory protein which is specific for calcium. It is a flexible protein, with 4 binding sites for Ca distributed in 4 domains.

Q.146 What is gluconeogenesis. Name few gluconeogenic substances.

- The process of formation of glucose from non carbohydrate substances is called gluconeogenesis.
- The gluconeogenic substances are pyruvic acid, propionate, lactic acid, glycerol and amino acids (Fig. 23.8).

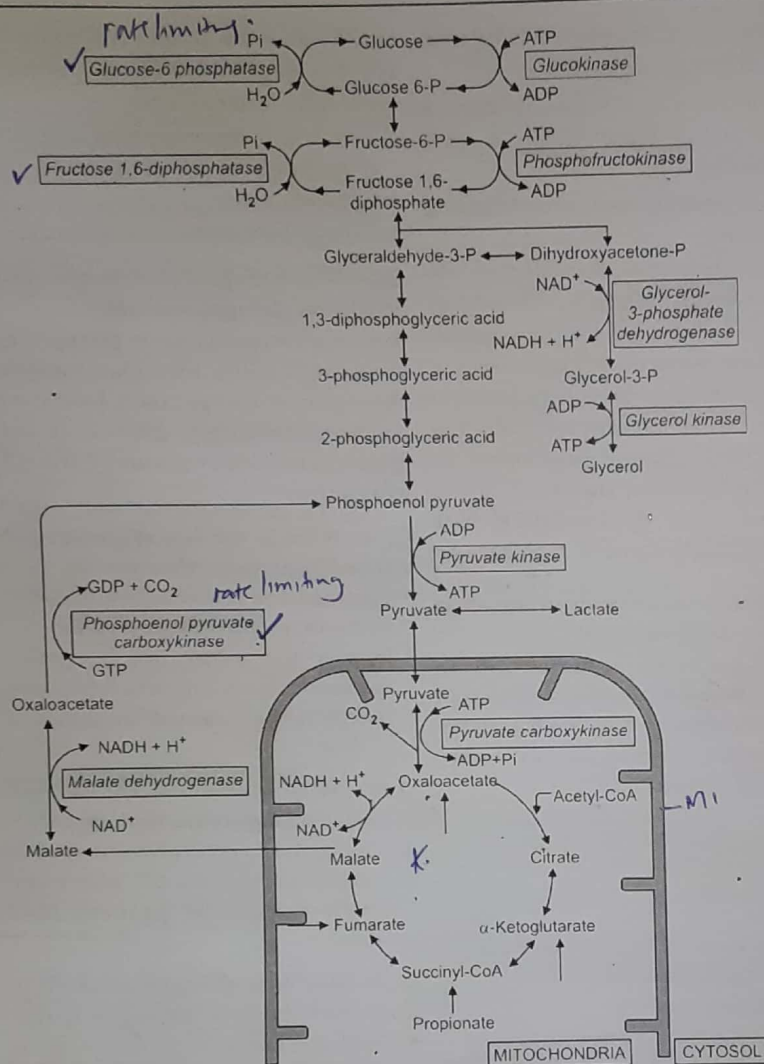


Fig. 23.8: Pathway of gluconeogenesis in the liver

Q.147 Name the rate-limiting enzymes of gluconeogenesis.

- Four rate-limiting enzymes are:
- Pyruvate carboxylase (mitochondrial).

- Phospho-enol pyruvate carboxykinase (cytosol).
- Fructose-1, 6-biphosphatase (cytosol).
- Glucose-6-phosphatase (cytosol).

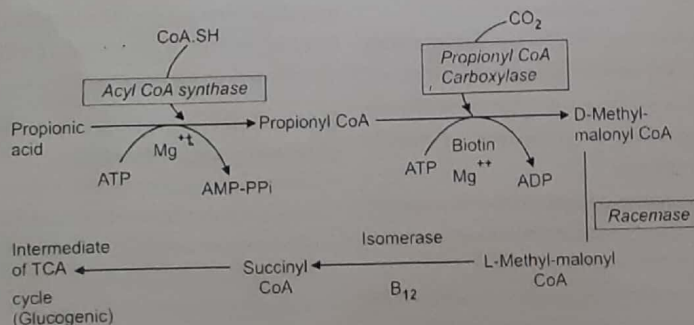


Fig. 23.9: Formation of glucose

Q.148 Name the irreversible steps in gluconeogenesis and enzymes used to circumvent the irreversible steps.

Irreversible steps (Energy barrier)	Enzymes used to circumvent
• Pyruvate → phosphoenol pyruvate.	• Pyruvate carboxylase (Mitochondrial)—conversion of PA to OAA by CO ₂ fixation reaction.
• Fructose-1-5-bi-P → fructose-6-P	• Phosphoenolpyruvate carboxykinase (cytosol) converts OAA to phosphoenol pyruvate.
• Glucose-6-P → glucose.	• Fructose-1, 6-bi-phosphatase (cytosol).
	• Glucose-6-phosphatase (cytosol).

Q.149 Name the tissues where gluconeogenesis occur and name one disease and one condition in which gluconeogenesis is significantly enhanced.

- Principally occurs in liver (85%) and kidney (15%).
- Uncontrolled diabetes mellitus and prolonged starvation.

Q.150 State how glucose is formed from glycerol?

- Glycerol is phosphorylated in presence of the enzyme "Glycerol kinase" and ATP to form α-Glycero P.
- α-Glycero-P is converted to Di-OH-acetone-P by dehydrogenase and NAD⁺.
- Di-OH-acetone-P and glyceraldehyde-3-P forms fructose, 1-6, biphosphate which by reversal of glycolysis form glucose.

Q.151 Mention the sources of propionyl-CoA in humans.

- From catabolism of L-methionine.
- Catabolism of isoleucine.
- Oxidation of odd-chain FA.
- Synthesis of bile acids.
- Non-oxidative deamination of L-threonine.

Q.152 Show schematically how glucose is formed from propionic acid. See Figure 23.9.

Q.153 What is the role of hormones in gluconeogenesis?

- Glucagon: Increases gluconeogenesis from lactic acid (LA) and amino acids.
- Gluco-corticoids: stimulate gluconeogenesis by increasing protein catabolism in the peripheral tissues and increasing hepatic uptake of amino acids. It increases the activity of transaminases and other key enzymes concerned in gluconeogenesis.

Q.154 What are the enzymes concerned with the reversal of glycolysis, i.e. gluconeogenesis?

1. Pyruvate carboxylase.
2. Phosphoenolpyruvate carboxy kinase.

3. Fructose 1,6-diphosphatase.
4. Glucose-6-phosphatase.

Q.155 How pyruvate is converted to glucose?

Pyruvate is converted to glucose by gluconeogenesis. Reaction is given as:
Pyruvate → Oxaloacetate
Oxaloacetate → Phosphoenolpyruvate
Phosphoenolpyruvate → Glucose.
(Fig. 23.10)

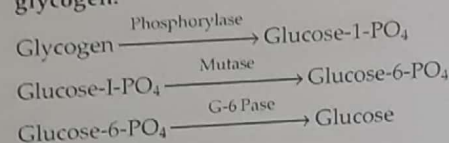
Q.156 What is Cori cycle?

Lactic acid produced in the muscle reaches the liver through blood where it is converted to glucose by gluconeogenesis, which again becomes source of energy for utilisation. This process continues and is called Cori cycle (Fig. 23.11).

Q.157 Can muscle glycogen be the source of blood glucose?

Muscle glycogen cannot be a source of blood glucose, because it lacks an enzyme glucose 6-phosphatase.

Q.158 Give the breakdown products of glycogen.



Q.159 Name the six classical types of GSDs and indicate the enzyme deficiencies.

Type/Name	Enzyme deficiency
Type I - von Gierke's disease	Glucose-6-Pase
Type II - Pompe's disease	Acid maltase
Type III - Forbe's disease (Limit dextrinosis)	Debranching enzyme
Type IV - Andersen's disease (Amylopectinosis)	Branching enzyme
Type V - McArdle's disease	Muscle phosphorylase
Type VI - Her's disease	Liver phosphorylase

Q.160 Explain von Gierke's disease.
In von Gierke's disease there is deficiency of Glucose-6-phosphatase.

Q.161 Explain McArdle's disease.
In McArdle's disease there is deficiency of muscle phosphorylase.

Q.162 What is hexose monophosphate pathway or shunt?

It is an alternate pathway of glucose oxidation which takes place in certain special tissues to serve certain special functions.

Q.163 How does HMP shunt differ from EM pathway?

Essential differentiating points are:

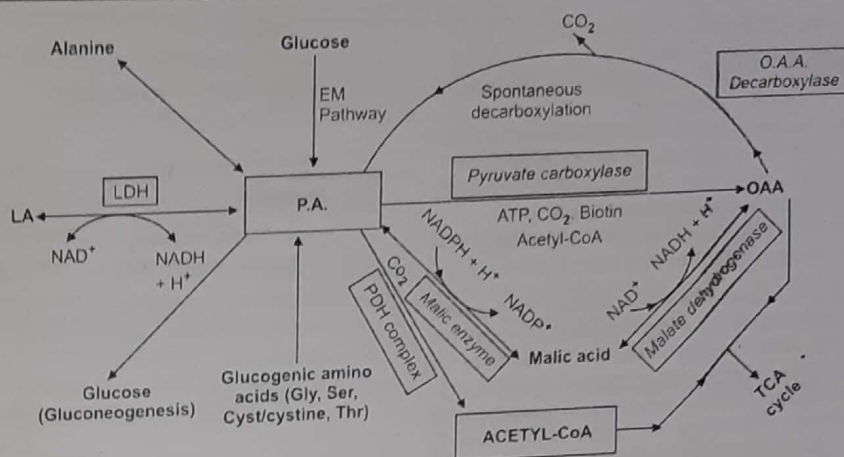


Fig. 23.10: Formation and fate of pyruvic acid

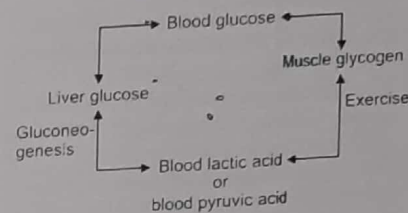


Fig. 23.11: Cori cycle

HMP shunt	EM pathway
• Occurs in certain special tissues only	• Occurs in all tissues
• A multicyclic process	• Not so
• NADP ⁺ acts as H-acceptor	• NAD ⁺ acts as H-acceptor
• ATP is not produced	• Energy producing
• Not meant for energy	• ATP is produced.
• CO ₂ is formed	• CO ₂ is never formed.

Q.164 In which part of the cell HMP shunt operates?
In the cytosol.

Q.165 Why HMP shunt is called as a multicyclic process and what are the products.
It is called a multicyclic process, as 3 mols of glucose-6-P enter the cycle, producing 3 mols of CO₂, 6 mols of NADHP and 3 mols of 5-C residues which rearrange to give 2 mols of glucose-6-P (re-enters the cycle) and one mol of glyceraldehyde-3-P.

Q.166 Which is the key and rate-limiting enzyme in HMP shunt?
Glucose-6-phosphate dehydrogenase (G-6-PD).

Q.167 What is the role of thiamine in HMP shunt?
TPP the active form of thiamine is required as a coenzyme with "transketolase" enzyme for two transketolation reactions.

Q.168 Name the tetroses formed in HMP shunt.
Erythrose-4-P.

Q.169 How does NADP⁺/NADPH ratio regulate HMP shunt?
• If the ratio is high, it enhances the rate-limiting reaction and shunt pathway.
• If the ratio is low, it inhibits G-6-PD and 6-phosphogluconate dehydrogenase enzymes and decreases shunt pathway.

Q.170 State the hormones that regulate HMP shunt.

• **Insulin:** It induces the synthesis of dehydrogenase enzymes and enhances the activity of the pathway.
• **Thyroid hormones:** It stimulates the activity of G-6-PD enzyme of the pathway.

Q.171 State the metabolic role of HMP shunt.

• Does not produce energy.
• **Provides NADPH:** required for various reductive synthesis.
• **Provides pentoses:** required for nucleic acid synthesis.
• **Role in lens metabolism:** Provides reduced glutathione (G-SH) necessary for maintenance of lens proteins.
• **Role in RB Cells:** Provides reduced glutathione (G-SH) necessary for RB cells membrane integrity.
• CO₂ is produced which is used in CO₂ fixation reaction.

Q.172 Mention some metabolic reactions where NADPH is used for reductive processes.

• NADPH is used in a number of metabolic processes where it provides hydrogen for reductive synthesis, e.g.
• Extramitochondrial *de novo* fatty acid synthesis.

4120 RD

- In synthesis of cholesterol.
- In synthesis of steroids.
- In conversion of oxidized glutathione to reduced glutathione.
- For conversion of phenylalanine to tyrosine.
- For conversion of methemoglobin → Hb.
- In synthesis of sphingolipids.
- In "microsomal" chain elongation of FA.
- In uronic acid pathway.

Q.173 Muscle tissues contain very small amounts of the dehydrogenases enzymes but still skeletal muscle is capable of synthesizing pentose sugars. How?

Probably this is achieved by *reversal of shunt Pathway* utilizing fructose-6-P, glyceraldehyde-3-P and the enzymes *transketolase* and *transaldolase* (by non-oxidative Pathway).

Q.174 What is Wernicke-Korsakoff syndrome?

- A genetically variant form of *transketolase* occurs which cannot bind TPP⁺ thus affecting transketolation reaction.
- The patient shows severe neuropsychiatric symptoms characterized by lesions and hemorrhages near IIIrd ventricle.
- The patient shows deranged mental function, loss of memory, depression, disorientation and mental confusions.

Q.175 What is galactosemia?

Excretion of galactose in urine due to the deficiency of enzyme galactose-1-PO₄ uridyl transferase leads to a condition known as galactosemia (Fig. 23.12).

Q.176 State the salient clinical features in galactosemia.

- **Infants** appear normal at birth but later:
 - fails to thrive
 - becomes lethargic and may vomit.
 - develops hypoglycemia and
 - may develop jaundice.
- **After 2 to 3 months:**
 - develops cataracts in both eyes.
 - shows mental retardation due to accumulation of galactose and galactose-1-P in cerebral cortex.
 - liver may have fatty infiltration and produces cirrhosis liver.

Q.177 What is the cause of cataracts in galactosemia?

- Excess of galactose in lens of the eye is reduced to "galactitol" (Dulcitol), an alcohol by the enzyme *aldose reductase*. Galactitol cannot escape from lens cells. Osmotic effect of sugar alcohol contributes to injury to lens proteins producing cataracts.
- Excess of galactose inhibits the enzyme G-6-PD of HMP shunt leading to less

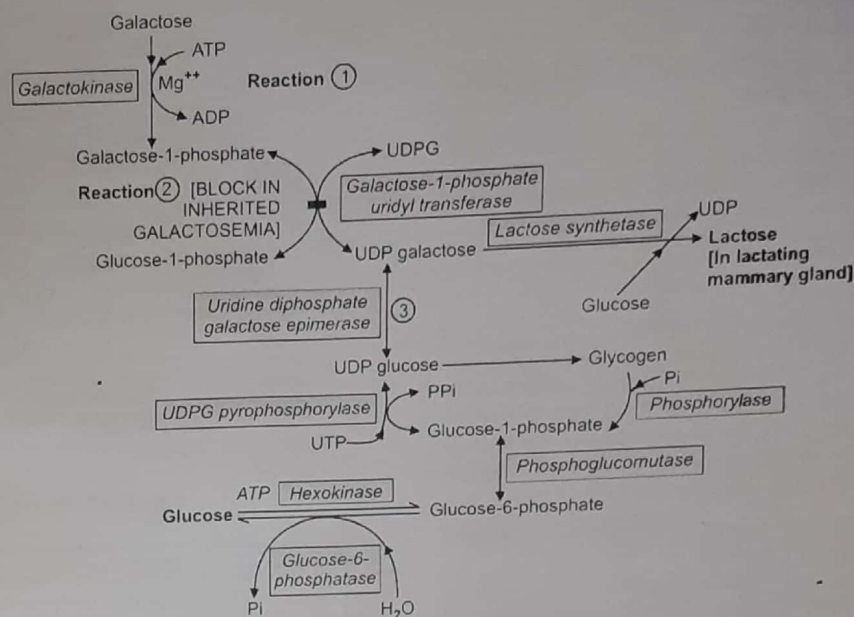


Fig. 23.12: Metabolism of galactose

NADPH which results to low reduced glutathione (G-SH).

Q.178 What are the enzymes deficient in galactosemia?

Enzymes deficient are:

- Galactose kinase.
- Galactose-1-PO₄-uridyl transferase.

Q.179 What is uronic acid pathway?

It is an alternative pathway for glucose oxidation. This pathway also does not produce energy (Fig. 23.13).

Q.180 What is the metabolic significance of uronic acid pathway.

- Formation of D-glucuronic acid which is used for detoxication.
- Produces vitamin C in lower animals, but cannot synthesize vitamin C in man, and other primates including G. pigs due to absence of enzymatic machinery.
- Inherited deficiency of the enzyme "L-xylitol dehydrogenase" produces essential pentosuria.

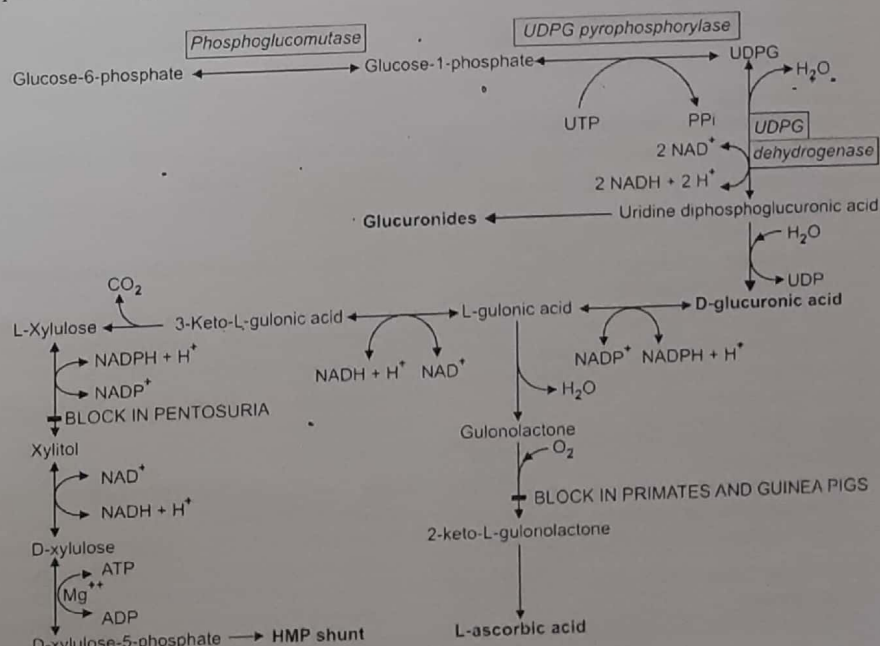


Fig. 23.13: Uronic acid pathway

- Excessive xylitol or parenteral administration of xylitol may lead to oxalosis.

Q.181 What are the metabolic functions of D-glucuronic acid produced in uronic acid pathway?

Detoxication: Detoxicates drugs, chemicals, antibiotics, hormones, etc convert them to corresponding soluble glucuronides which are excreted.

Examples:

- Aromatic acid like benzoic acid.
- Phenol and secondary/tertiary aliphatic alcohols.
- Drugs and other xenobiotics—they are first hydroxylated by mono-oxygenase cyt. P_{450} system and then conjugated with D-glucuronic acid.
- Antibiotics like chloramphenicol.
- Steroid hormones and thyroid hormones.
- Bile pigments: unconjugated bilirubin is conjugated with UDP-glucuronic acid and converted to soluble mono and diglucuronides.
- Synthesis of heteroglycans containing D-glucuronic acid, e.g. heparin, hyaluronic acid, chondroitin SO_4 .

Q.182 Name some drugs which increase the formation of D-glucuronic acid by uronic acid pathway.

- Barbiturates
- Amino Pyrine and
- Antipyrine.

Q.183 What is true glucose?

True glucose means glucose only without taking into account the presence of any other reducing substances in the blood.

Q.184 Name some dietary sources of fructose.

- Principal source is sucrose (cane sugar/table sugar), which on hydrolysis in intestine gives fructose.
- Other sources are fruit juices and honey.

Q.185 Name the specific enzyme that phosphorylates fructose and the product formed.

- Specific enzyme is *fructokinase*.
- Product is fructose-1-P.

Q.186 State some metabolic importance of fructose.

- Fructose is easily metabolized and a good source of energy.
- Seminal fluid is rich in fructose and spermatozoa utilizes fructose for energy.
- Excess dietary fructose is harmful, leads to increased synthesis of TG.

- In diabetes mellitus, fructose metabolism through *sorbitol pathway* may account for development of cataracts and neuropathy.
- Inherited deficiency of the enzyme "aldolase-B" produces inherited disorder *hereditary fructose intolerance*.

Q.187 How is glucose converted to fructose in seminiferous tubular epithelial cells?

- It is achieved by sorbitol pathway.
- Sorbitol pathway for conversion of glucose to fructose is shown schematically (Fig. 23.14).

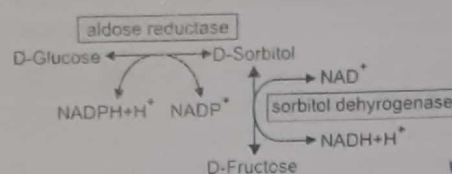


Fig. 23.14: Sorbitol pathway

Q.188 How do you explain biochemically the formation of cataract and neuropathies in diabetes mellitus?

- Formation of sorbitol from glucose by sorbitol pathway proceeds rapidly in the lens of the eye and Schwann cells of the nervous system.
- Elevated sorbitol concentration in these cells increase the osmotic pressure which may be responsible for the development of cataracts of lens of the eye and diabetic neuropathy.

Q.189 What is hereditary fructose intolerance?

- It is an inherited disorder, due to inherited deficiency of the enzyme *aldolase B*. It leads to excessive rises of fructose and fructose-1-P in blood.
- Blood glucose falls leading to hypoglycemia. The cause of hypoglycemia is probably due to:
 - excessive insulin secretion and
 - inhibition of *phosphoglucomutase* enzyme, by fructose-1-P
- Diets low in fructose and sucrose is beneficial.

Q.190 How will you estimate true glucose? True glucose is estimated by glucose oxidase method.

Q.191 What is use of glucose vial?

Glucose vial is used to collect blood for estimation of glucose level.

Q.192 Glucose vial constitutes?

It contains sodium fluoride and potassium oxalate in 1:3 ratio. Sodium fluoride inhibits

glycolysis and potassium oxalate inhibits coagulation of blood.

Q.193 What are different methods of measuring reducing sugars level?

1. Reduction method:

- King and Asatoor method
- Folin and Wil method
- Ferricyanide method.

2. Orthotoluidine method.

3. Enzymatic method.

4. Dextrostick method.

Q.194 What is the normal blood sugar level? 80-100 mg%.

Q.195 What is glucose postprandial level? Glucose postprandial level is 100-140 mg%.

Q.196 What are different types of glucose transporters?

GLUT-1	Brain, kidney RBC	(BKR ₁)
GLUT-2	Liver, pancreas	LP GROT 2
GLUT-3	Kidney, placenta	KP G3
GLUT-4	Heart, skeletal muscle,	HAS 4
	adipose tissue	→ S/Glutamine
GLUT-5	Small intestine	
SGLT-1	Small intestine and kidney	

Q.197 What are the conditions in which blood sugar level is low?

Blood sugar level is low in:

- Overdosage of insulin in the treatment of diabetes.
- Hypothyroidism.
- Addison's disease.

Q.198 What are the conditions in which blood sugar level is raised?

Conditions in which blood sugar level is raised are.

- Diabetes mellitus.
- Hyperthyroidism.
- Hyperadrenalism.
- Hyperpituitarism.
- Thyrotoxicosis.

Q.199 What is the kidney threshold for glucose? 180 mg.

Q.200 What is TM_C ?

The maximum rate at which glucose can be resorbed from the tubule is called TM_C .

Q.201 What are the hormones which keep the blood sugar level high?

- Glucagon.
- Epinephrine.
- Adrenal cortex hormones.
- Growth hormone and ACTH.
- Thyroid hormone.

Q.202 What is the hormone which regulates blood sugar level? Insulin.

Q.203 Why insulin cannot be given orally?

Insulin is polypeptide and is digested by the enzymes of digestive system into amino acids before it reaches in the blood. Hence, it cannot be given orally.

Q.204 What is renal glycosuria?

As a result of low kidney threshold, glucose appears in the urine. Blood sugar level remains normal.

Q.205 What is diabetes mellitus?

Diabetes Mellitus is a metabolic disorder due to the deficiency of insulin, resulting in high blood glucose level and glucose excretion in the urine.

Q.206 What are the types of diabetes mellitus?

Two types:

1. Insulin dependent diabetes mellitus (IDDM)
2. Non insulin dependent diabetes mellitus (NIDDM).

Q.207 What is IDDM?

In this disease there is autoimmune destruction of β -cells of pancreas and it is also known as juvenile diabetes.

Q.208 What is NIDDM?

In this disease there is resistance to insulin so in spite of normal or elevated levels of insulin hyperglycemia occurs.

Q.209 What is diabetes insipidus?

Lack of ADH (antidiuretic hormone) gives rise to a condition known as diabetes insipidus. Patient may excrete urine up to 30 liters/day.

Q.210 What is the abnormality in the urine sample of diabetic patient and the urine sample of starving patient?

In diabetic patient, urine will show the presence of glucose and ketone bodies; Whereas in case of starving patient only ketone bodies will be present in the urine.

Q.211 What are the different reducing sugars that appears in urine and under what conditions?

1. **Glucose:** Appears in urine in renal glycosuria and diabetes mellitus.
2. **Lactose:** During later stages of pregnancy and lactation.
3. **Galactose:** In galactosemia due to the deficiency of the enzyme galactose-1-PO₄ uridyl transferase. This condition is encountered in infants.
4. **Fructose:** Due to the consumption of lot of fruits containing fructose such as grapes, plums, cherry, etc.

5. **Pentoses:** Due to the consumption of lot of fruits containing pentoses. Also in congenital abnormality characterised by inability to metabolise L-xylulose.

Q.212 How the collection of blood specimen is done for estimating blood glucose?

The blood sample is collected in potassium oxalate: sodium fluoride bottle.

Q.213 What is the function of each?

Sodium fluoride: It prevents glycolysis by inhibiting the enzyme Enolase of the glycolytic pathway.

Potassium oxalate: It acts as an anticoagulant.

Q.214 What is the role of isotonic CuSO₄ in blood sugar estimation?

Isotonic CuSO₄ prevents the hemolysis of red blood cells so that glutathione which is present in the red blood cells does not come out, otherwise will also reduce the alkaline CuSO₄ and give rise to high blood sugar values.

Q.215 What are the functions of insulin?

1. Insulin promotes the entry of glucose in all the tissues of the body except liver.
2. Insulin helps in glycogenesis.
3. Insulin prevents glycogenolysis.
4. Insulin inhibits gluconeogenic enzymes.

Q.216 How glucose is removed from the blood?

- Oxidation of glucose in tissues to provide energy.
- Hepatic glycogenesis.
- Glycogenesis in muscles and other tissues.
- Conversion to fat (lipogenesis) in adipose tissue mainly.
- For synthesis of lactose, fructose, ribose sugars, glycoproteins, glycolipids, MPs etc.
- Excretion in urine when blood glucose exceeds renal threshold (abnormal condition).

Q.217 What is the role of insulin on carbohydrate metabolism?

- Net effect produces fall in blood glucose concentration.
- Diminished supply of glucose to blood due to:
 - decrease hepatic glycogenolysis
 - increased hepatic glycogenesis.
 - Decreased gluconeogenesis
- Increased in the rate of utilization of glucose by tissue cells:
 - increased uptake by tissues
 - increased oxidation for energy
 - increased lipogenesis

Q.218 Name the emergency hormones which increase blood glucose.

- Catecholamines viz. epinephrine and norepinephrine.
- Glucagon.

Q.219 What is hyperglycemia?

Increase in blood glucose level above normal is called hyperglycemia.

Q.220 Enumerate some causes of hyperglycemia.

- Diabetes mellitus.
- Hyperactivity of thyroids (thyrotoxicosis), anterior pituitary (acromegaly/gigantism) and adrenal cortex (Cushing's syndrome).
- In diffuse diseases of pancreas, e.g. in pancreatitis, carcinoma of pancreas.
- In sepsis and in some infectious diseases.
- In intracranial diseases, e.g. meningitis, encephalitis, intracranial tumors, and hemorrhage.
- In emotional 'stress'.

Q.221 What is hypoglycemia?

Decrease in blood glucose level below normal is called hypoglycemia.

Note: Hypoglycemia manifests clinically when the blood glucose is below < 40 mg % ("true" glucose).

Q.222 Enumerate some causes of hypoglycemia.

- Overdosage of insulin in treatment of DM—most common cause and clinically important.
- Hypoactivity of thyroids (myxoedema, cretinism), anterior pituitary (Simmond's disease), and adrenal cortex (Addison's disease).
- Insulin secreting tumor (insulinoma)—rare cause.
- In severe liver diseases.
- Leucine sensitive hypoglycemia.
- Glycogen storage diseases (GSDs).
- Idiopathic hypoglycemia in children.
- Nonendocrine tumors like retroperitoneal fibrosarcoma can produce insulin-like hormones.

Q.223 State important biochemical changes that occur in an untreated uncontrolled DM.

- **Hyperglycemia:** Increase in blood glucose.
- **Glycosuria:** Excretion of glucose in urine.
- **Hypercholesterolemia:** Increase in blood cholesterol.
- **Ketonemia:** Increased ketonebodies in blood—acetoacetic acid, β OH-butyric acid and acetone.
- Increased urea and nonprotein nitrogen in blood.

Dehydration.

Acidosis: Lowered pH, hyperventilation (Metabolic) and Kussmaul breathing. Lowered HCO_3^- and alkali reserve. Lowered sodium in blood—hyponatremia and disturbance in fluid and electrolyte balance.

Q.224 What is meant by glucose tolerance?

The ability of the body to utilize glucose is ascertained by measuring its glucose tolerance. It is indicated by the nature of blood glucose curve following the administration of a standard dose of glucose.

Q.225 Name some conditions where you find decreased glucose tolerance.

- Diabetes mellitus.
- Hyperthyroidism.
- Hyperactivity of adrenal cortex and anterior pituitary.

Q.226 Name some conditions where you find increased glucose tolerance.

- Hypothyroidism,
- Hypofunction of adrenal cortex (Addison's disease),
- Hypopituitarism,
- Hyperinsulinism,
- Decreased absorption of glucose like sprue/ celiac disease, etc.

Q.227 How will you perform a standard glucose tolerance test (GTT)?

- A 'fasting' sample of venous blood is collected in fluoride bottle ("fasting" sample of blood).
- The bladder is completely emptied and urine is collected for qualitative test for glucose and ketone bodies ("fasting" urine).
- The individual is given 75 gm of glucose dissolved approximately in 250 ml of water to drink orally. Time of oral glucose administration is noted.
- A total of five specimens of venous blood and urine are collected every ½ hr, after the oral glucose (½ hr, 1 hr, 2 hr, and 2½ hr samples of blood and urine).

- Glucose content of all six (including fasting sample) samples of blood are estimated. Corresponding urine samples are tested qualitatively for glucose and ketone bodies.
- The results of blood glucose values are plotted as a graph against time. The curve thus obtained is called *glucose tolerance curve*.

Q.228 Describe a typical normal glucose tolerance curve (GTC).

- Fasting glucose is within normal limits of 60 to 100 mg % ("true" glucose).
- The highest peak value is reached within one hour.
- The highest value does not exceed the renal threshold, i.e. 170 to 180 mg%.
- At 2 hr, there is a *hypoglycaemic dip* (10 to 15 mg lower than fasting value).
- The fasting level is reached by 2½ hr.
- No glucose or ketone bodies are detected in any specimens of urine.

Q.229 Describe a diabetic type of GTC.

- Fasting blood glucose is definitely raised >110 mg% ("true" glucose).
- The highest value is usually reached after 1 to 1½ hr.
- The highest value exceeds the normal renal threshold. The blood glucose does not return to the fasting level within 2½ hr, remains much higher than the fasting value. This is the most characteristic feature of DM.
- Urine samples show presence of glucose. Urine may or may not contain ketone bodies depending on the type of DM and its severity.
- According to severity, the GTC may be:
 - Mild diabetic curve.
 - Moderately severe diabetic curve.
 - Severe diabetic curve.

Q.230 What is primary lactose intolerance?

It is a hereditary deficiency of lactase, most commonly found in persons of Asian and African descent. Common symptoms of lactose intolerance include vomiting, bloating, cramps, watery diarrhea and dehydration. Diagnosis is based on positive

hydrogen breath test after an oral lactose load.

Q.231 What is secondary lactose intolerance?

It is precipitated at any age by gastrointestinal disturbances such as celiac sprue, colitis or viral-induced damage to intestinal mucosa.

Q.232 What is the treatment for lactose intolerance?

Dietary restriction of milk and milk products or using drug like lactase pills.

Q.233 What is galactokinase?

1. Galatosemia
2. Galatosemia
3. Cataract in early childhood.

Q.234 What is the treatment of galactokinase deficiency?

Eliminate the source of galactose from diet. Avoid dairy products.

Q.235 What is essential fructosuria?

It is a genetic deficiency of fructokinase and is usually benign. There is a presence of fructose in urine.

Q.236 What is hereditary fructose intolerance?

It is one of the serious diseases resulting due to the accumulation of fructose 1-phosphate in liver and in proximal tubule in kidney. Present with vomiting, apathy, liver damage can lead to jaundice. Renal damage can lead to Fanconi like syndrome and severe hypoglycemia.

Q.237 What is the treatment for hereditary fructose intolerance?

Eliminate fructose from diet.

Q.238 Explain Pompe's disease.

In this disease there is deficiency by lysosomal L-1,4 glucosidase. It is infantile onset and death usually occurs by 2 years.

Q.239 Explain Cori's disease.

In this disease there is deficiency of glycogen debranching enzyme.

DO YOU KNOW ?

- Muscle stores excess glucose as glycogen.
- Phosphoenolpyruvate (PEP) in glycolysis acts as a substrate level phosphorylation and works irrespective of oxygen and mitochondria hence work when ATP needed in anaerobic phase in myocardial infarction.
- Thiamine deficiency patients along with alcoholic also suffer from hypoglycemia. So if you administer only glucose in these alcoholic patients it would not enter the cell so you need to administer thiamine along with glucose to get the best result.
- Congestive heart failure may be a complication owing to insufficient ATP and accumulation of keto acid in the cardiac muscle.
- Vitamins are very important for the activity of PDH. Any deficiency of vitamins causes decrease function of PDH and decrease glycolysis or carbohydrate metabolism.

Q.1 What are lipids?

Lipids are ester-like compounds of fatty acids which are insoluble in water but are soluble in fat solvents.

Q.2 Give the classification of lipids.

Lipids are classified into:

1. Simple lipids.
2. Compound lipids.
3. Derived lipids.

Q.3 What is biological importance of lipids in the body?

1. Lipids are the most concentrated source of energy.
Their caloric value is 9 KCal/gm.
2. Lipids provides essential fatty acids.
3. Supplies fat-soluble vitamins.
4. As components of cell wall.
5. Lipids act as insulating material.

Q.4 What are fats?

Fats are the esters of fatty acids with glycerol.

Q.5 What is the difference between fat and an oil?

Fats are solid at ordinary temperature whereas oil are liquid, i.e. (liquid) fat at ordinary temperature.

Q.6 What are saturated fatty acids? Give examples.

Fatty acids which do not have any double bond in their structure are called saturated fatty acids.

Examples: Acetic acid (CH_3COOH), propionic acid ($\text{C}_2\text{H}_5\text{COOH}$), butyric acid ($\text{C}_3\text{H}_7\text{COOH}$), etc. Higher homologues like palmitic acid ($\text{C}_{15}\text{H}_{31}\text{COOH}$), stearic acid ($\text{C}_{17}\text{H}_{35}\text{COOH}$), etc.

Q.7 What are unsaturated fatty acids? What are the types?

- Fatty acids which contain double bonds in their structure are called unsaturated fatty acids (UFA).
- **Types:** Depending on the degree of unsaturation they are divided into two groups:

- Monounsaturated (monoethenoid): containing only one double bond.
- Polyunsaturated (polyethenoids): containing more than one double bond in their structure.

Q.8 Give an example of monosaturated FA (monoethenoid) found in our body fat.
Oleic acid $\text{C}_{17}\text{H}_{33}\text{COOH}$ (Formula 18: 1; 9) is found in abundance in our body fat.

Q.9 Name the three polyunsaturated fatty acids (polyethenoids).

- Three polyunsaturated fatty acids of biological importance are:
- **Linoleic acid:** Two double bonds between C_9 and C_{10} and another between C_{12} and C_{13} (formula: 18: 2; 9, 12).
- **Linolenic acid:** contains three double bonds between carbons 9 and 10, 12 and 13, and 15 and 16. (Formula: 18: 3; 9, 12, 15).
- **Arachidonic acid:** It is a 20 C Fatty acid and contains four double bonds between 5 and 6, 8 and 9, 11 and 12, and 14 and 15. (Formula: 20: 4; 5, 8, 11, 14).

Q.10 What are the saturated fatty acids most abundant in fats?

1. Palmitic acid.
2. Stearic acid.

Q.11 What are essential fatty acids?

Essential fatty acids are those which cannot be synthesised by the body and hence are supplied in the diet. They are also called 'polyunsaturated acids'.

They are:

	No. of double bonds	No. of carbon atoms
1. Linoleic acid	2	18
2. Linolenic acid	3	18
3. Arachidonic acid	4	20

Q.12 What are the functions of essential fatty acids?

1. Essential fatty acids prevent deposition of cholesterol in atherosclerosis.
2. In the synthesis of prostaglandins.

Q.13 Which EFA is important?

- Linoleic acid is most important as arachidonic acid can be formed in the body from linoleic acid.
- Biologically arachidonic acid is very important as prostaglandins (PGs) and leukotrienes (LTs) are formed in the body from it.

Q.14 What is saponification?

Hydrolysis of fat by an alkali is called saponifications.

Q.15 What is saponification number?

Saponification number is defined as the mg of KOH required to saponify 1 gm of fat.

Q.16 What does the saponification number of a fat indicates?

- Saponification number of a fat gives an idea about the average chain length of the fatty acids present in the fat.
- Higher the saponification number, the shorter (or smaller) will be the chain length of the fatty acid and vice versa.

Q.17 What is iodine number?

Iodine number is defined as the gms of iodine absorbed by 100 gm of the fat.

Q.18 What does the iodine number indicate?

Iodine number indicates the degree of unsaturation of the fat. Higher the iodine number, the more is the degree of unsaturation of the fat.

Q.19 What is an acid number?

The amount of KOH in mgs required to neutralise free fatty acids presents in 1 gm of fat.

Q.20 Which will have a higher acid number, a pure fat or rancid fat?

A rancid fat will have a higher acid number.

Q.21 What is glycerol?

Glycerol is commonly called as glycerine. Chemically it is a trihydric alcohol containing three -OH groups in the molecule.

Q.22 What are the sources of glycerol in the body?

- Sources of glycerol in the body:
- **Exogenous:** From dietary fats, approximately 22% of glycerol formed in the gut by lipolysis of dietary TG is absorbed directly to portal blood.
- **Endogenous:** From lipolysis of fats (TG) in adipose tissue.

Q.23 What is acrolein test?

The presence of glycerol in a compound is detected by acrolein test. Glycerol, when heated with KHSO_4 , two molecules of water are removed and it produces **acryl aldehyde** which has characteristic pungent or acrid odor.

Q.24 Is glycerol produced in the body by lipolysis a waste product or useful?

- Glycerol produced in the body is not a waste product. It is useful in that:
- Re-esterified to form TG again
- It has nutritive value. It can be converted to glucose/and glycogen by the process called gluconeogenesis.

Q.25 Can glycerol be used clinically in medicine?

Glycerol has been used orally or IV in cases of cerebrovascular diseases. It is nontoxic and it reduces cerebral edema with improvement in CS fluid. There is no rebound increase in intracranial pressure on discontinuation of therapy.

Q.26 What is rancidity?

Rancidity is a chemical resulting in the unpleasant odor and taste in the fat on storage when they are exposed to light, heat, and moisture.

Q.27 What are antioxidants?

Antioxidants are substances which prevents **rancidity**.

Q.28 Give classification of antioxidants?

1. **Primary antioxidants:** Prevent formation of new free radicals, e.g. Luperoxide dismutase, glutathione peroxidase, cellulo plasmin.
2. **Secondary antioxidants:** Remove pre-formed free radicals before they can initiate chain reaction, e.g. vitamin E and C, β carotene, uric acid, bilirubin.
3. **Tertiary antioxidants:** Repair cell structures damaged by free radical attack, e.g. DNA repair enzymes.

Q.29 Which vitamin prevents rancidity?
Vitamin E.

Q.30 What are phospholipids?

Phospholipids are compound lipids that contain in addition to fatty acids, and glycerol/or other alcohol, a phosphoric acid residue, nitrogen containing base and other constituents.

Q.31 State how phospholipids are classified?

- As per Celmer and Cartar's classification phospholipids are classified into 3 groups:
- **Glycerophosphatides:** Containing glycerol as alcohol group, e.g. phosphatidyl choline (lecithin), phosphatidyl ethanolamine (cephalin), phosphatidyl serine, plasmalogens, etc.
- **Phosphoinositides:** Containing inositols as alcohol, e.g. phosphatidyl inositol (lipositol).
- **Phospho sphingosides:** Containing an unsaturated 18 carbon aminoalcohol called sphingosine (sphingol), e.g. sphingomyelin.

Q.32 What are phospholipases? What are the types? Mention their site of action.

- Phospholipases are enzymes which hydrolyze phospholipid in a characteristic way. Five types described. Their specific site of action on lecithin are as follows:
- **Phospholipase A_2 :** Attacks β position and forms lysolecithin + one mole of FA.
- **Phospholipase A_1 :** Attacks ester bond in position 1 of lecithin.
- **Phospholipase B:** Substrate is lysolecithin. (lysophospholipase). It hydrolyzes ester bond in a position and forms glyceryl phosphoryl choline + one mole of FA.
- **Phospholipase C:** Hydrolyzes phosphate ester bond and forms α , β -diacyl glycerol + phosphoryl choline.
- **Phospholipase D:** Splits off choline and forms phosphatidic acid.

Q.33 What is sphingomyelin?

- It is a phospholipid. It does not contain glycerol, instead contains an 18 carbon unsaturated aminoalcohol called as sphingosine (sphingol).
- **On hydrolysis**, sphingomyelin yields one molecule FA + phosphoric acid + nitrogenous base choline + sphingosine (alcohol).

Q.34 What are plasmalogens? How does it differ from lecithin/cephalin?

- Plasmalogens are phospholipids. They are predominantly found in brain and nervous tissue.

- **On hydrolysis**, they yield one molecule of FA + glycerol + phosphoric acid + a nitrogenous base usually ethanolamine (sometimes choline) + one molecule of long chain aliphatic aldehyde. Thus it differs from lecithin/cephalin in having a long chain aliphatic aldehyde.

Q.35 What is cardiolipin?

It is a phospholipid found in mitochondrial inner membrane. Chemically, it is diphosphatidyl glycerol and is formed from phosphatidyl choline.

Q.36 What is dipalmityl lecithin (DPL)? What is its clinical importance?

- Dipalmityl lecithin (DPL), is secreted in lung alveoli by type II granular pneumocytes lining the alveolar wall. It acts as a surfactant and lowers the surface tension in lung alveoli and prevents collapse of lung alveoli.
- Absence of DPL, in premature infants, may produce collapse of lung alveoli producing difficulties in respiration and death. It is called as **respiratory distress syndrome (RDS)** or **hyaline membrane disease**.

Q.37 What are glycolipids? Give examples.

- Glycolipids are lipids that contain carbohydrate moiety in their molecule.
- They are mainly of 2 types:
 - Cerebrosides, and
 - Gangliosides.

Q.38 What are cerebrosides?

Cerebrosides are glycolipids. A cerebroside is built as follows:

Sphingosine (alcohol) $\begin{cases} \text{FA of high molecular weight} \\ \text{Usually galactose (may contain glucose sometimes)} \end{cases}$

Q.39 What are the different types of cerebrosides? How do they differ from each other.

- Cerebrosides are mainly 4 types. They differ from each other by the fatty acid content. They are:
- **Kerasin:** Contains FA lignoceric acid
- **Cerebron:** Contains hydroxy lignoceric acid called "cerebronic (phrenosin) acid".
- **Nervon:** Contains an unsaturated homologue of lignoceric acid called "nervonic acid".
- **Oxyneron:** Contains hydroxy derivative of nervonic acid.

Q.40 What is a ceramide?

Ceramide is formed by esterification of sphingosine with FA of high molecular weight.

- Principally found in white matter of brain, in myelin sheaths and medullated nerves.

Q.41 What is the composition of lecithins?

Lecithins contain glycerol, fatty acids, Phosphoric acid and choline.

Q.42 What is the composition of cephalins?

Cephalins contain glycerol, fatty acids, Phosphoric acid and ethanolamine.

Q.43 What is the sugar component present in cerebroside?

Galactose.

Q.44 What are sphingolipidoses? Describe some important sphingolipidoses.

Sphingolipidoses are a class of heterogeneous group of inherited disorders relating to sphingolipids and they primarily affect the central nervous system (Table 24.1).

Q.45 What are steroids?

Substances possessing cyclopenthydropenanthrene nucleus are called steroids. (Fig. 24.1)

Q.46 Give the structure of cholesterol.
See Figure 24.2.

Q.47 State the characteristic features of cholesterol structure.

- Characteristic features of cholesterol structure are:
- Possesses "cyclopentanoperhydrophenanthrene" nucleus.
- α -OH group at C_3
- an unsaturated double bond between C_5 and C_6
- has two $-CH_3$ groups at C_{10} and C_{13}
- and has an eight carbon side chain attached to C_{17} .

Q.48 What are the sources of cholesterol in the body?

Two sources:

- **Exogenous:** Dietary cholesterol approximately 0.3 gm/day. Diet rich in cholesterol are butter, egg yolk, milk, cream, meat, etc.
- **Endogenous:** Synthesized in the body from two carbon units, acetyl CoA

$$O$$

$$||$$

$$(CH_3C-S-CoA) \text{ Approximately } 1.0 \text{ gm/day.}$$

Q.49 What are the forms in which cholesterol exist in blood and tissues?

Cholesterol occurs both in free form in which -OH group on C_3 is free and in ester form in which -OH group is esterified with fatty acids

Table 24.1: Some examples of sphingolipidoses

Disease	Enzyme deficiency	Lipid accumulating (see key below)	Clinical symptoms
• Niemann-Pick disease	Sphingomyelinase	Cer + p-choline-sphingomyelin	Enlarged liver and spleen, mental retardation, fatal in early life
• Gaucher's disease	β -glucosidase	Cer + Glc glycosylceramide	Enlarged liver, massive splenomegaly, erosion of long bones, mental retardation in infants
• Tay-Sachs disease	Hexosaminidase A	Cer - Glc - Gal (NeuAc) + Gal NAC GM ₂ ganglioside	Mental retardation, muscle weakness, blindness
• Metachromatic leukodystrophy	Arylsulfatase A	Cer - Gal - Gal + O SO ₃ 3 sulfogalactosyl ceramide	Mental retardation, demyelination, psychologic disturbances in adults
• Fabry's disease	α -Galactosidase	Cer - Glc - Gal + Gal Globotriaosyl ceramide	Skin rash, renal failure (full symptoms only in males X-linked recessive)
• Krabbe's disease	β -Galactosidase	Cer + Gal galactosyl ceramide	Mental retardation myelin almost absent

Key: Cer - Ceramide

Glc - Glucose

Gal - Galactose

Neu-Ac - N-acetyl neuraminic acid

* Denotes site of deficient enzyme reaction

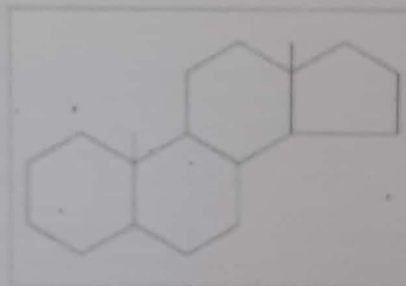


Fig. 24.1: Structure of steroids

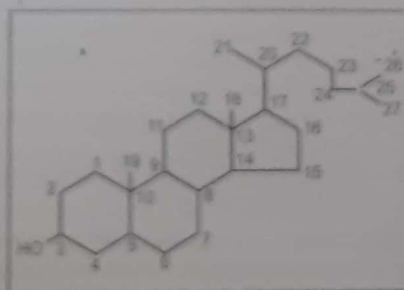
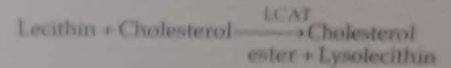


Fig. 24.2: Structure of cholesterol

in plasma by transfer of an acyl group, usually unsaturated, from β -position of lecithin to cholesterol by the enzyme lecithin-cholesterol acyl transferase (LCAT).



Q.51 What are bad effects of cholesterol?
Excessive cholesterol is harmful to body in that it gets deposited in arterial walls producing atherosclerosis. This can narrow the lumen of the blood vessel impeding blood flow which can cause thrombosis.

Q.52 Is cholesterol good for the body? What are the good effects?

- In normal quantities cholesterol has number of good effects. They are:
- Bile acids are produced from cholesterol by its oxidation in liver.
- Cholesterol is converted to steroid hormones in adrenal cortex and to sex hormones in gonads.
- Cholesterol forms 7-dehydrocholesterol and in skin it is converted to vit D₃ by UV rays.
- Cholesterol is poor conductor of heat and hence acts as an insulator.
- It is also a poor conductor of electricity and has a high dielectric constant.
- Cholesterol is in abundance in brain and nervous tissues where it functions as an insulating covering of structures which generate and transmit electrical impulses.

Q.50 How does cholesterol esterified?

Two ways:

- Some cholesterol esters are formed in tissues by the transfer of "acyl" groups from acyl-CoA to cholesterol by the enzyme acyl transferase.
- By interaction of lecithin and cholesterol. Plasma cholesterol esters are produced

Q.53 What are the tests by which cholesterol is detected?

1. Libermann-Burchard reaction.
2. Salkowski test.

Q.54 What is Liebermann-Burchard reaction?

A chloroform solution of cholesterol when treated with acetic anhydride and conc. H_2SO_4 gives a grass green color (cholesta polyene sylphonic acid is formed). This reaction is utilized in colorimetric estimation of cholesterol in blood by Sackett's method.

Q.55 What is Zak's reaction?

When solution of cholesterol is treated with glacial acetic acid, ferric chloride and conc H_2SO_4 , it gives purple red color (cholesta polyene carbonium ion). This reaction forms the basis for the colorimetric estimation of cholesterol by Zak's method.

Q.56 What is 7-dehydrocholesterol? Why it is called pro-vitamin D_3 ?

7-dehydrocholesterol is produced in the body from cholesterol and it is present in skin epidermis. Ultraviolet ray of sunlight changes 7-dehydrocholesterol to vitamin D_3 (cholecalciferol). Hence it is called as *pro-vitamin D_3* .

Q.57 What is pro-vitamin D_2 ?

Ergosterol is a plant sterol. When it is irradiated with UV rays [long wave 265 μm (millimicron)] it changes to vitamin D_2 . Over irradiation may produce toxic substances viz. Toxisterols and suprasterols.

Q.58 What are chylomicrons?

Chylomicrons are the central core of triglycerides, phospholipids and cholesterol combined with small amount of proteins.

Q.59 What are lipoproteins?

The combination of lipid with proteins are called lipoproteins.

Q.60 What are the various functions of plasma lipoproteins?

1. High density lipoproteins, i.e. α -lipoproteins.
2. Low density lipoproteins, i.e. β -lipoproteins.
3. Very low density lipoproteins, i.e. pre- α -lipoproteins.
4. Chylomicrons.

Q.61 Which fraction of the lipoprotein contains maximum cholesterol content?

Low density lipoprotein fraction (LDL).

Normal lipid profile in humans shown in Table 24.2.

Table 24.2: Normal values of lipoproteins (Normal lipid profile)

Lipid fraction	Normal values
• Total cholesterol	150 to 240 mg/dl
• Serum HDL—cholesterol	males—35 to 60 mg/dl female—40 to 70 gm/dl
• Serum TG (Triacylglycerol)	males—60 to 165 mg/dl females—40 to 140 mg/dl
• Serum chylomicrons	up to 28 mg/dl (14 hrs post-absorptive state)
• Serum pre- β lipoproteins (VLDL)	males—up to 240 mg/dl females—up to 210 mg/dl
• Serum β -lipoproteins (LDL)	up to 550 mg/dl
• Serum LDL—cholesterol	up to 190 mg/dl

**Serum LDL—cholesterol can be calculated by the Friedewald formula:

$$\begin{aligned} \text{LDL—cholesterol in mg/dl} &= \frac{\text{Total cholesterol—HDL cholesterol—TG}}{5} \\ \text{LDL—cholesterol in m.mol/L} &= \frac{\text{Total cholesterol—HDL cholesterol—TG}}{2.2} \end{aligned}$$

Note: The formula is not much reliable at TG concentration > 4.5 m.mol/L (> 400 mg/dl)

Q.62 LCAT is present on which lipoprotein?

LCAT is present on HDL molecule. It converts cholesterol and lecithin to cholesteryl ester and lysolecithin respectively.

Q.63 What is the other name for VLDL remnant?

Intermediate density lipoprotein (IDL).

Q.64 LDL receptor can be expressed by another which method?

APO B-100, E receptor.

Q.65 What are apolipoproteins?

The protein component of plasma lipoproteins are called apolipoproteins.

Q.66 What are bile salts?

They are sodium and potassium salts of glycocholates and taurocholates.

Q.67 What is the role of bile salts?

1. As powerful emulsification agents.
2. Bile salt lowers the surface tension of the media and thus aid in the absorption of fats.

Q.68 At which step bile acid synthesis is regulated?

7- α hydroxylase step.

Q.69 What do you understand by respiratory distress syndrome?

Deficiency of lung surfactant cause this disease.

Q.70 Name the substrates required in adipose tissue for TG synthesis.

Two substances are required:

- α -glycero-P and
- Acyl CoA-activated FFA (Fig. 24.3)

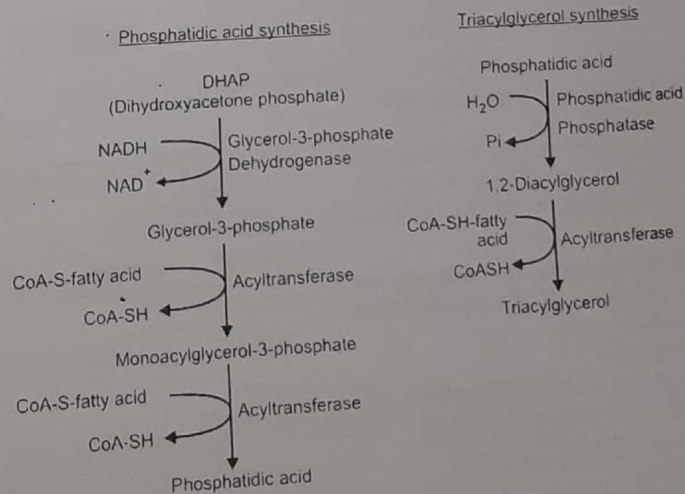


Fig. 24.3: Synthesis of triglycerides

Q.71 Name the enzymes required for breakdown of TG (lipolysis).

Three enzymes are required:

- **Triacyl glycerol lipase**-Hormone sensitive and key rate limiting enzyme. The enzyme can exist in "active" or "inactive" state.
- **Diacyl glycerol lipase** Both are non-lipase
- **Monoacyl glycerol lipase** hormone sensitive.

Q.72 How the action of TG lipase is regulated?

- TG lipase can exist in active "a" and inactive "b" forms.
- Increased cyclic AMP $\uparrow$ level in the cells converts "inactive" cyclic AMP dependent protein kinase (C₂R₂) to "active" protein kinase (C₂), which in turn phosphorylates "inactive" hormonesensitive TG lipase "b" to "active" TG lipase "a" which breaks down TG to form DG + FFA.
- Active TG lipase "a" is converted to "inactive" TG lipase "b" by dephosphorylation.

Q.73 What is the action of insulin on adiposetissue?

- Insulin increases esterification (TG formation) as it enhances the glucose uptake by adipose tissue cells. Also increases glucose oxidation which provides α -glycero-P through di-OH-acetone-P.
- Insulin inhibits lipolysis. This is brought about by decreasing the levels of cyclic AMP in the cells. This is achieved by:
 - inhibiting *adenyl cyclase* enzyme and
 - increasing the *phosphodiesterase* activity..
 Lowered cyclic AMP brings about dephosphorylation of TG lipase "a" to form TG lipase "b" (inactive) through protein kinase.

Q.74 What is the net effect of insulin on plasma FFA level?

Net result of insulin on adipose tissue is to inhibit the release of FFA from adipose tissue which results in fall of circulating plasma FFA level.

Q.75 List the hormones that increase the rate of breakdown of TG (lipolysis) in adipose tissue.

- Catecholamines:
 - Epinephrine and norepine phrine are the principal lipolytic hormones.
- Other hormones are:
 - Glucagon,
 - Growth hormones (GH),

- Glucocorticoids (GC),
- Thyroid hormones,
- ACTH, a and b MSH TSH and vasopressin.

Q.76 What is Lipoprotein lipase? Where it is found? What is the action of this enzyme?

- The enzyme *lipoprotein lipase* is located in the walls of the blood capillaries in various organs.
- TG of circulating chylomicrons and VLDL is acted upon by *lipoprotein lipase* which brings about progressive delipidation. The enzyme requires PL and apo-C II as cofactors

TG of chylomicrons PL, apo-C II
and VLDL
Glycerol

Lipoprotein lipase

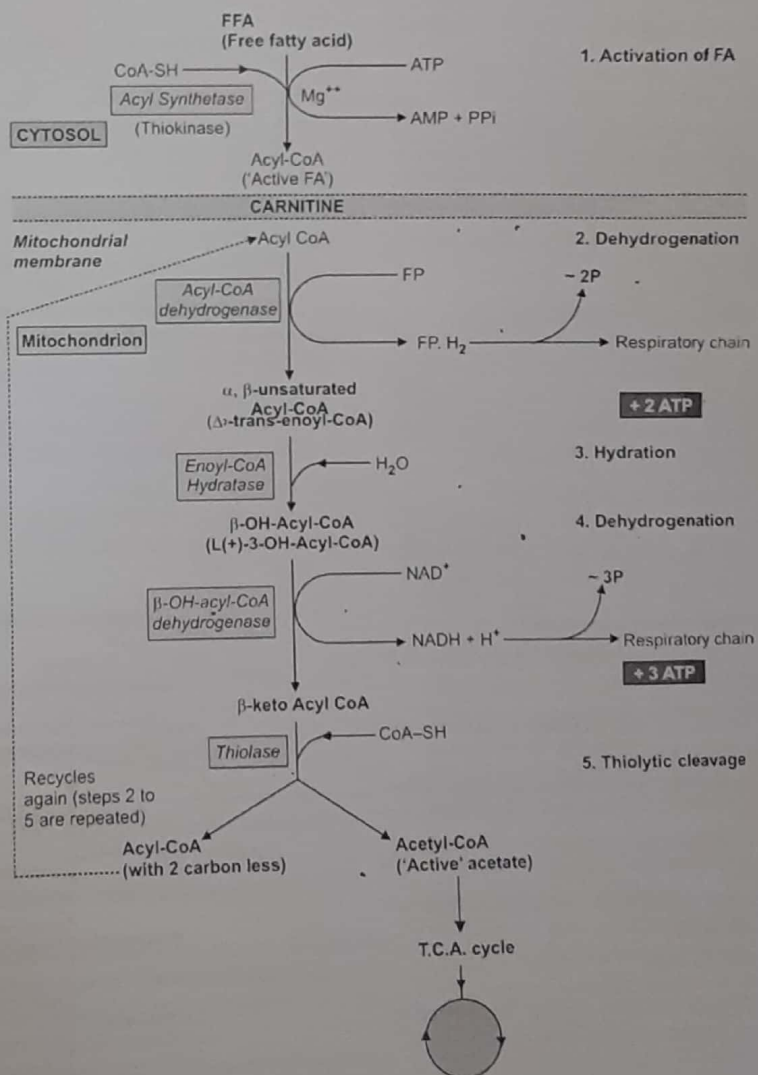


Fig. 24.4: Beta-oxidation of fatty acids

Q.77 What is brown adiposetissue? Where is it located?

- Brown adipose tissue is involved in metabolism particularly when heat generation is necessary.
- The tissue is extremely active:
 - in arousal from hibernation
 - in animals exposed to cold and
 - in heat production in new borns.
- Location site: It is present particularly in the thoracic region.

Q.78 What is β -oxidation?

Oxidation takes place at the β -carbon atom and the β -carbon is oxidised to carboxyl group (Fig. 24.4).

Q.79 Where does β -oxidation take place?
Mitochondria.

Q.80 What is the site of β -oxidation of FA in a cell? Mention the enzyme responsible.

- **Site** In mitochondrion of the cell.
- **Enzyme** Several enzymes known collectively as *FA oxidase enzyme system* are found in the mitochondrial matrix, adjacent to the respiratory chain (ETC). These enzymes catalyze β -oxidation of a long chain FA to acetyl CoA.

Q.81 Activation of long chain FA occurs in cytosol, but β -oxidation occurs in mitochondrial matrix. Activated long chain FA (acyl CoA) is impermeable to mitochondrial membrane. Explain how acyl CoA reach mitochondria?

Acyl CoA penetrate to the inner mitochondrial matrix only in combination with a substance called *carnitine* present in mitochondrial membrane.

Q.82 What is carnitine?

- Carnitine is chemically " β -OH- γ -trimethyl ammonium butyrate". It is widely distributed in liver, and other tissues, particularly in large quantities in muscles.
- Carnitine level of tissues is considerably influenced by dietary methionine and choline levels.
- It is synthesized from lysine and methionine principally in liver and in kidney.

Q.83 What are the functions of carnitine?

- Carnitine does not penetrate the mitochondrial membrane. It is considered as a *carrier molecule* and acts as a *fatty boat* to transport long chain Acyl CoA across mitochondrial membrane for its oxidation in mitochondrial matrix.
- Also facilitates exit of acetyl-CoA and acetoacetyl CoA from mitochondria to cytosol where FA synthesis occurs.

Q.84 State the enzymes involved in transportation of acyl CoA to mitochondrial matrix by carnitine.

- Carnitine-palmitoyl transferase I
- Carnitine-acyl carnitine translocase
- Carnitine-palmitoyl transferase II.

Q.85 Name the steps of β -oxidation.

Once acyl CoA is transported by carnitine in the mitochondrial matrix, it undergoes β -oxidation by the enzyme complex *fatty acid oxidase system*. The steps of β -oxidation are:

- **Dehydrogenation** by the enzyme *acyl CoA dehydrogenase*. H-acceptor is FAD.
- **Hydration** by the enzyme *enoyl hydratase*, addition of one molecule of H_2O .

• **Dehydrogenation** by the enzyme *3-OH acyl CoA dehydrogenase*. H-acceptor is NAD^+ .

• **Thiolytic cleavage** by the enzyme *thiolase* and CoA SH.

Q.86 How many ATPs are formed by one turn of β -oxidation of FA?

Five ATPs:

- Two from oxidation of reduced FAD in ETC.
- Three from oxidation of reduced NAD in ETC.

Q.87 How many acetyl CoA are produced from β -oxidation of one molecule of palmitic acid?

- Palmitic acid is $C_{15}H_{31}COOH$.
 - For complete oxidation by β -oxidation, it will undergo 7 cycles producing 7 acetyl CoA in each turn + one acetyl CoA extra will be produced in last cycle.
- $\therefore$ β -oxidation of one molecule of palmitic acid will form 8 acetyl CoA.

Q.88 How much energy will be produced by β -oxidation of palmitic acid? What is the efficiency?

Each cycle produces 5 ATP, hence
7 cycles will produce $7 \times 5 = 35-P$
Total 8 molecules of acetyl CoA $96-P$
when oxidized in TCA, cycle will form
(12 $\times$ 8) Total = 131-P

In initial activation of palmitic acid $\sim P$ bonds utilized. = -2-P
Total = 129-P

$\therefore$ Energy production = $129 \times 7.6 = 980$ KC
Caloric value of palmitic = 2340 KC/mole acid

$\therefore$ Efficiency = $\frac{980}{2340} \times 100 = 41\%$ of total energy of combustion of palmitic acid.

Q.89 How propionyl CoA enters the citric acid cycle?

Propionyl CoA $\xrightarrow[CO_2, \text{Biotin}]{\text{Propionyl CoA Carboxylase}}$ D-Methyl malonyl CoA.
D-Methylmalonyl CoA $\xrightarrow{\text{Recemase}}$ L-Methylmalonyl CoA.
L-Methylmalonyl CoA $\xrightarrow[B_{12}]{\text{Isomerase}}$ Succinyl CoA.

Q.90 What is α -oxidation?

- α -oxidation is another alternative pathway for oxidation of FA which involves decarboxylation of the $-COOH$ group

after hydroxylation and the formation of a FA containing an "odd-number" of carbon atoms which subsequently undergoes repeated β -oxidation producing (acetyl CoA) $_n$ + propionyl CoA.

- It occurs in microsomes of brain and liver, an aerobic process requires molecular O_2 .

Q.91 What is ω -oxidation?

In ω -oxidation, fatty acids undergo oxidation at ω -carbon, producing a dicarboxylic acid, which is then subjected to β -oxidation and cleavage to form successively smaller dicarboxylic acids.

Q.92 What is precursor of cholesterol biosynthesis?

Acetyl CoA.

Q.93 Where the enzyme system for cholesterol biosynthesis located?

Enzyme system of cholesterol biosynthesis are associated with:

- Cytoplasmic particles (microsomes)
- Soluble fraction of cytosol.

Q.94 Show schematically the steps of cholesterol biosynthesis.

See Figure 24.5.

Q.95 What is the key and rate-limiting enzyme in cholesterol biosynthesis?

HMG-CoA reductase which converts HMG-CoA to mevalonate.

Q.96 How cholesterol biosynthesis is regulated by HMG-CoA reductase?

- Cholesterol itself inhibits the enzyme HMG-CoA reductase by "feed-back" inhibition.
- Fasting/starvation inhibits the enzyme and decreases cholesterol synthesis.

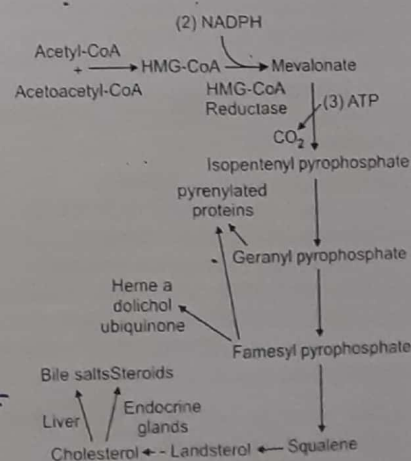


Fig. 24.5: Biosynthesis of cholesterol

- Increased dietary intake reduces the endogenous hepatic biosynthesis of cholesterol by reducing the activity of HMG-CoA reductase.

Hormones:

- Insulin and thyroid hormones increases HMG-CoA reductase activity
- Glucagon and corticosteroids decreases the activity of the enzyme.

Q.97 What are the function of cholesterol?

- Cholesterol is an important tissue component. Because of its conductivity, cholesterol plays a role in insulating nerves and brain structure.
- Cholesterol through the formation of ester of fatty acids appears to play a role in the transport of fatty acids in the body.
- Cholesterol neutralises the hemolytic action of a number of agent such as snake venom, bacterial toxins, etc.
- Cholesterol gives rise to provitamin D.
- Cholesterol is a precursor of cholic acids in the body.
- Cholesterol gives rise to sex hormones.

Q.98 What is the relationship between polyunsaturated acid and cholesterol?
Polyunsaturated acids tends to lower the plasma cholesterol level.

Q.99 What are the source of acetyl CoA?

- Carbohydrate metabolism, i.e. glycolysis.
- Fat metabolism, i.e. β -oxidation of fatty acids.
- Protein metabolism, i.e. transamination.

Q.100 What are the various pathways by which acetyl CoA is utilised?

- Acetyl CoA gives rise to CO_2 and H_2O by citric acid cycle.
- Acetyl CoA gives rise to fatty acid synthesis via malonyl CoA.
- Acetyl CoA gives rise to cholesterol synthesis.
- Acetyl CoA gives rise to acetoacetic acid, i.e. ketone bodies.
- Acetyl CoA undergoes acetylation reactions.

Q.101 What is Refsum's disease?

An inherited disorder due to deficiency of the enzyme *phytate a-oxidase*, as a result phytanic acid cannot be converted to pristanic acid which accumulates in blood and tissues. Principal manifestation is neurological-polyneuropathy with muscle atrophy.

Q.102 What is Zellweger's syndrome (hepatorenal syndrome)?

A rare inherited disorder in which there is inherited absence of peroxisomes in all

tissues. Due to absence of peroxisomes and its enzymes fail to oxidize long chain FA in peroxisomes, resulting to accumulation of long-chain FA (C_{26} to C_{38}) in brain and other tissues like liver/kidney.

Q.103 What is the enzyme system present in higher animals including mammals for "de novo" synthesis?

In higher animals, the synthesis is carried out by a multienzyme complex called *fatty acid synthase* (or *synthetase*), which also incorporates ACP. This multienzyme complex is a dimer, i.e. made up of two identical monomeric units (I and II) aligned "head to foot" on either side. One end has the 4-phosphopantetheinyl group of ACP (Pan-SH), while the other end has cysteinyl SH (cys-SH). Each monomeric unit has seven enzymes required for total synthesis of fatty acid palmitic acid from acetyl CoA (Fig. 24.6).

Q.104 Name the tissues in which "de Novo" synthesis of FA take place.

FA synthesis occurs in adipose tissue, liver, brain kidney, mammary gland, and lungs in which the multienzyme complex have been found in soluble cytosolic fraction of these tissues.

Q.105 What are the sources of NADPH in FA synthesis?

- HMP shunt is the principal source of NADPH,
- Other alternative sources are:
 - Isocitrate dehydrogenase
 - Malic enzyme.

Q.106 State the role of hormones in lipogenesis.

- Glucagon:** Inhibits FA synthesis by inhibiting the key enzyme *acetyl CoA carboxylase*.
- Insulin:** Increases FA synthesis—in several ways:
 - By decreasing lipolysis

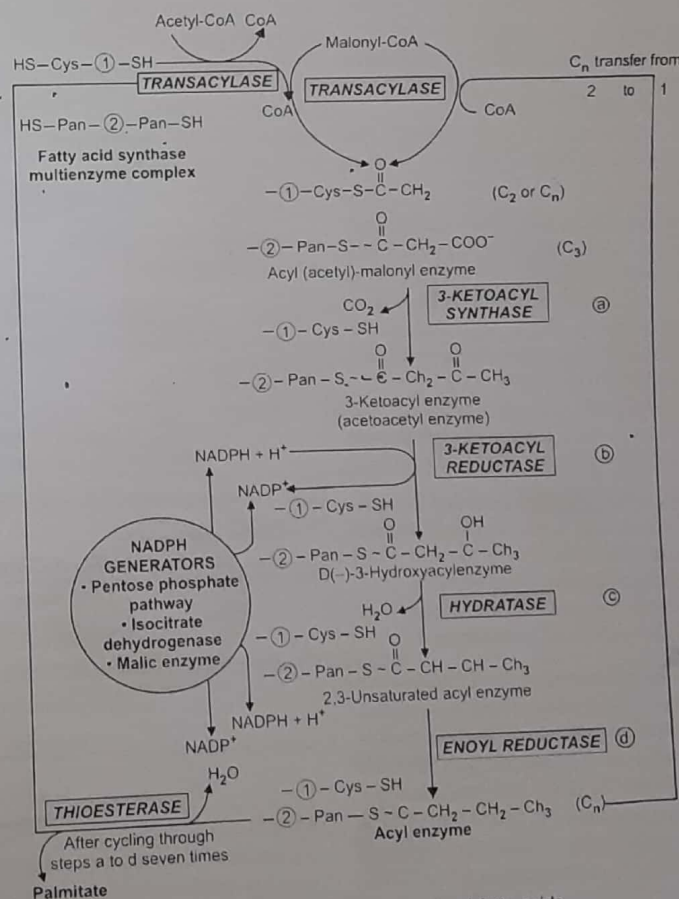


Fig. 24.6: De novo biosynthesis of fatty acids

- Activation of protein phosphatase
- Stimulating synthesis and activation of citrate lyase.
- Enhancing formation of acetyl CoA by stimulating glycolysis.

Q.107 Differentiate mitochondrial and microsomal chain elongation system. Mention the salient points.

Microsomal system	Mitochondrial system
• Usual common pathway. • Operates in endoplasmic reticulum (ER) of microsomes • Acyl CoA of saturated C₁₀ to C₁₆ FA are the starting material. • End product is next higher homologue • Requires O₂ (aerobic)	• Not a common pathway • Operates in mitochondria • Palmityl CoA is usually the starting material. • Stearic acid is produced. • Operates under relative anaerobiosis. High NADH/NAD⁺ ratio favors. • Acetyl CoA is directly incorporated in palmityl CoA. • Pyridoxal-P is required. • NADPH is required.

Q.108 How short chain fatty acids are transported through mitochondrial membrane?

Short chain fatty acids directly enter into the mitochondrial membrane.

Q.109 How long chain fatty acids are transported through mitochondria membrane?
Long chain fatty acids are transported through mitochondrial membrane by continue palmitoyl-transferase and carnitine acylcarnitine translocase.

Q.110 What is a fatty liver?

Significant accumulation of triglycerides in the liver give rise to fatty liver.

Q.111 What are lipotropic factors?

Substances which prevent the deposition of triglycerides in the liver are called lipotropic factors.

Q.112 Name lipotropic substances?

Lipotropic substances are choline, methionine.

Q.113 What are ketone bodies?

Ketone bodies are:

1. Acetoacetic acid.
2. β -hydroxybutyric acid.
3. Acetone.

Q.114 What is the nature of ketone bodies?

Ketone bodies are acidic.

Q.115 Who is the net producer of ketone bodies?

Liver.

Q.116 Can liver utilise the ketone bodies? If not why?

Liver cannot utilise the ketone bodies because the activating enzyme required for ketone bodies utilisation is absent in the liver.

Q.117 What are the tissues which prefer ketone bodies for utilisation?

Extrahepatic tissue prefer ketone bodies for utilisation because they possess the activating enzymes.

Q.118 What is ketosis?

Significant accumulation of ketone bodies in blood (Ketonemia) and their excretion in the blood (ketonuria) leads to a condition known as ketosis.

Q.119 What is the normal excretion of ketone bodies?

1 mg per day.

Q.120 How ketone bodies are formed?

Ketone bodies are formed as intermediary breakdown products of fat metabolism. If carbohydrate metabolism is defective, more fat breaks for energy purpose. Hence more of ketone bodies are formed.

Q.121 Name the ketone bodies.

- Acetoacetic acid
- β -OH butyric acid
- Acetone

Q.122 Show schematically the interrelationship of three ketone bodies.

See Figure 24.7.

Q.123 How ketone bodies are formed in the liver?

- **Principal pathway** is by HMG-CoA formation (Fig. 24.8)
- Acetyl CoA + Acetyl CoA $\rightarrow$ Acetoacetyl CoA.
- Acetoacetyl CoA + Acetyl CoA

Q.124 How are ketone bodies utilized by extrahepatic tissues?

Ketone bodies once formed in the liver flow to the blood from where ketone bodies are taken up by extrahepatic tissues, and utilized as "fuel".

- **Main pathway:** uses succinyl CoA
Acetoacetate + Succinyl CoA
Thiophorase

Acetoacetyl CoA + Succinic acid

Note: Thiophorase enzyme also called as CoA transferase.

- **Minor pathway** (Fig. 24.9)

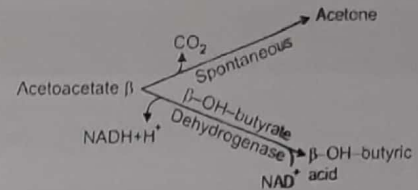


Fig. 24.7: Interrelationship of three ketone bodies

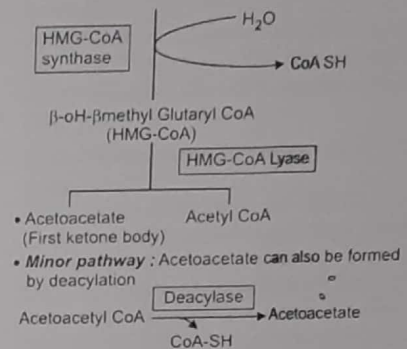


Fig. 24.8: Principal pathways

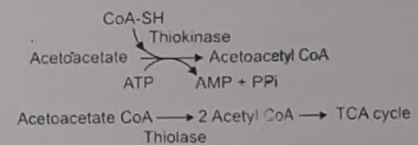


Fig. 24.9: Minor pathway

Q.125 What is Tay-Sachs disease?

Deposition of Ganglioside Gm₂ in brain and eye (macula) gives rise to Tay-Sachs disease. Death usually occur in <2 yr.

Q.126 What is I cell Disease?

Accumulation of inclusion bodies in the cell due to defect in phosphorylation of mannose.

Q.127 What is sphingolipids?

They are the important constituent of cell membrane. They contain no glycerol but are similar in structure to glycerophospholipid.

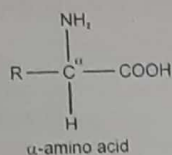
Q.128 What is MCAD (Medium chain acyl CoA dehydrogenase) deficiency?
MCAD deficiency include profound fasting hypoglycemia and lethargy, coma and even death if not treated.

Q.129 What is the Rx (treatment) of MCAD deficiency?
IV glucose.

Amino Acids and Proteins

Q.1 What is an α -amino acid?

Amino acid containing amino group and carboxyl group on the α -carbon atom.



Q.2 What are glycogenic amino acids?

Those amino acids which give rise to the intermediates of carbohydrate metabolism are called glycogenic amino acids.

Q.3 Name the glycogenic amino acids?

Alanine, arginine, aspartic acid, cysteine, glutamic acid, glycine, histidine, proline, methionine, serine, leucine, threonine.

Q.4 What are ketogenic amino acids?

Amino acids which give rise to acetyl CoA during its course of metabolism are called ketogenic amino acids.

Q.5 Name some amino acids which are both glycogenic and ketogenic.

1. Phenylalanine.
2. Tryptophan.
3. Tyrosine.

Q.6 What are essential amino acids?

Essential amino acids are those amino acids which can not be synthesised by the body and hence have to be supplied in the diet.

Q.7 Name ketogenic amino acids?

Leucine.

Q.8 Name the essential amino acids?

Isoleucine, Leucine, Tryptophan, Threonine, Phenylalanine, Valine, Methionine, Lysine. The essential amino acids can be remembered using the formula "MTTVILPL".

Q.9 What are semi-essential amino acids?

Those amino acids which are partly synthesised by the body are called semi-essential amino acids, e.g. arginine and Histidine.

Q.10 What are non-essential amino acids?

Those amino acids which are synthesised by the body.

Q.11 How are amino acids classified?

- Amino acids are classified mainly into three groups depending on their reaction in solution as follows:
- Neutral amino acids
- Acidic amino acids
- Basic amino acids

Q.12 What are neutral amino acids? Give examples.

- Neutral amino acids form the largest group and they can be further classified as follows:
- **Aliphatic amino acids:** Having an aliphatic side chain and are called simple monoamino monocarboxylic acids, e.g. glycine (gly), alanine (ala), valine (val), leucine (leu) and isoleucine (ile).
- **Hydroxyamino acids:** Having an —OH gr, e.g. serine (ser), threonine (thr).
- **Aromatic amino acids:** Contain a benzene ring, e.g. phenylalanine (phe), tyrosine (tyr).
- **Heterocyclic amino acids:** Containing heterocyclic group like indole ring or imidazole ring, e.g. tryptophan (trp), histidine (his).
- **Sulphur containing amino acids:** Containing Sulphur, e.g. methionine (met), cysteine (cys) and cystine.
- **Imino acids:** They do not have a free —NH_2 group but have a basic, pyrrolidone ring, e.g. proline (pro), hydroxyproline (hyp).

Q.13 What is the relationship between cysteine and cystine?

Two molecules of cysteine are joined together by S-S-bond to form one molecule of cystine. One molecule of cystine on oxidation gives 2 molecules of cysteine.

Q.14 What are acidic amino acids? Give examples.

Acidic amino acids have two —COOH groups and one NH_2 group and they are mono-amino dicarboxylic acids, e.g. aspartic acid (asp) glutamic acid (glu).

Q.15 What are basic amino acids? Give examples.

Basic amino acids have one —COOH group and two —NH_2 groups and they are di-aminomonocarboxylic acids, e.g. arginine (arg), lysine (lys) and hydroxy-lysine (hyl). **Note:** Histidine (his) can also be classified as basic amino acid due to presence of imidazole ring.

Q.16 Which is the simplest of the amino acids?

Glycine is the simplest of the amino acids.

Q.17 What is the chemical name of glycine?

Chemically glycine is called as aminoacetic acid.

Q.18 Which amino acids do not have an asymmetric carbon and do not show optical activity?

Glycine is the only amino acid which does not have an asymmetric carbon and does not exhibit optical activity.

Q.19 Name the amino acids that do not occur in proteins. Give 5 examples.

- There are compounds similar to basic skeletal structure of amino acid but do not occur in proteins. **Examples:**
- β -alanine found in coenzyme A
- α -amino butyrate—a neurotransmitter
- Taurine—present in bile salt
- Ornithine and citrulline—metabolites of urea cycle
- DOPA—precursor of pigment melanin.

Q.20 Give examples of three amino acids which produce specific biologic compounds.

- Tyrosine: produces
– Thyroid hormones,

- Catecholamines, and
- Melanin pigments

• **Glycine:** Required for—

- Heme synthesis
- Formation of creatine
- Formation of glutathione

• **Tryptophan:** Necessary for

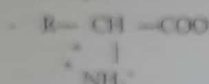
- Formation of serotonin
- Forms vitamin nicotinic acid.

Q.21 How does an amino acid behave in an acidic and basic media?

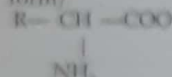
- Every amino acid has atleast two ionizable groups:

- The $-NH_3^+$ group and
- The $-COOH$ group.

- In acidic medium: The $-NH_3^+$ group behaves as a base and accepts a proton and becomes positively charged (cationic form).



- In basic medium: The $-COOH$ group acts as proton donor and the amino acid becomes negatively charged (anionic form)



Q.22 What is isoelectric pH and what are zwitterion (or hybrid ion)?

At a specific pH the amino acids carry both the charges in equal number and thus exists as *dipolar ion or zwitterion*. At this point the net charge on the amino acids is zero, i.e. the positive and negative charges on the amino acids equalize. The pH at which the amino acid or protein is in zwitterion form is called *isoelectric pH (pI)*.

Q.23 What is the isoelectric pH (pI) for albumin, hemoglobin and casein?

- Isoelectric pH (pI) of:
- Albumin is 4.7
- Hemoglobin is 6.7
- Casein is 4.6

Q.24 Name the sulphur containing amino acids.

1. Methionine.
2. Cysteine.
3. Cystine.

Q.25 Which sulphur containing amino acid is an essential amino acid?

Methionine.

Q.26 Deficiency of essential amino acid causes what?

Deficiency of essential amino acids leads to negative nitrogen balance

Q.27 What are the sources of amino acids in the blood?

1. Tissue protein breakdown.
2. Dietary proteins.
3. Intracellular synthesis.

Q.28 What is the role of amino acids?

1. As building block of proteins.
2. As precursors of:
 - a. Hormones
 - b. Purines
 - c. Pyrimidines
 - d. Porphyrins
 - e. Certain vitamins such as pantothenic acid, folic acid, etc

Q.29 What are the amino acids containing aromatic ring?

1. Phenylalanine.
2. Tyrosine.
3. Tryptophan.

Q.30 Guanidino group is present in which amino acid?

Arginine.

Q.31 Imidazole group is present in which amino acid?

Histidine.

Q.32 Indole group is present in which amino acid?

Tryptophan.

Q.33 Name an imino acid.

Proline.

Q.34 How amino acids are detected?

1. Aromatic amino acids containing benzene ring by Xanthoproteic reaction.
2. Hydroxy aromatic amino acids by Millon's reaction.
3. Tryptophan by Hopkin's Cole reaction.
4. Arginine by Sakaguchi reaction.

Q.35 What is active methionine?

S-adenosyl methionine is called the active methionine.

Q.36 What is the role of methionine?

As methyl group donor.

Q.37 Which amino acid give rise to nicotinic acid during its metabolism?

Tryptophan.

Q.38 What are proteins?

Proteins are polypeptides, formed by the linking of amino acids through peptide bonds.

Q.39 Give the classification of proteins.

Proteins are classified into:

1. Simple proteins.
2. Conjugated proteins.

Q.40 What is the configuration of amino acids present in the naturally occurring proteins?

L-configuration.

Q.41 What are the general tests for proteins?

1. Biuret reaction.
2. Ninhydrin test.

Q.42 Give one common reaction for all amino acids.

All amino acids on heating with a solution of ninhydrin in acetone to about 100 °C give a purple-blue colored compound (*Ninhydrin reaction*).

Q.43 Which amino acids do not give purple-blue color in ninhydrin reaction?

Proline and hydroxyproline are exceptions and they give yellow color.

Q.44 What is biuret reaction?

Protein solution in water, two or three drops of dilute copper sulphate and excess of NaOH (about one ml) produces a pink or purple-violet color.

Q.45 What is the mechanism of biuret reaction?

The color depends upon the presence of 2 or more peptide bonds. It is due to coordination of cupric ions with the unshared electron pairs of peptide nitrogen and the oxygen of water to form the colored co-ordinate complex.

Note: Thus di-peptides and free amino acids do not give the biuret test.

Q.46 Name one amino acid which can give +ve biuret reaction.

Only Histidine can give a +ve biuret reaction.

Q.47 What is xanthoproteic reaction?

The aromatic amino acids like phenylalanine, tyrosine and tryptophan present in protein molecule give yellow precipitate when heated with conc HNO₃. The reaction is due to nitration of aromatic ring.

Note: Collagen and gelatin do not give a positive reaction.

Q.48 Give a specific color test for tyrosine in protein.

Millon's test: It is a specific color test for tyrosine of protein. Proteins having tyrosine give a white precipitate with

Millon's reagent (10% mercurous chloride in H_2SO_4) on heating. On addition of NaNO_2 the precipitate turns pink-red.

Q.49 Mention a specific color test for arginine in protein.

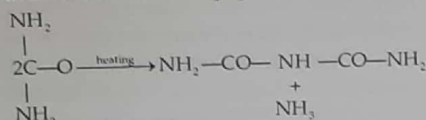
Sakaguchi test: It is a specific color test for arginine of protein. Sakaguchi reagent consists of alcoholic α -naphthol and a drop of sodium hypobromite. Guanidine group of arginine reacts to give the red color.

Q.50 Mention a specific color test for sulphur containing amino acids.

Lead acetate test: It is specific for sulphur containing amino acids. The proteins having S-containing amino acids when boiled with strong alkali split out sulphur as sodium sulphide which reacts with lead acetate to give black precipitate of lead sulphide (Pbs).

Q.51 What is Biuret?

Biuret is a compound formed by heating urea. It contains two peptide linkages.



Q.52 What are the methods by which proteins can be estimated?

1. Buret method.
2. Lowry method.
3. Microkjaldahl method.
4. Spectroscopy.

Q.53 What is the factor used in the conversion of nitrogen to protein?

6.25.

Q.54 How does the factor 6.25 come?

The average nitrogen content of proteins is 16 gm%, i.e. 16 gm of nitrogen is present in 100 gm protein. 1 gm of nitrogen is present in 6.25 gm protein.

Q.55 Name two dipeptides found in our body.

The two dipeptides present in muscle tissues are:

- **Carnosine:** β -alanine + histidine
- **Anserine:** β -alanine + methyl-histidine

Q.56 Name tripeptide of immense biological importance.

- Glutathione is a tripeptide consisting of three amino acids—Glutamic acid + Cysteine + and Glycine.
- It functions in the body in oxidation-reduction system.

Q.57 What are fibrous proteins? Give two examples.

When the axial ratio of length: Width of a protein molecule is more than 10, it is called a fibrous protein.

Examples:

- α -keratin from hair
- Collagen of connective tissues.

Q.58 What are globular proteins? Give two examples.

When the axial ratio of length: width of a protein molecule is less than 10, it is called a globular protein.

Examples:

- Hemoglobin
- Myoglobin

Q.59 What are chromoproteins? Give suitable examples.

Chromoproteins are conjugated proteins that contain a simple protein with colored substance as the prosthetic group.

Examples:

- All hemoproteins are chromoproteins which contain "heme" as prosthetic group, e.g. hemoglobin, cytochromes, catalase, peroxidase.
- Flavoproteins contain riboflavin, a yellow colored substance.
- Visual purple (rhodopsin) contains protein 'opsin' + prosthetic group 11-cisretinal.

Q.60 What are phosphoproteins? Give two examples which have dietary importance.

Phosphoproteins are conjugated proteins containing phosphoric acid as the prosthetic group. The phosphoric acid is esterified through the -OH groups of serine and threonine.

Examples:

- Caseinogen of milk
- Vitellin of egg yolk

Q.61 What are metalloproteins? Give three examples.

Metalloproteins are conjugated proteins which contain a metal ion as their prosthetic groups.

Examples:

- **Carbonic anhydrase** contains zinc
- **Ceruloplasmin** contains copper
- **Ferritin** contains Fe

Q.62 What is the essential difference between glycoproteins and mucoproteins?

- Both are conjugated proteins and contain carbohydrate moiety as prosthetic group.
- They differ by carbohydrate content in that glycoproteins contain less than 4%

of carbohydrates whereas mucoproteins contain more than 4% of carbohydrates.

Q.63 How does primary derived proteins differ from secondary derived proteins? Give examples of each group.

Primary derived proteins are synonymous with denatured proteins. In this, primary structure is not disturbed and peptide bonds remain intact.

Examples: Proteans, metaproteins-acid/alkaline, coagulated proteins.

Secondary derived proteins are formed by progressive hydrolysis of the peptide bonds. Protein is gradually hydrolyzed to produce smaller molecules.

Examples: Proteoses (albumoses), peptones, peptides.

Q.64 What are the various levels of protein structure?

1. Primary structure.
2. Secondary structure.
3. Tertiary structure.
4. Quaternary structure.

Q.65 What is primary structure?

Primary structure indicates the sequence of amino acids linked through peptide bonds. It has a N-terminus and for the other end C-terminus.

Q.66 What is an α -helix?

The coiling of polypeptide chain into a right hand helix.

Q.67 How will you determine the N-terminal amino acid of a protein?

1. Sanger's method.
2. Edman's method.

Q.68 Which reagent is used in Sanger's method?

2,4 dinitrofluorobenzene (2,4 DNFB).

Q.69 Which reagent is used in Edman's method?

Phenyl isothiocyanate.

Q.70 Which method is superior, Sanger's or Edman's?

Edman's method is superior over Sanger's method because Edman's method involves the removal of one amino acid at a time from the N-amino acid of a peptide, leaving the remaining peptide chain intact. The process can be repeated and sequence of amino acid is obtained from N-terminal end. Whereas in Sanger's method only the N-terminal amino acid is identified because after the removal of N-terminal amino acid with DNFB, the remaining peptide chain breaks into amino acid mixture.

Q.71 What is secondary structure?

- Secondary structure refers to the twisting of the peptide chain into a helical form. Hydrogen bonds are responsible for the secondary structure.
- Hydrogen bonds in secondary structure may form either:
 - α -helix or
 - β -pleated sheet.

Q.72 Name different secondary structures of proteins.

1. α -helix.
2. β -pleated sheath.
3. β -bend.
4. Super secondary (motifs).
5. Non-repetitive secondary structures.

Q.73 Which amino acid is conspicuously absent in α -helix?

Proline is never found in α -helix and is conspicuously absent.

Q.74 Which amino acids tend to destabilize α -helix?

More polar residues such as arginine, glutamic acid and serine may repel and destabilize α -helix.

Q.75 Which protein tends to form triple helix?

Collagen is rich in proline and hydroxyproline hence it cannot form α -helix or β -pleated sheet; it forms a triple helix. The triple helix is stabilized by both non-covalent as well as covalent bonds.

Q.76 What is tertiary structure? Which bonds are found in tertiary structure?

- Tertiary structure arises due to the folding of the helical structure into globular, ellipsoidal or any other conformations.
- The folding is brought about by:
 - Disulfide bonds (S-S)
 - Polar or/salt linkages between atoms with positive and negative charges.
 - Hydrophobic bonds, and
 - Van der Waal forces.

Q.77 What is quaternary structure?

When proteins consists of 2 or more peptide chains held together by non-covalent interactions or by covalent cross-links, it is referred to as the quaternary structure.

Examples:

- **Hemoglobin:** A tetramer having 4 polypeptide chains
- **Lactate dehydrogenase (LDH):** Four polypeptide chains.

Q.78 What is meant by +ve ion precipitation?

The +ve ions most commonly used are those of heavy metals like Pb^{2+} , Zn^{++} , Ca^{++} , Hg^{++} , etc. The metals precipitate proteins at the pH alkaline to its isoelectric pH (pI). At this pH, proteins behave as anions and +ve metal ions combine with $-COO$ group to give insoluble precipitate of metal proteinate.

Q.79 What is meant by -ve ions precipitation?

Negative ions like tungstic acid, trichloroacetic acid, picric acid, tannic acid, etc. combine with proteins when the pH of the medium is on acidic side of its isoelectric pH (pI). In acidic pH, proteins exist as Pr^+ and forms precipitate with -ve ions.

Q.80 What is electrophoresis?

Migration of charged molecules under the influence of electricity is called electrophoresis.

Q.81 What are the fractions obtained in paper electrophoresis of serum?

Albumin being lighter moves fast. It is followed by α_1 globulin, α_2 globulin, β -globulin and last γ -globulin near the origin.

Q.82 What is Rf value?

The ratio of the distance moved by a compound/amino acid in paper chromatography to the distance moved by the solvent front is known as Rf value.

Q.83 Name six important plasma proteins.

- Six plasma proteins are:
- Albumin, α_1 -globulin, α_2 -globulin, β -globulin, fibrinogen and γ -globulin.

Q.84 Which fraction will be absent in serum proteins and why?

Fibrinogen is absent in serum protein because fibrinogen is used up in clot formation in separation of serum.
Serum = Plasma - Fibrinogen.

Q.85 Where is albumin synthesized in the body?

Albumin is mainly synthesized in the liver. Rate of synthesis is approximately 14.0 gm/day.

Q.86 State the important functions of albumin.

- Nutritive function.
- Exerts low viscosity

- Contributes 70 to 80% of osmotic pressure.
- Plays important role in exchange of fluids between blood and tissue.
- Helps in transport of several substances viz. FFA, unconjugated bilirubin, Ca^{++} and steroid hormones.
- Binding of certain drugs: Sulphonamides, aspirin, penicillin are bound to albumin and carried in the blood.

Q.87 What is haptoglobin? What is its clinical importance?

Haptoglobin (Hp) is an α_2 -globulin synthesized in Liver. Haptoglobin binds "free" Hb if present in blood. Average binding capacity of Hp irrespective of phenotype is approximately 100 mg/dl. Hp-Hb complex circulates in the blood and ultimately destroyed by RE cells.

Q.88 What is methemalbumin? What does it signify?

Normally blood does not contain methemalbumin. When the degree of intravascular (IV) hemolysis is rapid and severe, as may happen in incompatible blood transfusion, free Hb released exceeds the binding capacity of Hp. The free Hb combines with albumin to form *met-hemalbumin*, which can be detected by a sensitive test called *Schumm's test*. Detection of methemalbumin points to IV hemolysis.

Q.89 What is Bence Jones protein? What is the clinical significance?

Bence Jones protein is an abnormal protein which occurs in blood and urine of people suffering from a disease called *multiple myeloma* (a plasma cell tumor). It is monoclonal light chain either " κ " or " λ " and excreted in urine of multiple myeloma patient.

Q.90 How Bence Jones protein can be detected in urine in clinical laboratory?

Presence of Bence Jones protein in urine can be identified easily by a simple "heat test". Take urine in a clean test-tube and heat upto 50° to $60^\circ C$, when Bence Jones protein if present, are precipitated. But when heated further the precipitate dissolves again. Reverse occurs on cooling.

Q.91 What are the normal values of total proteins and differential proteins?

- Total proteins: 7.0 to 7.5 gm%
- Differential protein:

Gm% (by precipitation)	% of total proteins (by electrophoresis)
• Albumin—5.7 to 5.9	50 to 70%
• Globulins—1.8 to 3.6	29.5 to 54%
• α_1 -globulins—0.1 to 0.4	2.0 to 6.0%
• α_2 -globulins—0.4 to 0.8	5.0 to 11.0%
• β -globulins—0.5 to 1.3	7.0 to 16%
• γ -globulins—0.6 to 1.5	11.0 to 22.0%

Q.92 What is the normal level of fibrinogen in plasma?

0.2 to 0.4 gm% (200 to 400 mg%).

Q.93 What is the normal A:G ratio? In which condition the A:G ratio is reversed?

- Normal A:G ratio is 2.5 to 1.0 (usually 2:1)
- It is reversed characteristically in cirrhosis liver.

Q.94 What are the types of hypergamma-globulinemias? Give a few examples.

- **Polyclonal gammopathies:**
 - Chronic infections, e.g. TB, kala-azar
 - Chronic liver disease—cirrhosis liver
 - Sarcoidosis
 - Autoimmune diseases, e.g. rheumatoid arthritis, systemic lupus erythematosus (SLE)
- **Monoclonal gammopathies:**
 - Multiple myeloma
 - Macroglobulinemia, etc.

Q.95 What is hypoproteinemia?

Decrease of total proteins below normal is called hypoproteinemia.

Q.96 State some causes of hypoproteinemia.

- Hypoproteinemia can occur as follows:
- **Hemodilution:** Water intoxication (overload), IV infusion of fluids. In this both albumin and globulins are decreased and A:G ratio remains unaltered.
- **Hypoalbuminemia:** Conditions resulting in low albumin level, accompanied either by no increase in globulin or by an increase which is less than the fall in albumin. A:G ratio is decreased.

Examples:

- Loss through kidney: Nephrotic syndrome
- Loss through G.I Tract: Protein losing enteropathy, etc.
- **Hypergamma-globulinemia:**
 - Conditions due to decrease in γ globulin.

Q.97 Name four transport binding proteins.

- **Transferrin (siderophilin):** Binding and transport of Fe.

- **Thyroxine binding globulin (TBG):** Binds and transport thyroxine (T_4)

- **Transcortin:** Binding and transport of cortisol.

- **Retinol binding protein (RBP):** Binding and transport of retinol (Vit A).

Q.98 Which carbamoyl phosphate synthase is there in urea cycle?

Carbamoyl phosphate synthase I.

Q.99 Name the hormones which favor tissue uptake of amino acids.

- Insulin
- Growth hormone (GH) and
- Testosterone favor the uptake of amino acids by tissues. Hence these hormones are called "anabolic" hormones.
- Estradiol stimulates selectively the uptake of amino acids by uterus.
- Epinephrine and glucocorticoids (GC) stimulate uptake of amino acids by the liver only.

Q.100 What is meant by nitrogen balance or nitrogen equilibrium?

- In an adult healthy individual maintaining a constant weight, the amount of intake of 'N' in food (as dietary proteins mainly) will be balanced by an excretion of an equal amount of 'N' in urine (in the form of urea mainly, uric acid, creatine and to a small extent by amino acids) and
- In faeces (mainly as unabsorbed N)
- Such an individual is then said to be in nitrogen balance or in nitrogen equilibrium.

Q.101 What is meant by positive nitrogen balance?

- If the amount of nitrogen consumed exceeds the amount of nitrogen lost, the person is said to be in *positive nitrogen balance*. This happens in:
 - Growing children
 - Pregnancy
 - Convalescence after illness/surgery, and
 - After administration of anabolic hormones.
- In positive nitrogen balance, the body puts on weight and N-intake will be greater than N-output since new cells are formed and N is retained as tissue proteins.

Q.102 What is meant by negative nitrogen balance?

A subject whose intake of N is less than the output of N is said to have a *negative nitrogen balance*. The person loses weight and the tissue proteins are catabolized to a

greater extent than they are formed. Such a situation occurs in:

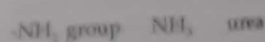
- Old age
- Starvation
- Wasting diseases like *tuberculosis*
- Cancer
- Prolonged illness
- Postoperative conditions and burns.

Q.103 What non-protein nitrogenous substances are synthesized from amino acids? List them.

- **Creatine** from glycine, arginine and methionine.
- **Heme** from glycine and succinyl CoA.
- **Melanins** from tyrosine.
- **Choline** from serine through ethanolamine.
- **Purine nucleotides** from glycine with 5-Phosphoribosylamine.
- **Pyrimidine nucleotides** from aspartate and carbamoyl phosphate.

Q.104 What happens to the N-part of an amino acid?

In mammalian tissues, α -NH₂ group of amino acids, derived either from the diet or breakdown of tissue proteins, is first converted to NH₃ and then to urea and excreted in urine.



excreted in urine

Humans are therefore called *ureotelic*.

Q.105 Why proteins are amphoteric in nature?

Proteins contain some free amino group and carboxyl group. It will give the tests of both acid and basic groups. Hence they are amphoteric in nature.

Q.106 Why proteins behave as buffers?

On either side of pI, Proteins behave as cations or anions depending upon the pH of the media and hence they behave as buffers.

Q.107 What is denaturation of proteins?

Denaturation is defined as loss in secondary, tertiary and quaternary (if present) structure of proteins on heating. Denaturation results in loss in biological activity of the proteins.

Q.108 What is a nitrogen balance?

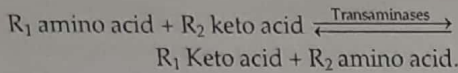
If the ratio of N (protein) intake to N output is equal to 1, the person is in a *Nitrogen balance*.

$$\frac{\text{N intake}}{\text{N out take}} = 1$$

Q.109 What is transamination?

Transamination is an intermolecular transfer of amino group of an amino acid to a keto group of keto acid resulting in the formation of corresponding keto acid and amino acid.

Transaminases



Q.110 What is the co-factor required in transamination?

Pyridoxal phosphate.

Q.111 Name the tissues in which transamination reaction occur.

Transamination takes place principally in

- Liver
- Kidney
- Heart
- Brain, but to a small extent transamination can take place in all tissues.

Q.112 What are the amino acids which participate most in transamination?

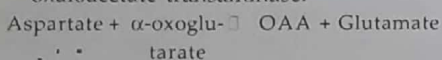
Alanine, aspartic acid, glutamic acid.

Q.113 What are the amino acids which does not participate in transamination?

Lysine, threonine.

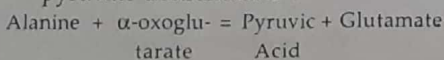
Q.114 Name two specific transaminases which are of clinical importance. What are the substrates and end-products?

- **Aspartate transaminase** (or aspartate aminotransferase): Previously used to be called as "S-GOT"— serum glutamate oxaloacetate transaminase.



Donor (amino acid)	recipient (keto acid)	(New keto acid)	(New amino acid)
Aspartate	α-oxoglutarate	OAA	Glutamate

- **Alanine transaminase** (or Alanine aminotransferase): Previously used to be called as "S-GPT"— serum glutamate "pyruvate transaminase".



Donor (amino acid)	recipient (keto acid)	(New keto acid)	(New amino acid)
Alanine	α-oxoglutarate	Pyruvate	Glutamate

Note: In both the cases, glutamic acid is produced as new amino acid.

Q.115 What are normal levels of S-GOT and S-GPT?

- **S-GOT** (aspartate transaminase): Normal serum activity is 4 to 7 i.u/L (7 to 35 Karmen units/ml).

- **S-GPT** (alanine transaminase): Normal serum activity is 3 to 15 i.u/L (6 to 32 Karmen units/ml).

Q.116 What is deamination? What are the types?

Deamination is the process by which N of amino acid is removed as NH_3

2 types of deamination:

- Oxidative deamination
- Non-oxidative deamination.

Q.117 What is the site of oxidative deamination?

- Liver and
- Kidney are the main organs where oxidative deamination takes place.

Q.118 What is the nature of enzyme and coenzyme that take part in oxidative deamination?

- **Enzyme:** L-amino acid oxidase which acts specifically on L-amino acids and oxidatively deaminate to form NH_3 .
- **Coenzyme:** is flavoprotein (FMN) which acts as H-acceptor.
- **Nature of the enzyme:** The L-amino acid oxidases are auto-oxidizable flavoproteins. The reduced flavoproteins FMN. H_2 are re-oxidized at substrate level directly by molecular O_2 forming H_2O_2 which is converted to H_2O by the enzyme *catalase*.

Q.119 What are the toxic substances that can form in oxidative deamination of L-amino acids by "L-amino acid oxidase"?

- **Superoxide anion O_2^- :** This is scavenged by the enzyme *superoxide dismutase*.
- **H_2O_2 :** Scavenged by the enzyme *catalase*. Both superoxide anion O_2^- and H_2O_2 are toxic and they can oxidize membrane proteins, lipids, -SH groups and methionine sulphur and produce harmful effects.

Q.120 What is the function of presence of D-amino acid oxidase in cells, when it is known there is absence of D-amino acids in cells?

There is no D-amino acids in the body tissues, hence it is not clear about the presence of D-amino acid oxidase in cells and its functions. Probably the dietary deamino acids may be oxidized by D-amino acid oxidase and be isomerized through their ketoacids to L-amino acids.

Q.121 Does oxidative deamination fulfill a major role in formation of NH_3 ?

- L-amino acid oxidase does not fulfill a major role in mammalian amino acid catabolism and formation of NH_3 as:

- It is restricted only to liver and kidney.
- Activity of the enzyme in these tissues is low.
- It does not have any effect on glycine or the L-isomers of the dicarboxylic acid or β-OH-α-amino acids.
- It does contribute to formation of NH_3 to some extent.

Q.122 What is non-oxidative deamination? Give two examples.

- Certain amino acids like OH-amino acids, histidine and S-containing amino acids can be deaminated non-oxidatively by specific enzymes to form NH_3 . They also do not fulfill major role in NH_3 formation.
- Two examples (Fig. 25.1).

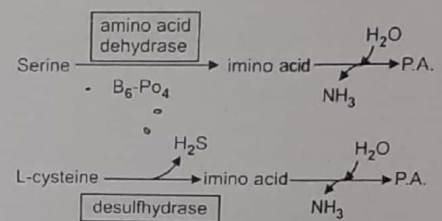


Fig. 25.1: Examples of nonoxidative deamination

Q.123 State the principal method by which NH_3 is formed from L-amino acids.

- **By transdeamination:** It is combination of transamination + Deamination.
- The NH_2 group of most L-amino acids are transferred to α-oxoglutarate by specific transaminases by the process of transamination forming L-glutamate as end-product. L-glutamate is oxidatively deaminated by a specific Zn-containing enzyme *glutamate dehydrogenase*, of high activity and wide distribution, which requires NAD^+ or NADP^+ as coenzyme, to form NH_3 .
- As the process involves first transamination and coupled with oxidative deamination, it is called as **transdeamination**.

Q.124 List the sources of NH_3 in the body.

- Transdeamination—the major pathway.
- Oxidative deamination by L-amino acid oxidase
- Non-oxidative deamination
- Absorption of NH_3 from gut produced by intestinal bacterial flora:
 - From dietary proteins
 - From urea present in fluids secreted into the gut

- From uric acid secreted in bile (This assumes importance in intestinal obstruction and portocaval shunt)
- Pyrimidine catabolism (small fraction).

Q.125 How albumin and globulin are separated?

- Albumin can be separated by full saturation of ammonium sulphate.
- Globulin can be separated by half saturation of ammonium sulphate.

Q.126 Name some biologically active peptides.

1. Glutathione.
2. Oxytocin.
3. Vasopressin.
4. Gramicidin-S.

Q.127 Which amino acid give rise to thyroxine?

Phenylalanine.

Q.128 What is the fate of ammonia?

1. Glutamine synthesis.
2. Synthesis of urea.
3. Reanimation process.

Q.129 Describe the urea cycle. What is another name for urea cycle?

This is a cycle through which the toxic NH_3 in the body is detoxicated to form non-toxic urea in the liver. An alternative name for urea cycle is Krebs' Henseleit cycle. Urea cycle is shown in Figure 25.2. In this process, one molecule of NH_3 and the molecule of CO_2 are converted into urea through five sequential enzymatic reactions in each turn of cycle. Ornithine is regenerated in the last reaction which acts as a catalytic agent and repeats the cycle again.

Q.130 What are non-protein nitrogenous substances?

Nitrogen containing substances other than amino acids or proteins are called non-protein nitrogenous substances. They are urea, uric acid, creatine, purine, pyrimidine, amino acids, ammonia, etc.

Q.131 Which organ is associated with urea formation?

Liver is the only organ which can form urea and all the enzymes involved have been isolated from the liver tissue.

Q.132 Can other organs like kidney and brain synthesize urea?

Kidney: Can form upto arginine but cannot form urea as enzyme "arginase" is absent in kidney tissues.

- **Brain:** Can synthesize urea if citrulline is available but lacks the enzyme *ornithine transcarbamoylase* which forms citrulline from ornithine.

Q.133 State the five steps of urea cycle indicating the enzymes involved and location of the enzyme.

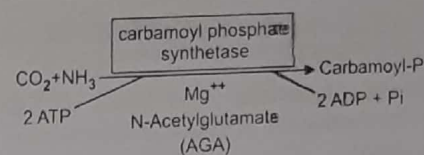
	Enzyme	Location
• Reaction 1 : Synthesis of carbamoyl-P	Carbamoyl synthetase I	Mitochondria
• Reaction 2 : Synthesis of citrulline	Ornithine transcarbamoylase	Mitochondria
• Reaction 3 : Synthesis of arginino-succinate	Arginino-succinate synthetase	Cytosol
• Reaction 4 : Cleavage of arginino-succinate	Arginino-succinase	Cytosol
• Reaction 5 : Cleavage of arginine to form urea and ornithine	Arginase	Cytosol

Q.134 What is the role of N-acetyl glutamate (AGA) in the first reaction in formation of carbamoyl-P?

N-acetyl glutamate (AGA) acts as an allosteric activator of the enzyme *carbamoyl synthetase I*, it brings about conformational changes in the enzyme so that the second molecule of ATP can bind.

Q.135 Show schematically the first reaction, synthesis of carbamoyl-P. What are the starting materials? Is biotin necessary?

- Starting materials: CO_2 and NH_3
- Biotin is not required.



Q.136 How many amino acids take part in the urea cycle?

5 amino acids participate in urea cycle. They are:

- Ornithine
- Citrulline
- Glutamic acid
- Aspartic acid
- Arginine.

Q.137 What is the energy requirement in formation of one molecule of urea in urea cycle?

3 molecules of ATP are utilized for the formation of one molecule of urea. As one ATP is converted to AMP, it is equivalent to utilizing 4 high energy bonds.

Q.138 Urea contains two nitrogen (NH_2 groups), what are the sources of the two nitrogens?

- One nitrogen of NH_2 group is derived from the NH_4^+ ion (Reaction 1)
- Other nitrogen of NH_2 group is provided by aspartic acid (Reaction 3).

Q.139 What is the normal blood level of urea in the body?

- Normal level ranges from 20 to 40 mg%.
- Indians take less proteins in diet, hence in Indians normal level is taken as 15 to 40 mg%.

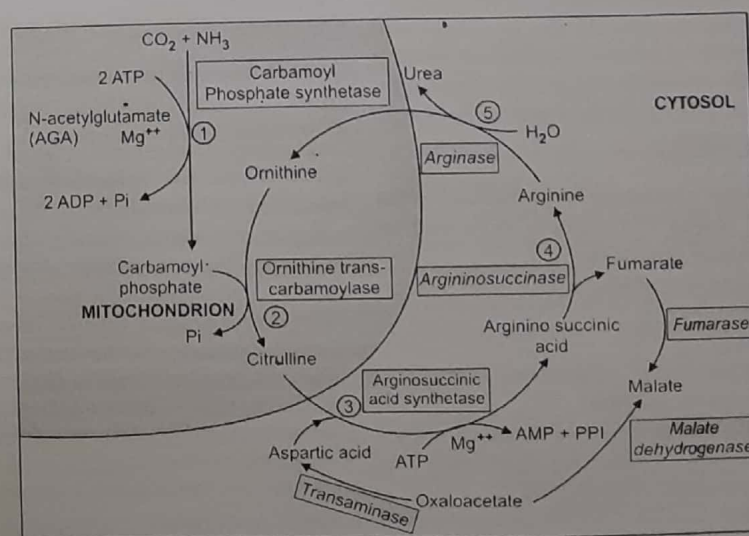


Fig. 25.2: Biosynthesis of urea or ornithine (urea cycle)

Q.140 In what conditions the blood level of urea increase:

Causes may be:

- Prerenal
- Renal
- Postrenal.

Q.141 Mention some prerenal causes.

Conditions in which plasma volume/body fluid are reduced, e.g.:

- Salt and water depletion
- Severe and protracted vomiting as in pyloric/intestinal obstruction,
- Severe and prolonged diarrhea,
- Hemorrhage and shock,
- In burns.

Q.142 State some renal causes:

- In acute glomerulonephritis.
- In type II nephritis (nephrotic syndrome)—in later stages.
- Chronic pyelonephritis.
- Malignant nephrosclerosis.
- Mercurial poisoning.

Q.143 Mention some post-renal causes.

- Enlargement of prostate—benign and malignant
- Stone in urinary tract (urinary lithiasis)
- Stricture of urethra
- Tumors of the bladder affecting urinary flow.

Q.144 What is glutamine?

- Glutamine is chemically δ -amide of α -amino glutaric acid. Glutamine formation helps in removal of NH_3 which is toxic (detoxification of NH_3).
- Glutamine serves as an important reservoir of nitrogen in tissues which is readily available and can be drawn upon for various synthetic processes, when the body needs.

Q.145 How glutamine is synthesized in the body? Name the tissues involved in synthesis of glutamine.

Glutamine is synthesized in tissues like:

- Liver
- Kidney
- Brain and retina. It is formed from:
 - Glutamic acid and
 - NH_3 by the action of a mitochondrial enzyme "Glutamine synthetase" in presence of ATP and Mg^{++} .

- The reaction is "irreversible", so that glutamine cannot reform glutamic acid and NH_3 by the same enzyme.

Q.146 How NH_3 can be obtained from glutamine?

- Glutamine is reservoir of NH_3 and readily available source of NH_3 .
- Glutamine is hydrolyzed readily by a specific enzyme glutaminase. The reaction is 'irreversible' and can occur in various tissues like liver, kidney, brain and retina.

Q.147 State some important functions of glutamine in the body.

- **Transamidation:** The amide N of glutamine can be transferred to keto-group of fructose-6-P to form glucosamine-P.
- Formation of guanylate (GMP).
- **Role of glutamine in kidney:** in conservation of Na^+ NH_4^+ - Na^+ exchange
- **Role of glutamine in brain:** Brain tissues cannot detoxicate NH_3 to form urea. Excess NH_3 entering brain, e.g. in hepatic failure is detoxicated by glutamine formation.
- Role of glutamine in conjugation reaction (Detoxication).
- Role in cancer: Certain tumors exhibit abnormally high requirement of glutamine/asparagine for their growth.

Q.148 What are the clinical manifestations of ammonia intoxication?

The symptoms of NH_3 intoxication include:

- A peculiar flapping tremor
- Slurring of speech
- Blurring of vision
- In severe cases lead to coma and death.

Q.149 How much urea is excreted daily in urine?

20 to 30 gm in 24 hours urine.

Q.150 Name the inherited disorders associated with urea cycle.

- Hyperammonemia type I—due to deficiency of enzyme carbamyl-P-synthetase I
- Hyperammonemia type II, (also called ornithinaemia)—due to the deficiency of enzyme ornithine transcarbamoylase
- Citrullinemia—due to deficiency of the enzyme argininosuccinate synthetase
- Argininosuccinic aciduria—due to deficiency of enzyme argininosuccinase

- Hyperargininemia—due to deficiency of enzyme arginase.

Q.151 What are the amino acids involved in creatine synthesis?

Glycine, arginine, methionine (as S-Adenosyl methionine).

Q.152 What is the role of creatine phosphate?

As a source of energy in the contraction of muscles.

Q.153 What are the uses of glycine?

Glycine is involved in the synthesis of

1. Hemoglobin.
2. Creatine phosphate.
3. Purines.
4. Glutathione.
5. Proteins.
6. Phospholipids through ethanolamine, choline and serine.
7. In detoxification reactions.
8. Bile acids.

Q.154 Which is the major non-protein nitrogenous constituent of the urine?

Urea.

Q.155 Where carbamoyl phosphate synthase II is required?

In pyrimidine de novo biosynthesis.

Q.156 What is the role of N-acetyl glutamic acid in the first step of urea cycle?

N-acetyl glutamic acid is the modifier of the enzyme carbamoyl phosphate synthetase. It keeps the enzyme in the correct information.

Q.157 How much protein is potentially glucogenic?

It has been shown by experimental studies that 60 gm (approximately 58 gm) of glucose are formed and excreted in urine for 100 gm of proteins metabolized. Thus 60% of proteins (amino acids) is potentially glucogenic.

Q.158 Name at least ten amino acids which are glucogenic.

Glycine, alanine, serine, cysteine/cystine, threonine, methionine, proline, valine, arginine, glutamate.

Q.159 Name the only amino acid which is ketogenic.

L-leucine.

Q.160 Name at least three amino acids which are both glucogenic and ketogenic. Phenylalanine/tyrosine, tryptophan, isoleucine.

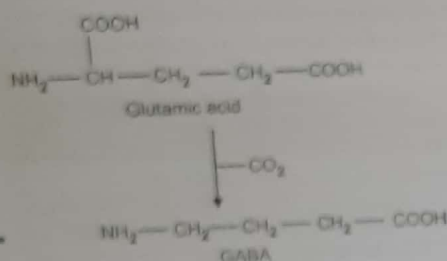
Q.161 1. What is GABA?

2. How it is formed?

3. What is the coenzyme of this reaction?

1. GABA is γ -aminobutyric acid, important in brain metabolism.

2. It is formed by the decarboxylation of glutamic acid.



3. Coenzyme of the reaction is pyridoxal phosphate.

Q.162 Define albinism.

Deficiency of enzyme tyrosinase in the tyrosine metabolism leads to an inborn error of metabolism called albinism.

Q.163 What is tyrosinosis?

Deficiency of enzyme p-hydroxyphenyl pyruvate hydroxylase in tyrosine metabolism leads to tyrosinosis.

Q.164 What is alkaptonuria?

Deficiency of enzyme homogentisic acid oxidase in tyrosine metabolism leads to alkaptonuria.

Q.165 What is phenylketonuria?

Deficiency of the enzyme phenylalanine-4 hydroxylase in tyrosine metabolism leads to phenylketonuria.

Q.166 Name some biological important amines.

1. Histamine
2. γ -aminobutyric acid (GABA).
3. Epinephrine (adrenaline).
4. Norepinephrine (noradrenaline).

Q.167 What are catecholamines?

Catecholamines are

1. Adrenaline
2. Noradrenaline
3. Dopamine

Q.168 Name at least five biogenic amines of clinical importance. State the amino acid from which it is derived and mention biological importance in the body.

from which it is derived and mention biological importance in the body.

Biogenic amine	Name of the amino acid	Biologic importance
• Histamine	Histidine	• Vasodilator, BP↓ • HCl↑ • Pepsin- • Liberated in anaphylactic reaction
• γ -amino butyric acid (GABA)	Glutamic acid	• Presynaptic inhibitor in brain • Forms a by-pass in TCA cycle (GABA shunt)
• Taurine	Cysteic acid (derived from cysteine)	• Constituent of bile acid (taurocholic acid)
• Tryptamine	Tryptophan	• Tissue hormone, a derivative of 5-OH tryptamine (serotonin) • Vasoconstriction, BP↑ • Forms choline
• Ethanolamine	Serine	• Constituent of PL like cephalin

Q.169 What is GABA shunt? Show schematically.

GABA by its conversion to succinic acid can form a "by-pass" in TCA cycle and this is called as GABA shunt (Fig. 25.3).

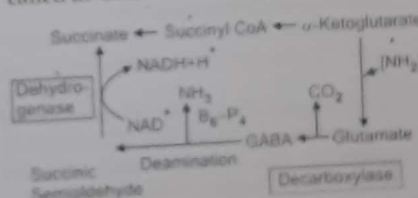


Fig. 25.3: GABA shunt

Q.170 Name the polyamines.

- Spermidine
- Spermine and
- Their precursor putrescine.

Q.171 What is carbamoyl-P synthetase II? What is its function?

Carbamoyl-P-synthetase II is an enzyme present in cytosol. It can form carbamoyl-P from CO_2 and amide-N of glutamine, which is used in pyrimidine synthesis.

Q.172 What are the aromatic amino acids? Name them, which is essential and which is not.

- Aromatic amino acids are:
 - Phenylalanine and
 - Tyrosine
- Phenylalanine is essential and must be provided in the diet

• Tyrosine is not essential as it can be formed from phenylalanine in the liver.

Q.173 State how phenylalanine is converted to tyrosine in the body.

Reactions occur in two stages

• **Stage 1:** Reduction of molecular O_2 to H_2O and conversion of phenylalanine $\rightarrow$ to tyrosine. Reduced form of folic acid F.H_2 acts as H-donor to molecular O_2 and is converted to F.H_2 . In this stage, there is incorporation of one atom of molecular O_2 to p-position of phenylalanine to form tyrosine, while other atom of O_2 is reduced to H_2O .

• **Stage 2:** Reduction of F.H_2 to reform F.H_4 takes place by NADPH, acting as a H-donor, catalyzed by the enzyme *dihydrobiopterin reductase*.

Note: Both the reactions are irreversible.

Q.174 What is meant by sparing action?

The feeding of tyrosine/or cysteine decreases the need of phenylalanine/or methionine in the diet respectively. This is called "sparing" action. Phenylalanine/or methionine is spared from synthesis of tyrosine/or cysteine respectively.

Q.175 Tyrosine is a non-essential amino acid. In which condition it becomes an essential amino acid?

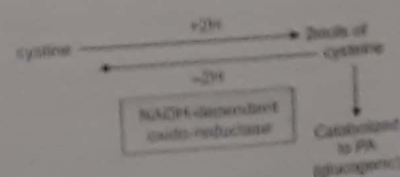
In phenylketonurias, where phenylalanine cannot be converted to tyrosine in the body due to inherited absence of the enzyme *phenylalanine hydroxylase*, tyrosine becomes an essential amino acid to the patient.

Q.176 What is albinism?

It is an inherited disorder of tyrosine metabolism in which the enzyme *tyrosinase* is lacking resulting in non-formation of melanin pigment for which the hair, skin and retina of eye are not black.

Q.177 What is the relationship between cysteine and cystine?

- Two molecules of L-cysteine oxidized to give one molecule of cystine which is joined by S-S-bond.
- One molecule of cystine is reduced to form two molecules of cysteine.



Q.178 What is active methionine? What is it known as chemically?

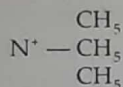
- "Active methionine" is an active compound in which CH_3 group is *biologically labile* and it acts as a methyl (CH_3) group donor to a suitable "acceptor" so that it can form biologically important compounds.
- Chemically "active methionine" is *S-adenosyl methionine*.

Q.179 What is meant by transmethylation reaction?

Certain compounds of the body, with structures containing CH_3 group attached to an atom other than carbon can take part in enzymic reactions, whereby these CH_3 groups are transferred to a suitable "acceptor" which have no methyl group. Such reactions are called as *transmethylation reaction*.

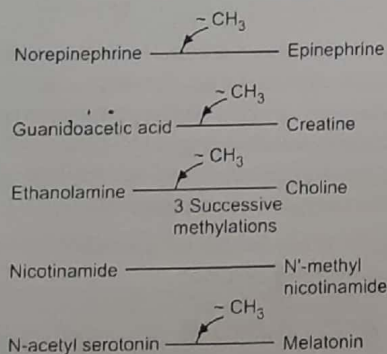
Q.180 Name the compounds which possess biologically labile methyl group.

- Active methionine (S-adenosyl methionine): containing $-\text{S}-\text{CH}_3$ group.
- Choline: containing $-\text{CH}_3$ groups attached to N as follows:



- Betaine: an oxidation product of choline.

Q.181 Give Five examples of transmethylation reactions, where active methionine acts as methyl donor.



Q.182 Enumerate the metabolic role or biomedical importance of L-methionine.

- Methionine is glucogenic.
- Methionine is converted to L-cysteine in the body.
- *Lipotropic function*: provides choline, a lipotropic agent, by methylations of ethanolamine.
- Transmethylation reactions: by forming active methionine.

- In polyamine synthesis.
- Formation of methyl mercaptan in liver diseases, which accounts for foul odor in breath (fetor hepaticus).

Q.183 Show schematically how L-cysteine is formed from L-methionine.

See Figure 25.4.

Q.184 What is homocystinuria? What is the enzyme deficiency?

- An inherited disorder of metabolism of L-methionine or more specifically its metabolic intermediates homocysteine/ or homocystine.
- *Enzyme deficiency*: Cystathionine synthetase enzyme deficiency leads to accumulation of homocystine which is excreted in urine.

Q.185 State five metabolic role/or biomedical importance of glutathione in the body.

- Reduced glutathione (G-SH) is necessary for maintaining integrity of red cell membranes and lens protein.
- It helps to destroy H_2O_2 and other peroxides in cells. The reaction is catalyzed by selenium containing enzyme *glutathione peroxidase*.
- Acts as coenzyme with liver enzyme *glutathione-insulin transhydrogenase* which helps in catabolism of insulin.
- Glutathione takes part in "V-glutamyl cycle" for absorption of amino acids from gut.
- Reduced glutathione (G-SH) is required as coenzyme/cofactor for *PG synthetase system* in PG synthesis and also for formation of "active methionine".

Q.186 What is serotonin?

Serotonin is chemically 5-OH tryptamine (5-HT), and it is formed from amino acid tryptophan. It is present in blood and is produced in tissues like gastric mucosa, intestine, brain, mast cells and platelets. (Probably stored in Platelets?)

Q.187 How serotonin is synthesized by the serotonin-producing cells in the body?

- Tryptophan is first hydroxylated to form 5-OH tryptophan in liver by *hydroxylase* enzyme in presence of O_2 and NADPH.
- 5-OH tryptophan is next decarboxylated by the enzyme *decarboxylase* to form 5-OH tryptamine (5-HT), called serotonin, the reaction requires B_6P as coenzyme.

Q.188 Name the enzyme which degrades serotonin. What is the degradation product?

- Enzyme is *monoamine oxidase* (MAO)
- Degradation product is 5-OH-Indole acetic acid (5-HIAA) which is excreted in urine.

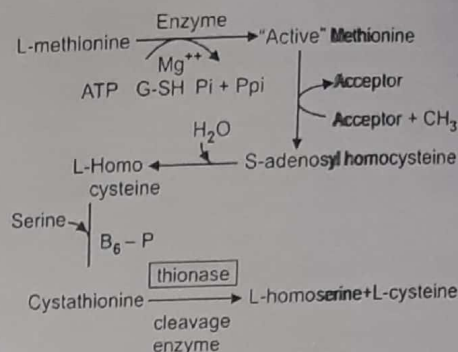


Fig. 25.4: Formation of L-cysteine

Q.189 In which condition excessive amount of serotonin is produced in the body?

- Excessive amount of Serotonin is produced in *carcinoids*, a malignant tumor of serotonin-producing cells. It is also called as *argentaffinoma*.
- The clinical features associated with it is called as *carcinoid syndrome*.
- In carcinoids, 60% of tryptophan is utilized for serotonin formation as compared to 1% in normals.

Q.190 What is melatonin?

- Chemically melatonin is "N-acetyl-5-methoxy serotonin". It is a hormone produced by the pineal gland and peripheral nerves of man and some other higher animals.
- The hormone lightens the color of melanocytes in the skin of frog and blocks the action of MSH (Melanocyte stimulating hormone) and ACTH.

Q.191 What is Hartnup's disease?

- It is an inherited disorder associated with the metabolism of tryptophan. The clinical symptoms are mental retardation, intermittent cerebellar ataxia and skin rash.
- Urine of patients with Hartnup's disease contains increased amounts of indoleacetic acid and tryptophan.

Q.192 What is the metabolic defect in Hartnup's disease?

- An inherited deficiency of the enzyme *tryptophan dioxygenase* in liver.
- Could also be due to a defective indole transport and consequent impairment of tryptophan metabolism.

Q.193 What is the relationship between creatine-P and creatinine?

- Both are nonprotein nitrogenous compounds.
- Creatinine-P is chemically methyl guanidinoacetic acid.
- Creatinine is *anhydride of creatine*, the reaction is irreversible and non-enzymatic. Creatinine has a ring structure.

Q 194 Show schematically the biosynthesis of creatine.
See Figure 25.5.

Q 195 Where does the formation of creatine occur?
In Liver and in Kidney.

Q 196 What is meant by creatinine clearance test.

Endogenous creatinine clearance is used as a renal function test. At normal levels of creatinine in the blood, this metabolite is filtered at the glomerulus but neither

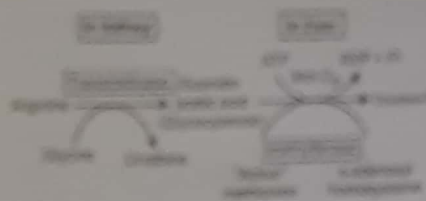


Fig. 25.5 Synthesis of creatine

secreted nor reabsorbed by the tubules. Hence its clearance measures the glomerular filtration rate (GFR).

Q 197 What is the normal endogenous clearance?

Normal values for creatinine clearance varies from 95 to 125 ml/min.

Q 198 What are the biological importance and metabolic role of arginine?

- Semiessential amino acid required in growing children, in pregnancy and lactation
- Required for creatine synthesis
- Required in urea formation (as intermediate metabolite)
- Tissue protein formation
- Glucogenic amino acid

DO YOU KNOW?

- When excess nitrogen is converted to urea in the liver and goes through the blood to the kidney, where it is eliminated in urine.
- Methotrexate inhibits DHF reductase (dihydrofolate reductase) making it a very useful antineoplastic drug.

Q.194 Show schematically the biosynthesis of creatine.

See Figure 25.5.

Q.195 Where does the formation of creatinine occur?

In Liver and in Kidney.

Q.196 What is meant by creatinine clearance test.

Endogenous creatinine clearance is used as a renal function test. At normal levels of creatinine in the blood, this metabolite is filtered at the glomerulus but neither

secreted nor reabsorbed by the tubules. Hence its clearance measures the glomerular filtration rate (GFR).

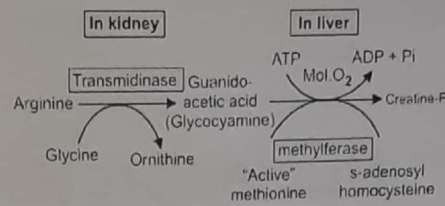


Fig. 25.5: Synthesis of creatinine

Q.197 What is the normal endogenous clearance?

Normal values for creatinine clearance varies from 95 to 105 ml/min.

Q.198 What are the biological importance and metabolic role of arginine?

- Semiessential amino acid required in growing children, in pregnancy and lactation
- Required for creatine synthesis
- Required in urea formation (an intermediate metabolite)
- Tissue protein formation
- Glucogenic amino acid.

DO YOU KNOW?

- Most excess nitrogen is converted to urea in the liver and goes through the blood to the kidney, where it is eliminated in urine.
- Methotrexate inhibits DHF reductase (Dihydrofolate reductase) making it a very useful antineoplastic drug.

Nucleoproteins

Q.1 What is a nucleoside?

Sugar-base combination is called nucleoside.

Q.2 What is a nucleotide?

Sugar-base-phosphate combination is called nucleotide.

Q.3 What are nucleic acids?

Nucleic acid are polynucleotides.

Q.4 What is the difference between adenine, adenosine and adenylic acid?

- Adenine is a purine base.
- Adenosine is a nucleoside. It is a adenine-ribose combination. Adenylic acid is a nucleotide. It is a adenine-ribose-phosphoric acid.

Q.5. What are the types of nucleic acid?

Two types of nucleic acid are:

1. Deoxyribose acid (DNA).
2. Ribose nucleic acid (RNA).

Q.6 What are the bases present in DNA?

Purine bases: Adenine, guanine.
Pyrimidine bases: Cytosine, thymine.

Q.7. What are the bases present in RNA?

Purine bases: Adenine, guanine.
Pyrimidine bases : Cytosine, uracil.

Q.8 How are the bases united in nucleic acid?

Bases are united through hydrogen bonding.

Q.9. What is the number of hydrogen bonds present between?

- i. Adenine—Thymine
Adenine—Thymine.....Two hydrogen bonds
- ii. Cytosine—Guanine
Guanine—Cytosine.....Three hydrogen bonds.

Q.10 What is the difference between RNA and DNA?

DNA	RNA
1 Bases present are adenine, guanine, cytosine, thymine	1 Bases present are adenine, guanine, cytosine, uracil
2 Sugar present is deoxyribose.	2 Sugar present is ribose.
3. It is double strand molecule.	3. It is a single strand molecule.

Q.11 What are the various types of RNA?

Three types of RNA are:

1. Transfer RNA.
2. Messenger RNA.
3. Ribosomal RNA.

Q.12 What is the function of transfer RNA?

Transfer RNA carries a specific amino acid to the site of protein synthesis on the m-RNA template.

Q.13 What are functions of mRNA?

It carry message from DNA to cytoplasm and is having codons on it in sequential manner.

Q.14 What are functions of rRNA?

It consists of 80% of total RNA and helps in protein synthesis by finding to ribosomes.

Q.15. What is replication?

Copying of DNA to form identical daughter molecules is called the replication.

Q.16 What is transcription?

Transcription is the process by which the genetic message in DNA is transcribed into the form of messenger RNA to be carried to the ribosomes.

Q.17. What is translation?

Translation is the process by which the genetic message is decoded on the ribosomes.

Q.18 What is genetic code?

Genetic code refers to the language used by the genetic material, i.e. the DNA to transfer genetic information to the site of new product formation. Genetic code is universal.

Q.19 What is a triplet or codon?

The sequence of three non-overlapping nucleotides is called a triplet.

Q.20 What is the chain initiating codon in protein synthesis?

Chains initiating codon is AUG.

Q.21 Chain initiating codon AUG codes for which amino acid?

AUG codes for:

- i. N-formylmethionine in prokaryotes.
- ii. Methionine eukaryotes.

Q.22 Which is the other initiator of protein synthesis in addition to AUG?

In some lower animals GUG serves the function of initiation.

Q.23 What are the chain terminating codons?

UAA, UAG, UGA.

Q.24 What are the sources of C and N of purine skeleton?

- N₁-aspartic acid
- C₂, C₈-formate.
- C₄, C₅, N₇-glycine.
- C₆-Respiratory CO₂.

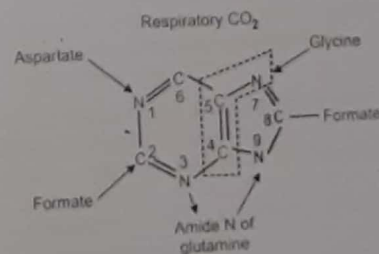


Fig. 26.1: Purine skeleton

Q.25. What is the fate of purine bases?
Purine base are converted to uric acid.

Q.26 What is the normal uric acid level?
2-6 mg%.

Q.27. What is a gout?
Increased uric acid level in the blood given rise to a condition known as gout.

Q.28 How are pyrimidines catabolised?
Pyrimidines are catabolised to β -alanine and amino-isobutyric acid.

Q.29. What are the unusual bases present in nucleic acid?

1. 5-methyl cytosine.
- N⁶-methyl adenine.
- N⁶, N⁶-dimethyl adenine
4. N⁶, N⁷-dimethyl guanine
5. 5-hydroxy methyl cytosine.

Q.30 Enumerate some plant derived methylated purines.

- Caffeine (1,3,7-trimethyl xanthine)
- Theophylline (1,3-dimethyl xanthine)
- Theobromine (3,7, dimethyl xanthine).

Q.31 Enumerate some adenosine derivatives.

- ATP, cAMP, adenosine-3-phosphate-5-phosphosulfate, S-adenosyl, methionine.

Q.32 Name some synthetic nucleotide analogs.

- 5-fluorouracil
- 5-iododeoxyuridine
- 6-thioguanine
- 6-mercaptopurine
- 6-azauridine
- 8-azaguanine
- Allopurinol
- Azathioprine.

Q.33 What is the main difference between de novo and salvage pathways of purine production?

Salvage reactions require for less energy than *de novo* synthesis of purines.

Q.34 What is mode of action of methotrexate?

Methotrexate, an anticancer drug, inhibits the enzyme dihydrofolate reductase.

Q.35 Name some medicinally important nucleotides with uses?

1. Allopurinol G or gout and hyperuricemia.
2. Azathioprine-immunosuppressive in organ transplantation.
3. 5-iododeoxyuridine for herpetic keratitis.

Q.36 Explain Lesch Nyhan syndrome?

An overproduction of uric acid with arthritis and bizzare syndrome of self mutilation is due to defect in HGPRTase, an enzyme of purine salvage, this causes rise in intracellular PRPP, results in purine over-production.

Q.37 Who demonstrated the DNA transformation experiment?

DNA-transformation experiment was conducted by Griffith. But the demonstration that DNA contains genetic information was first made by Avery Macleod Mc Carty.

Q.38 What is hyperchromatiaty of denaturation?

By denaturation of DNA strands there is increase in optical absorbance of purine and pyrimidine bases.

Q.39 What is the cap of mRNA?

Cap is present on 5' end of mRNA and consists of 7-methyl guanosine triphosphate.

Q.40 Which Histones are present in nucleosome?

Histone is octameric molecule consisting of H₁, H₂A, H₂B, H₃, H₄.

Q.41 Which diseases are related with defective DNA repair?

1. Xeroderma pigmentosum.
2. Ataxia telangiectasia.
3. Fanconi anemia.
4. Bloom's disease.

Q.42 Name some antibiotics that inhibits protein synthesis.

1. Puromycine.
2. Cycloheximide.
3. Diphtheria toxin.

Q.43 Who proposed operon model?
Jacob and Monod in year 1961.

Q.44 What are restriction endonuclease?
Restriction endonuclease are enzymes that cut the DNA segment in specific sequence,

e.g. Eco RI cut the DNA strand at $\frac{\text{GAATTC}}{\text{CTTAAG}}$

Q.45 Define intron.

Intron is a sequence of gene that is transcribed out excised before translation.

Q.46 What is southern blotting?

A method for transferring DNA from an agarose gel to nitrocellulose filter, on which DNA can be detected by a suitable probe (e.g. cDNA).

Q.47 Explain polymerase chain reaction (PCR).

An enzymatic method for the repeated copying of two strands of DNA that make up a particular gene sequence.

Q.48 Define probe.

A molecule used to detect the presence of a specific fragment of DNA/RNA.

Enzymes

Q.1 What are enzymes?

Enzymes are biological catalysts.

Q.2 What is the nature of enzymes?

Enzymes are protein in nature.

Q.3 Name the fastest acting enzyme?

Carbonic anhydrase (CA).

Q.4. What is Km (Michaelis constant)?

Km is defined as that substrate concentration which produces half of the maximum velocity.

Q.5 Km is defined in terms of what?

It is defined in terms of substrate concentration.

Q.6. Which Km value is preferred, low or high?

Low Km value is preferred because at low substrate we can achieve the maximum velocity.

Q.7 Give Michaelis Menten equation?

$$V_1 = \frac{V_{max} [s]}{K_m + [s]}$$

Q.8 Some enzymes don't follow Michaelis Menten equation then they follow which reaction?

Hill equation.

Q.9 Give example which follows Hill equation?

Hemoglobin.

Q.10 Why hemoglobin follows Hill equation not Michaelis Menten equation?

Due to property of heme-heme interaction, hemoglobin does not follow Michaelis Menten equation as no straight line is followed.

Q.11 Why Lineweaver-Burk curve or equation is preferred?

Lineweaver-Burk equation is in the form of a straight line. For a straight line, we need two points, i.e. by selecting few different concentrations, we can plot the curve and hence find out the Km.

Q.12 What are the factors which influence enzyme activity?

Factors which influence enzyme activity are:

1. Substrate concentration.
2. Enzyme concentration.
3. pH.
4. Temperature.
5. Effect of activators and coenzymes.

Q.13 What are competitive inhibitors?

Competitive inhibitors are those which has structural similarity with the substrate molecules for the active site of the enzyme.

Q.14 What is the characteristics of line weaver Berk plot in competitive inhibition?

In competitive inhibition Vmax remains the same but Km value increases.

Q.15 What is the characteristics of lineweaver-Burk plots in non-competitive inhibition?

Reverse of above is followed in non-competitive inhibition.

Q.16. What are non-competitive inhibitors?

Non-competitive inhibitors are those which are attached not to the active site but to some other site of the enzyme.

Q.17 Give the example of competitive inhibitors.

Inhibition of succinate dehydrogenase by malonate. L-lactate dehydrogenase (substrate lactate) by oxamate. Cis-aconitase by fluorocitrate.

Q.18 Give the example of non-competitive inhibition.

1. Inhibitions of acetyl choline esterase, trypsin, chymotrypsin by diisopropyl-fluorophosphate.
2. Inhibition by heavy metal ions like Ag⁺ and Hg²⁺ ions.

Q.19. What is the difference between prosthetic group and co-enzyme?

- Prosthetic group is tightly bound to the enzyme.
- Whereas co-enzymes exist in the solution in free state and contact the enzyme only at the site of reaction.

Q.20 What is an allosteric site?

Site other than the active site is called the allosteric site.

Q.21 What are isoenzymes? Give two examples.

- Isoenzymes are the multiple forms of the same enzymes
- Lactate dehydrogenase,
- Alkaline phosphatase.

Q.22 Name the isoenzymes of LDH?

- LDH has 5 iso enzymes
- HHHH, HHHM, HHMM, HMMM, MMMM.

Q.23 Name the diagnostic enzymes in MI?
SGOT (AST), LDH.

Q.24. Name the diagnostic enzyme in viral hepatitis?

SGPT (ALT).

Q.25 Which enzyme increases in serum in acute pancreatitis?
 α Amylase.

Q.26. In Wilson disease, which enzyme plays diagnostic role?
Ceruloplasmin.

Q.27 Name the diagnostic enzyme for muscle disorders.
Creatine kinase.

Q.28 Acid phosphatase is diagnostic enzyme for which disease?
Carcinoma of prostate.

Q.29 Enzyme that is diagnostic tool for bone and liver disease.
Alkaline phosphatase.

Q.30 What are anti-enzymes?

Anti-enzymes are the substances produced as a result of repeated injection of certain enzymes in the serum, which prevents the normal action of the enzyme injected.

Q.31 How will you differentiate whether the given reaction is enzyme catalysed or not?

We can differentiate the reaction by two test.

1. Heat sensitive test.
2. Acid test.

Q.32 What are constitutive enzymes?

Constitutive enzymes are those enzymes which are present in constant concentration during the life of the cell.

Q.33 What are inductive enzymes?

Inductive enzymes are those enzymes whose amount present in the cell is variable and depending upon the requirement.

Q.34. What is feedback inhibition?

In multienzyme reaction, the end-product of the reaction sequence may act as a specific

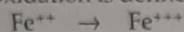
inhibitor of an enzyme at or near the beginning of the sequence with the result the rate of entire sequence of reactions is determined by the steady state concentration of the end-product. This type of inhibition is called feedback inhibition.

Q.35 Give an example of feedback inhibition.

Cholesterol inhibits cholesterol synthesis in the liver by inhibition of HMG-CoA reductase.

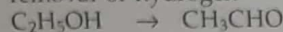
Q.1 What is oxidation?

Oxidation is defined as loss of electrons



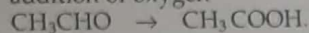
or

removal of hydrogen



or

addition of oxygen

**Q.2 What do you mean by biological oxidation?**

The stepwise degradation of metabolite for the liberation of energy carried out in the system.

Q.3 Give Gibbs free energy equation.

$$\Delta G = \Delta H - T\Delta S$$

ΔH = enthalpy of reaction

T = temp (K)

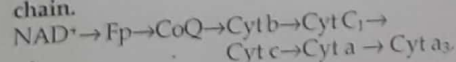
ΔS = entropy change.

Q.4 What inference is drawn from this reaction?

If ΔG is +ve then the reaction is endothermic while -ve implies, reaction to be exogenic.

Q.5 Who give the concept of high energy phosphate bond?

Lippmann introduced the concept of high energy phosphate bond.

Q.6 Name the components of respiratory chain.**Q.7 In which order the components of respiratory chain are arranged?**

The components of respiratory chain are arranged in order of increasing redox potential. NAD^+ has the lowest redox potential where as Cyt a_3 has the highest redox potential in the respiratory chain.

Q.8 Name the various ATP sites in the respiratory chain.

1. NAD^+ to FAD.
2. Cyt b to Cyt C_1 .
3. Cyt a to Cyt a_3 .

Q.9 What should be the minimum redox potential difference for the generation of ATP?

0.2 volts.

Q.10 What is the ATP yield in respiratory chain?

32 molecules of ATP (28 + 2 + 2).

Q.11 Which part of the NAD^+ accepts electrons?

Nicotinamide.

Q.12 Which part of the FAD accepts electrons?

Flavin.

Q.13 What is oxidative phosphorylation?

Oxidative phosphorylation is the process in which ATP is formed as electrons are transferred from NADH or FADH_2 to oxygen by a series of electron carries.

Q.14 Name the inhibitors of various energy sites in respiratory chain?

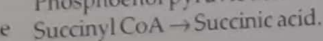
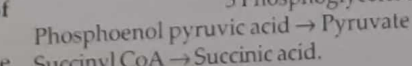
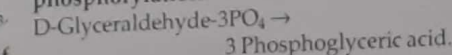
Inhibitors of site I = rotenone, amobarbital, piericidin.

Inhibitors of site II = Antimycin, BAL.

Inhibitors of site III = H_2S , CO_2 , CN^- .

Q.15 What is substrate level phosphorylation?

In substrate level phosphorylation, high energy phosphate bond formation takes place on the substrate without under going into the respiratory chain.

Q.16 Give the examples of substrate level phosphorylation.**Q.17 What is P/O ratio?**

P/O ratio is defined as the number of inorganic phosphate taken to phosphorylate ADP per atom of oxygen consumed.

$$\frac{P}{Q} = \frac{\text{Number of inorganic phosphate to phosphorylate ADP}}{\text{No. of atom of oxygen consumed}}$$

Q.18 Name few reactions where P/O ratio is 2.

1. Succinate $\longrightarrow$ Malate (TCA cycle)
2. Acyl CoA $\longrightarrow$ β unsat. Acyl CoA (β -oxidation of fatty acids).

Q.19 Name few high energy compounds.

Adenosine triphosphate, adenosine diphosphate, Ionosine triphosphate, etc.

Q.20 What are uncouplers?

Uncoupler are those substances which separates oxidation from phosphorylation, e.g. Dinitro phenol.

Q.21 Name the different classes of oxidoreductases.

1. Oxidases.
2. Dehydrogenases.
3. Hydroperoxidases.
4. Oxygenases.

Q.22 Give some examples of oxidases.

1. Cytochrome oxidase.
2. FMN, FAD.
3. Xanthine oxidase.
4. Glucose oxidase.
5. L-amino acid oxidase.

Q.23 Which enzyme is responsible for lowering of superoxide ion?

Superoxide dismutase (SOD).

Q.24 Superoxide dismutase (SOD) contains which ions?

- Superoxide dismutase enzymes is of two types:
 1. Up to plasmic SOD.
 2. Mitochondrial SOD.
- Up to plasmic SOD contains copper and uric ions where as mitochondrial SOD contains maganese.

Q.25 Who proposed the chemiosmotic theory?

Peter Mitchelle was the person to propose chemiosmotic theory.

Q.26 Name the inhibitor of adenine nucleotide transporter.

- Atractyioside is inhibitor of adenine nucleotide transporter.

Q.1 What are vitamins?

Vitamins are defined as organic substances present in natural occurring food and are required in small amounts for maintenance, growth and reproduction.

Q.2. What are the types of vitamins?

Vitamins are two types:

1. Fat-soluble vitamins.
2. Water-soluble vitamins.

Q.3 What are fat-soluble vitamins?

Fat-soluble vitamins are:
Vitamin A,D,E and K.

Q.4. What are water soluble vitamins?

Water-soluble vitamins are:

- i. Vitamin B complex.
- ii. Vitamin C.

Q.5 What are the members of vitamin B complex?

1. Thiamine (B₁)
2. Riboflavin (B₂)
3. Pyridoxine (B₆)
4. Cyanocobalamin (B₁₂)
5. Niacin (B₃)
6. Pantothenic acid (B₅)
7. Biotin (B₇)
8. Folic acid
9. Lipoic acid
10. Para-amino-benzoic acid (PABA)
11. Choline
12. Inositol.

Q.6 Name the vitamins which are the components of electron transport chain.

1. Niacinamide
2. Riboflavin
3. Ubiquinone.

Q.7 Name the vitamins which are partly synthesised in the intestinal flora.

1. Vitamin K
2. Thiamine
3. Riboflavin
4. Pyridoxine
5. Niacin.

Q.8 What are provitamins?

Provitamins are substances which as such does not possess vitamin activity but on conversion give rise to vitamins. Provitamins of vitamin A are, a, b, g, carotenes, etc.

Q.9 What are the sources of vitamin A?

Butter, milk, cod liver oil, halibut liver oil, egg yolk, carrots.

Q.10 What is provitamin form of vitamins A?

Carotenes.

Q.11 How many molecules of vitamins A are formed from?

- a. α -carotenes
 - b. β -Carotenes
 - c. γ -carotenes.
- a. 1 molecule
 - b. 2 molecule
 - c. 1 molecule.

Q.12. What are the functions of vitamin A?

1. Visual function.
2. Growth and reproduction.
3. Proper healing of epithelium.
4. Bones and teeth.
5. Necessary for synthesis of mucopolysaccharides.
6. It is also involved in nucleic acid metabolism.
7. It is also involved in the electron transport chain and in oxidative phosphorylation.

Q.13 What is the daily requirement of vitamin A?

5000 IU.

Q.14. What is the deficiency disease associated with vitamin A?

Night blindness.

Q.15 What are the active forms of vitamin D?

The active forms are:

1. 25-hydroxycholecalciferol (25 HCC).
2. 1,25 dihydroxycholecalciferol (1,25-DHCC).

Q.16 What are the organs which synthesise the active forms of vitamin D?

Formation of 25-HCC takes place in liver whereas formation of 1,25 DHCC takes place in kidney.

Q.17. What are the precursors of vitamin A?

1. Ergosterol: It gives rise to calciferol (vitamin D₂).
2. 7-dehydrocholesterol: It gives rise to ergocalciferol (vitamin D₃).

Q.18 What are functions of vitamin D?

1. In the absorption of calcium and phosphorus from the intestines.
2. Growth.
3. Mineralisation of bones.
4. Normal functioning of parathormone.

Q.19 What is the daily requirement of vitamin D?

400 IU.

Q.20 What are the deficiency symptoms of vitamin D?

- Rickets in children
- Osteomalacia in adults.

Q.21 What is hypervitaminosis?

Excessive intake of vitamin A or D gives rise to condition known as hypervitaminosis.

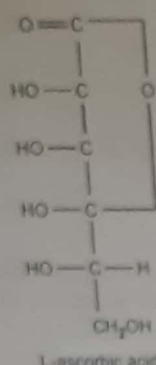
Q.22. What are the functions of vitamin E?

1. As an antioxidant.
 - a. Prevents the autoxidation of vitamin A and carotenes.
 - b. Prevents the formation of fatty acid peroxidases in tissues due to the auto-oxidation of unsaturated fatty acids with oxygen.
 - c. Protects the lipids of biological membranes against oxygen by acting as antioxidants.
2. Prevents rancidity.

Q.23 What are the functions of vitamin K?

1. In the synthesis of prothrombin.
2. In the oxidative process taking place in the photosynthesis in plant kingdoms.

Q.24 What is the structure of vitamin C?



Q.25 What are the functions of vitamin C?

1. In the synthesis of collagen.
2. In the synthesis of steroid hormones both in adrenal cortex and corpus luteum.
3. As cofactor in the following reactions:
 - a. In phenylalanine metabolism
p-hydroxy phenyl pyruvic acid $\rightarrow$ homogentisic acid.
 - b. Dopamine $\rightarrow$ Norepinephrine.
 - c. Folic acid $\rightarrow$ Folinic acid.
4. In the synthesis of carnitine in the liver.
5. Necessary for the absorption of iron by reducing ferric form to ferrous form.
6. In tissue respiration.

Q.26 What are the sources of vitamin C?
Citrus fruits such as lemon, orange, pineapple, etc. Indian gooseberry, green-pepper, cauliflower, tomatoes, spinach, potatoes.

Q.27 What is the daily requirement of vitamin C?
60 mg.

Q.28 What is the normal level of vitamin C in blood?
0.6-1.5 mg per 100 ml of blood.

Q.29 What is the deficiency disease of vitamin C?
Scurvy.

Q.30 Which animal can synthesise vitamin C?
Rat, rabbit, dog, and birds.

Q.31 What is biological active form of vitamin B₁?
Thiamine pyrophosphate (TPP).

Q.32 What are the functions of vitamin B₁?
Thiamine pyrophosphate participates as coenzymes
1. In oxidative decarboxylation of α -keto acids.

- a. Pyruvic acid $\rightarrow$ Acetyl CoA.
 - b. α -ketoglutaric acid $\rightarrow$ Succinyl CoA.
2. In transketolation reaction of hexose monophosphate shunt pathway.
D-xylulose 5-PO₄ + D-sedopheptulose 7-PO₄
 $\rightarrow$ D-ribose 5PO₄ + D-glyceraldehyde-3-PO₄

Q.33 What is the deficiency symptom of vitamin B₁?

Human: Beriberi

Rats: Bradycardia.

Q.34 What are the sources of vitamin B₁?
Germinating seeds, legumes, wheat, pork, eggs.

Q.35 What is the daily requirement of vitamin B₁?
1.4 gm.

Q.36 Why vitamin B₂ is also called lactoflavin?
Vitamin B₂ was first of all isolated from milk, hence it is called lactoflavin.

Q.37 What are the coenzymes forms of vitamins B₂?

1. Flavin mononucleotide (FMN).
2. Flavin adenine dinucleotide (FAD).

Q.38 What are the various active forms of vitamin B₆?

1. Pyridoxine
2. Pyridoxal
3. Pyridoxamine.

Q.39 What is the biologically active form of vitamin B₆?
Pyridoxal phosphate and pyridoxamine phosphate.

Q.40 What are the reactions mediated by vitamin B₆?

1. In transamination reaction
2. In decarboxylation reaction
 - i. Histidine $\rightarrow$ Histamine
 - ii. Tyrosine $\rightarrow$ Tyramine
 - iii. Glutamic acid $\rightarrow$ γ -amino butyric acid (GABA)
 - iv. α -amino- β -keto adipic β -keto $\rightarrow$ adipic acid $\rightarrow$ δ -amino levulinic acid.
3. In dehydrases reaction
 - i. Serine $\rightarrow$ Pyruvic acid
 - ii. Threonine $\rightarrow$ α -ketobutyric acid
4. In transsulphurase reaction
Homocysteine $\rightarrow$ Serine.
5. In desulphuration reaction
Cystine $\rightarrow$ pyruvic acid

Homocysteine $\rightarrow$ α -Keto butyric acid.

6. In tryptophan metabolism
Kynurenine $\rightarrow$ Anthranilic acid
7. In the transport of amino acids and in the absorption of amino acid.
8. In essential fatty acid synthesis.

Q.41 What are the functions of biotin?

Biotin is required as cofactor in carbon dioxide fixation reaction.

Carbon dioxide fixation reactions are:

1. Acetyl CoA $\rightarrow$ Malonyl CoA.
2. Pyruvate $\rightarrow$ Oxaloacetate.
3. Propionyl CoA $\rightarrow$ Methyl malonyl CoA.
4. Formation of Carbamoyl phosphate in urea cycle.
5. In purine skeleton, i.e. C₆.

Q.42 What are carbon dioxide fixing or carboxylation reactions?

Carboxylation reactions are those in which a molecule of carbon dioxide is added to produce a carboxyl group.

Q.43 What is the active form of folic acid?
Tetrahydrofolic acid.

Q.44 What are the functions of folic acid?
As a carrier of one carbon moiety.

Q.45 Name one carbon moiety.

1. Methyl group ($-\text{CH}_3$).
2. Hydroxy methyl group ($-\text{CH}_2\text{OH}$).
3. Formyl group ($-\text{CHO}$).
4. Formimino group ($-\text{CH}=\text{NH}$).

Q.46 Name the reactions mediated by one carbon moiety.

1. Ethanolamine $\rightarrow$ Choline.
2. Glycine $\rightarrow$ Serine.
3. Norepinephrine $\rightarrow$ Epinephrine.
4. Guanidoacetic acid $\rightarrow$ Creatine.
5. Uracil $\rightarrow$ Thymine.
6. Ribonucleotides $\rightarrow$ Deoxyribonucleotides.
7. Formation of N-formylmethionine transfer RNA.
8. In purine synthesis (i.e. C-2 and C-8 positions in purine skeleton comes from one carbon moiety).

Q.47 What are the sources of folic acid?

Green leafy vegetables, cauliflower, liver, kidney etc.

Q.48 What are the 3 D's of niacin deficiency?

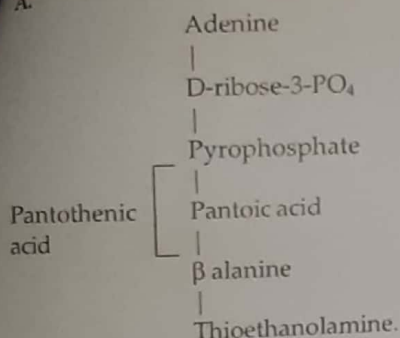
1. Diarrhea.
2. Dermatitis.
3. Dementia.

What are the functions of vitamin

glutamic acid $\rightarrow$ β Methyl aspartic acid.
 Methyl malonyl CoA $\rightarrow$ succinyl CoA.
 Ribonucleotides $\rightarrow$ Deoxy ribonucleo-
 tides.

Q.50 Name the components of coenzyme

A.



Q.51 The increased excretion of the following in the urine is a measure of the deficiency of which vitamins?

- Formiminoglutamic acid
- Methyl malonic acid
- Xanthurenic acid

d. Homogentisic acid

e. Pyruvic acid.

a. Folic acid.

b. Vitamin B₁₂.

c. Vitamin B₆.

d. Ascorbic acid.

e. Thiamine.

Q.52 What is the deficiency disease of vitamin B₁₂?

Pernicious anemia.

Q.53 What are the sources of vitamin B₁₂?

Liver, kidney, meat, milk, cheese.

Q.54 Which vitamin is present in coenzyme A?

Pantothenic acid.

Q.55 What is scurvy?

It is poor wound healing, easy bruising, bleeding gums, bleeding time and painful glossitis. It can ultimately lead to anemia.

Q.56 What are the deficiency of folic acid?

Megloblastic anaemia.

Hemocystinemia

Deficiency in early pregnancy causes neural tube defects in fetus.

Q.57 Who are the people more affected by Folic acid deficiency?

Alcoholics

Pregnant women in their early pregnancy.

Q.58 Name the proteins undergoing Vit K-dependent carbonylation?

Coagulation factors II, VII, IX, X and protein C and S.

Q.59 What are the deficiency of vit K?

- Fat malabsorption which can lead to bile duct occlusion.
- Prolong treatment with broad spectrum antibiotics supply of vit K by killing intestinal bacteria.
- Breast-fed newborns as mother milk is very low in vit K.
- Home births where babies were not given vit K. injections
- Infants whose mothers were treated with anticonvulsants during pregnancy.

Q.60 Name some anticoagulants?

Warfarine, dicumarol, heparine.

Q.61 Sources of vit D?

Sunlight is a very good source of vit D. Vit D₃ is found in salmon (saltwater fish) and egg yolks.

VITAMINS

Fat-soluble Vitamins

Vitamins	Functions	Sources	Deficiency diseases
A	- Visual cycle. - Maintenance of proper health of epithelium tissues - Stability of cellular and subcellular membranes - Synthesis of mucopolysaccharides. - Growth and reproduction - Bones and teeth. - Nucleic acid metabolism. - Electron transport chain and in oxidative phosphorylation	- Ready made sources : Fish liver oils such as shark, cod, halibut fish liver oils, milk products, egg yolk. - Provitamin sources: carrot, papaya, tomatoes	Night blindness
D	- In the absorption of calcium and phosphorous from the intestines. - Growth. - Mineralisation of bones.	Fish liver oils, egg yolk, milk products	Rickets
E	- As powerful antioxidants - Prevent autooxidation of vitamin A and carotenes - Prevent formation of fatty acids peroxidases in tissues due to autooxidation of unsaturated fatty acids with oxygen. - Protect lipids of biological membranes against oxygen - Prevent rancidity	Wheat germ oil, corn oil, peanut oil, soyaben oil, sunflower oil: egg yolk, spinach, alfalfa	Sterility in rats
K	- In the synthesis of prothrombin. - In oxidative process taking place in photosynthesis in plant kingdom. - In electron transport chain and in oxidative phosphorylation	Alfalfa, spinach, cabbage, cauliflower, egg yolk, liver.	- Hemorrhage conditions - Prolongation clotting time

Water-soluble Vitamins

Vitamins	Functions	Sources	Deficiency diseases
Thiamine (B ₁)	1. Oxidative decarboxylation of α -Ketoacids i.e. i. Pyruvic acid-acetyl CoA ii. α -keto gluta-succinylric acid CoA 2. Transketolation reaction	Yeast, outercoating of Seeds, cereals, legumes, wheat, pork, egg.	Human: Beriberi Rats: Bradycardia Beriberi
Riboflavin (B ₂)	Forms the part of FMN and FAD which functions as prosthetic group of various enzymes I. FMN i. Warburg yellow enzyme ii. L-Amino acid oxidase iii. Cytochrome C reductase. II FAD i. Acyl CoA dehydrogenase ii. D-Amino acid oxidase.	Yeast, milk, eggs, meat, fish liver, kidney, green leafy vegetables.	
Pantothenic acid (B ₃)	Forms the part of coenzyme A which serves as a carrier of acyl group in enzymatic reactions. They are: 1. Fatty acid oxidation. 2. Fatty acid synthesis 3. Pyruvic acid oxidation. 4. Biological acetylations 5. Cholesterol biosynthesis. 6. In acyl carrier proteins.	Yeast, liver, kidney, egg yolk, molasses.	
Niacin (B ₅)	As components of NAD and NADP ⁺ which acts as coenzymes for many anaerobic dehydrogenases reactions. Enzymes requiring NAD ⁺ or NADH coenzymes are: i. glyceraldehyde-3-PO ₄ dehydrogenase. ii. Lactate dehydrogenase. iii. Malic dehydrogenase. Enzymes requiring NADP ⁺ or NADPH as coenzymes are: i. Isocitrate dehydrogenase. ii. Glucose-6-PO ₄ dehydrogenase. iii. Aldolase reductase.	Yeast, meat, liver, kidney, eggs, legumes.	Man: Pellagra tongue
Pyridoxal (B ₆)	1. Transamination reaction. 2. Decarboxylation reaction. a. Histidine $\rightarrow$ Histamine b. Tyrosine $\rightarrow$ Tyramine c. 5 hydroxy $\rightarrow$ 5 hydroxy Tryptophan tryptamine d. Glutamic acid $\rightarrow$ γ amino butyric acid GABA e. α -amino β - δ -amino Keto adipic acid $\rightarrow$ levulinic acid 3. In dehydrase reaction a. Serine $\rightarrow$ Pyruvic acid b. Threonine $\rightarrow$ α keto butyric acid 4. Transsulfurase reaction Homocysteine $\rightarrow$ Serine 5. Desulphuration reaction a. Cystine $\rightarrow$ Pyruvic acid b. Homocystine $\rightarrow$ α -keto butyric acid 6. In tryptophan metabolism 7. In transport of amino acids and in absorption of amino acids. 8. In essential fatty acid synthesis.	Yeast, liver, egg yolk, rice polishings	
Biotin (B ₇)	In carboxylation reactions i.e. carbon dioxide fixing reactions: a. Acetyl CoA $\rightarrow$ Malonyl CoA b. Pyruvic acid $\rightarrow$ Oxaloacetic acid c. Propionyl CoA $\rightarrow$ D Methyl malonyl CoA	Yeast, egg yolk, milk, molasses, Peanuts.	

Contd.